16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime

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A 16-component Dirac field with a mass term and a quartic self-interaction is studied on a spacetime with seven space-like dimensions and one time-like dimension in which three of the space-like coordinates deflate exponentially as time advances. The Dirac equation is solved exactly: the deflation creates particles in every mode that depends on a deflating coordinate, with a Fermi-Dirac spectrum whose temperature is the rate divided by 2π, and fields that do not depend on the deflating coordinates obey the flat equations with couplings that grow with the deflation. Of the nine couplings of the self-interaction only four are independent. Matter and vacuum are analyzed with the Kohn-Sham method and with the effective action of Cornwall, Jackiw and Tomboulis, which agree. A gas becomes self-bound matter, a massless field acquires a mass, and fermion pairs bound by the Vorton-Scalar form a superfluid whose gap grows, each at a definite moment of the deflation. With equal rates the spacetime solves Einstein-Gauss-Bonnet gravity without matter; the solution is unstable, and matter accelerates the deflation. Bounds on the rate from cosmology and from atomic clocks are given.

I. INTRODUCTION

A. The question

This paper studies a single fermion field with sixteen complex components on a spacetime of eight dimensions. Two such spacetimes are of interest. One has four space-like and four time-like dimensions, and is written 4+4; the other has seven space-like dimensions and one time-like dimension, and is written 7+1. On both, the eight Dirac matrices can be chosen real [1–3], and the field supports a superfluid of fermion pairs of spin zero whose order parameter points along one of the four coordinates that are not the coordinates of ordinary spacetime. This paper treats the 7+1 spacetime.

The eight coordinates are written $x0, x1, \ldots, x7$. Throughout, $x1, x2, x3$ are the coordinates of ordinary space and $x4$ is the ordinary time. The coordinate $x0$ is an additional space-like coordinate, called the hidden coordinate. The remaining three, $x5, x6, x7$, are three additional *space-like* coordinates.

On a flat spacetime nothing explains why the fields of ordinary physics do not depend on $x5$, $x6$ and $x7$. On a flat 7+1 spacetime it has to be assumed. The present paper examines a definite proposal for what happens to these three coordinates, and it does so for a field with a mass and with a self-interaction.

B. The hypothesis of deflation

The hypothesis is that *the coordinates* $x5$*,* $x6$ *and* $x7$ *deflate exponentially*: as the ordinary time $x4$ advances, the proper extent of every interval along each of these coordinates shrinks as an exponential of $x4$. Flat spacetime is replaced by a curved spacetime with a diagonal metric, in which the three components of the metric along the deflating coordinates are

$$\begin{aligned} g_{55}^{(7+1)} &= +\mathrm{const5}^2\, e^{-2\,\mathrm{g55}\,x4}, \\ g_{66}^{(7+1)} &= +\mathrm{const6}^2\, e^{-2\,\mathrm{g66}\,x4}, \\ g_{77}^{(7+1)} &= +\mathrm{const7}^2\, e^{-2\,\mathrm{g77}\,x4}. \end{aligned} \tag{1}$$

Here g55, g66 and g77 are three positive constants, the *rates* of deflation; const5, const6 and const7 are three positive constants that set the scale at $x4 = 0$ and may differ from one another. The names of the rates are written upright and without a subscript, to distinguish them from the components $g_{55}^{(7+1)}$, $g_{66}^{(7+1)}$, $g_{77}^{(7+1)}$ of the metric, which carry subscripts and the superscript $(7+1)$. The other five components of the metric keep their flat values, $+1$ for $x0, \ldots, x3$ and -1 for the time $x4$. The sign convention is that space-like components of the metric are positive and time-like components negative: the plus signs in Eq. (1) say that $x5, x6, x7$ are space-like.

C. Mass term, self-interaction, and the two methods

The Lagrangian of the fermion field has three parts (Sec. III): the kinetic term; a *mass term*, bilinear in the field; and a *self-interaction*, quartic in the field. The self-interaction is the most general one without derivatives: nine invariants, one for each number $r = 0, \ldots, 8$ of Dirac matrices in the bilinears from which it is built, with nine couplings g_r. The field equation is a nonlinear Dirac equation. The paper answers the following questions about it. How many of the nine couplings are independent? What do the mass term and the self-interaction do to matter made of these fermions, on flat spacetime and while three coordinates deflate? What do they do to the pairing of the fermions, which in this model is mediated by a second field, the Vorton-Scalar? And what

does such matter do to the spacetime itself, if it is the source of gravity?

Two methods are used throughout, and each checks the other.

- The *Kohn-Sham method*, known from density-functional theory: the interacting fermions are replaced by independent fermions that move in fields which are determined self-consistently by their own densities. For uniform matter the fields are a shift of the mass and a shift of the energy.
- The *effective potential for composite operators* of Cornwall, Jackiw and Tomboulis (CJT): a function of the full propagator of the fermions whose stationary points are the possible states of matter and whose value there is minus the pressure.

The two methods are equivalent at the level of approximation used here, and the paper verifies this numerically (Sec. XIII).

The paper also examines three hypotheses about the superfluid.

H1. The fermion field forms a superfluid state of spin 0 in which the fermions are paired by the exchange of a new scalar boson, the *Vorton-Scalar*, through a neutral current of the kind known from the weak interaction.

H2. This superfluid state is the Higgs field.

H3. This superfluid state explains the observations of dark matter, or the Hubble tension, or both.

The purpose is to carry out every calculation that these hypotheses call for under the hypothesis of deflation, with the mass term and the self-interaction, and to report what follows for H1, H2 and H3, favorable or not.

D. Summary of results

Geometry (Sec. II). The deflating spacetime is homogeneous and has 14 symmetries, or 15 or 20 if two or three of the rates are equal, instead of the 36 of flat spacetime. It is not a solution of Einstein's equations without matter. With three equal rates it *is* an exact solution of the equations of Einstein-Gauss-Bonnet gravity without matter, for one definite relation between the rate and the two constants of that theory.

The Dirac field (Secs. III, V). A field that does not depend on $x5, x6, x7$ obeys, after multiplication by the square root of the volume factor of the deflating coordinates, exactly the flat equation with the mass unchanged and with every coupling of the self-interaction growing as the exponential of the time. A field that does depend on the deflating coordinates is affected directly. The linear equation is solved exactly. *The deflation creates particles*: a mode that starts empty ends with the occupation of a Fermi-Dirac distribution at a temperature equal to the rate divided by 2π, the Gibbons-Hawking temperature of the de Sitter space that the deflating coordinates and the time form when the rates are equal.

The self-interaction (Sec. IV). For anticommuting fields the nine invariants obey five linear relations: only four couplings are independent, a scalar, a vector, an axial-vector and a pseudoscalar one. Uniform matter feels two combinations of them, pairs of particles at rest feel a third.

Matter (Secs. VII, XIII). The Kohn-Sham method and the CJT effective potential give the same states. For matter that has momentum along all seven space coordinates the deflation raises the density and the anisotropy of the pressure, and matter that is self-bound at one moment is so only for about one e-folding of the deflating volume. In the sector of fields that do not depend on the deflating coordinates a scalar attraction grows with the deflation and turns a gas into self-bound matter at a definite moment. With the filled states of negative energy taken into account, the same growth gives a massless field a mass at a critical coupling. A bare mass together with the Vorton-Scalar gives the phase mode of that field a mass.

The superfluid (Secs. VI to XIV). For fields that do not depend on the extra coordinates the pairing calculation of flat spacetime holds at each moment with the couplings of that moment. The masses do not change. Every threshold is passed at a definite time. The gap of the paired state grows much faster than the coupling; a scalar contact attraction raises it, a vector contact repulsion lowers it. Two fermions become bound at a scattering resonance. The paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength; before that it lags behind and breaks pairs. Matter of a given density loses its fermion mass at a definite time. A vortex line carries four fermion zero modes.

Gravity (Sec. XV). With Kohn-Sham matter as the source of the Einstein-Lovelock equations the deflating spacetime is not stationary: the three coordinates deflate faster than exponentially and the four flat coordinates expand. Even without matter the exactly exponential deflation is an unstable solution of the gravity in which it is a solution.

Cosmology (Sec. XVI). If gravity is the Einstein theory in eight dimensions, Newton's constant grows at the total rate of deflation; lunar laser ranging then bounds that rate to 3×10^{-3} of the Hubble rate, and the Hubble constant inferred from the microwave background falls. The deflation does not relieve the Hubble tension. Whatever keeps the rates constant must carry a tension and must take part in the perturbations of ordinary space; the spectra of the microwave background at all but the largest angular scales do not depend on how it does so, and the growth of structure does.

The Higgs hypothesis (Sec. XVII). The rotations of the four extra space-like coordinates, under which the pair condensate has the pattern of the Higgs vacuum, are no longer symmetries of the spacetime; the pattern survives

only as a hidden symmetry. A vector or an axial-vector contact coupling breaks one factor of that symmetry. If the condensate were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would limit the rate of deflation to about 10^{-6} of the Hubble rate. The hypothesis H2 and a deflation of any consequence today exclude each other.

E. Method, and the limits of what is claimed

Every statement of this paper about geometry, matrices and operators is an identity that a program has verified in exact rational arithmetic, and that an independent program has verified again by computer algebra or with Grassmann-valued fields. Every numerical statement is the output of a program, with a stated accuracy, and the central ones were computed twice by independent programs. The programs and their output are public [4]; Appendix D says how each number of this paper is reproduced. The program performs 295 checks and the independent verification 80, all of which pass. The metric of Eq. (1) is a hypothesis. The paper derives its consequences; it does not claim that the hypothesis is true. Section XVIII lists what has been established and what has not.

Units are natural, $\hbar = c = k_B = 1$. Symbols that carry the tag $1671d$ refer to the deflating 7+1 spacetime. A general coordinate among the three deflating ones is written xt with $t \in \{5,6,7\}$ and its rate rate_t; a general coordinate among $x0, x1, x2, x3$ is written xj with $j \in \{0,1,2,3\}$. The letter a is kept for the four extra axes, $a \in \{0,5,6,7\}$ (Sec. VIII). Every symbol is defined where it first appears, and Appendix A lists all of them.

II. THE DEFLATING SPACETIME

This section derives the geometry that the hypothesis implies: the curvature, the field equations that the metric does and does not solve, its symmetries, and the motion of a classical particle. Every formula of this section was obtained twice by exact computer algebra, by the program of this paper and by an independent script (Appendix D); the steps are short enough to be repeated by hand, and the text says how.

A. Metric, frame and volume

Throughout, the metric is written in the sign convention of the hypothesis (1): *space-like components are positive and time-like components are negative*. The flat metric is

$$\eta^{(7+1)} = \text{diag}(+1,+1,+1,+1,-1,+1,+1,+1), \qquad (2)$$

with the coordinates in the order $x0,\ldots,x7$: seven space-like coordinates and the one time-like coordinate $x4$. Define three *scale factors*, one for each deflating coordinate,

$$b_t(x4) = \text{const}_t\, e^{-\text{rate}_t\, x4}, \qquad t = 5,6,7, \qquad (3)$$

with $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$. The metric of the deflating spacetime is

$$g^{(7+1)}_{\mu\nu} = \text{diag}\big(+1,+1,+1,+1,-1,\, +b_5^2,\, +b_6^2,\, +b_7^2\big), \qquad (4)$$

whose last three components are those of Eq. (1). The superscript $(7+1)$ marks the metric of this spacetime in this sign convention; a component of the metric always carries it, so that $g^{(7+1)}_{55}$ cannot be taken for the rate g55. Greek indices label the coordinates. A coordinate interval dxt has the proper extent $b_t\,|dxt|$, which shrinks by the factor e in the time $1/\text{rate}_t$. Because $x5, x6, x7$ are space-like, what shrinks is a length.

Capital indices A, B label an orthonormal frame, eight vector fields $e_A = e_A{}^\mu \partial_\mu$ with $g^{(7+1)}_{\mu\nu} e_A{}^\mu e_B{}^\nu = \eta^{(7+1)}_{AB}$. We take the frame along the coordinates, $e_A{}^\mu = \text{diag}(1,1,1,1,1, b_5^{-1}, b_6^{-1}, b_7^{-1})$. The volume element is $V_d\, d^8x$ with

$$V_d(x4) = b_5 b_6 b_7 \propto e^{-\text{gsum}\, x4}, \quad \text{gsum} = \text{g55} + \text{g66} + \text{g77}. \qquad (5)$$

V_d is the proper volume of a unit coordinate cell of the deflating coordinates, and the *total rate* gsum is the rate at which it shrinks. Most results below depend on the three rates only through gsum.

B. Curvature

The only components of the metric that depend on a coordinate are the three $g^{(7+1)}_{tt}$, and they depend on $x4$ alone. The Christoffel symbols $\Gamma^\rho{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$ that do not vanish are therefore

$$\Gamma^t{}_{4t} = -\text{rate}_t, \qquad \Gamma^4{}_{tt} = -\text{rate}_t\, b_t^2, \qquad (6)$$

without summation over t. The Riemann tensor has, in the orthonormal frame, the components

$$R^t{}_{4t4} = -\text{rate}_t^2, \quad R^t{}_{sts} = +\text{rate}_t\, \text{rate}_s \;\; (t \neq s), \qquad (7)$$

and those related to them by symmetry. No component carries an index of $x0,\ldots,x3$: these four coordinates span a flat space, and the curvature lives in the four coordinates $x4,\ldots,x7$. With $\text{gsq} = \text{g55}^2 + \text{g66}^2 + \text{g77}^2$ and $\text{gprod} = \text{g55g66} + \text{g55g77} + \text{g66g77}$, the curvature scalar is the constant $R = \text{gsq} + \text{gsum}^2$, and the Einstein tensor $\mathcal{G}^\mu{}_\nu = R^\mu{}_\nu - \frac{1}{2} R\, \delta^\mu_\nu$ is diagonal with

$$\begin{aligned} \mathcal{G}^4{}_4 &= -\text{gprod}, \qquad \mathcal{G}^j{}_j = -\tfrac{1}{2}(\text{gsq} + \text{gsum}^2), \\ \mathcal{G}^t{}_t &= \text{rate}_t\, \text{gsum} - \tfrac{1}{2}(\text{gsq} + \text{gsum}^2), \end{aligned} \qquad (8)$$

with $j = 0,1,2,3$. In the sign convention used here these mixed components are the same on the 4+4 and on the

7+1 spacetime. (They follow from the sectional curvatures of Appendix C: ${\mathcal{G}^A}_A$ is minus the sum of the sectional curvatures of the planes that do not contain the direction A.) The Einstein tensor does not vanish, and it is not a multiple of the unit matrix unless all rates vanish. *The deflating spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant.*

C. A solution of Einstein-Gauss-Bonnet gravity

In more than four dimensions the general field equations of second order for the metric are those of Lovelock [5]. Their first term beyond Einstein's is the Lanczos tensor ${\mathcal{H}^\mu}_\nu$ of the Gauss-Bonnet term [6], and the equations without matter read

$$ {\mathcal{G}^\mu}_\nu + \Lambda_8\, {\delta^\mu}_\nu + \alpha_{\rm GB}\, {\mathcal{H}^\mu}_\nu = 0, \tag{9} $$

with a cosmological constant Λ_8 and a coupling $\alpha_{\rm GB}$. For the metric (4) we find

$$ \begin{aligned} &{\mathcal{H}^j}_j = -4\,\mathrm{g55}\,\mathrm{g66}\,\mathrm{g77}\,\mathrm{gsum} \quad (j=0,1,2,3),\\ &{\mathcal{H}^4}_4 = {\mathcal{H}^t}_t = 0, \end{aligned} \tag{10} $$

and the Lovelock tensor of the third order, the last that can differ from zero in eight dimensions, vanishes identically. The components of Eq. (9) along $x4,\dots,x7$ therefore contain no Gauss-Bonnet term and require the four corresponding components of the Einstein tensor to be equal, which holds if and only if the three rates are equal. The components along $x0,\dots,x3$ then fix the coupling. The result:

$$ \mathrm{g55} = \mathrm{g66} = \mathrm{g77}: \qquad \Lambda_8 = 3\,\mathrm{g55}^2, \qquad \alpha_{\rm GB} = -\frac{1}{4\,\mathrm{g55}^2}. \tag{11} $$

With three equal rates the deflating spacetime is an exact solution of Einstein-Gauss-Bonnet gravity without matter, in a theory whose two constants satisfy $4\alpha_{\rm GB}\Lambda_8 = -3$; the rate is then $\mathrm{g55} = \sqrt{\Lambda_8/3}$. With unequal rates it is not a solution for any Λ_8 and $\alpha_{\rm GB}$. Solutions of this theory in which every scale factor is an exponential of the time have been classified in five and six dimensions [7]; the present one, with four static and three contracting directions in eight dimensions, is of that kind. In the notation of the Einstein-Lovelock equations of Sec. XV, Eq. (11) reads $\bar\Lambda = 3\,\mathrm{g55}^2$, $\alpha_2 = 1/(4\,\mathrm{g55}^2)$. Section XV examines the stability of this solution, with and without matter: it is unstable.

D. Symmetries

A Killing vector is a vector field along which the metric does not change. Flat spacetime in eight dimensions has 36. The deflating spacetime has, for any rates, 14: the seven translations along all coordinates but $x4$; the six rotations of $x0,x1,x2,x3$; and the scaling

$$ \mathcal{X}_{\rm scale} = \partial_4 + \mathrm{g55}\,x5\,\partial_5 + \mathrm{g66}\,x6\,\partial_6 + \mathrm{g77}\,x7\,\partial_7, \tag{12} $$

which shifts the time and stretches the deflating coordinates so as to undo the shrinking. Two equal rates add the rotation of the two coordinates concerned. Three equal rates add the three rotations of $x5,x6,x7$ and three further vectors, 20 in all; the four coordinates $x4,\dots,x7$ then form a space of constant curvature of radius $1/\mathrm{g55}$: de Sitter space in the half of its flat slicing in which space contracts. These lists are complete. (A Killing vector that vanishes at a point is a Lorentz transformation of the tangent space that leaves the Riemann tensor (7) unchanged; the dimension of the space of such transformations is 6, 7 or 12, which bounds the number of Killing vectors by 14, 15 or 20.)

Three consequences are used below. The spacetime is homogeneous: no place and no moment is distinguished. The constants const_t are not observable by themselves, since a rescaling of xt changes them; the deflation is observable only through an object that fixes a coordinate scale along xt, such as a wave of given coordinate wavelength. And there is no conserved energy, because a translation in time alone is not a symmetry.

Of the 28 rotations and boosts of flat spacetime, only the six rotations of $x0,\dots,x3$ survive for general rates (Fig. 1). Nothing that involves the time survives, in particular not the boosts of ordinary special relativity: the deflation defines a preferred state of rest, as the expansion of the universe does.

E. Geodesics

The translations give seven conserved momenta P_μ ($\mu \neq 4$) of a classical particle. Let $E_\perp = (m^2 + \sum_j P_j^2)^{1/2}$, with the sum over $j=0,1,2,3$, be the energy that a particle of mass m would have without momentum along the deflating coordinates. The conservation of the norm of the momentum gives the velocity along the time, with an affine parameter ς along the path:

$$ \left(\frac{dx4}{d\varsigma}\right)^2 = E_\perp^2 + \sum_t \frac{P_t^2}{b_t(x4)^2}. \tag{13} $$

The last term grows as $e^{2\,\mathrm{rate}_t\,x4}$: the momentum P_t belongs to a fixed coordinate wavelength, and the proper wavelength shrinks. With the plus sign the energy of a particle that moves along a deflating coordinate grows without bound, the reverse of the redshift in an expanding universe. For momentum along $x5$ alone the proper time that elapses between $x4$ and $x4=\infty$ is

$$ \frac{m}{\mathrm{g55}\,E_\perp}\,\mathrm{arsinh}\Big(\frac{E_\perp\,\mathrm{const5}\,e^{-\mathrm{g55}\,x4}}{|P_5|}\Big), \tag{14} $$

which is finite [Fig. 2(a)]: the coordinates of Eq. (4) do not cover the whole future of such a particle. For equal

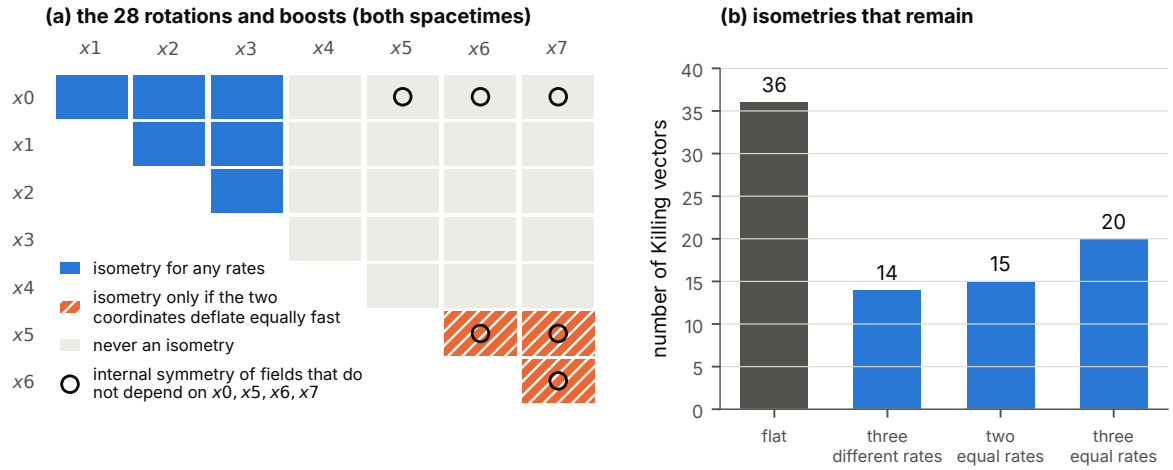


FIG. 1. Symmetries of the deflating spacetime. (a) The 28 rotations and boosts of flat spacetime, one square for each pair of coordinates: those that remain symmetries for any rates, those that remain symmetries only if the two coordinates deflate equally fast, and those that are never symmetries. Circles mark the six pairs whose rotations act as internal symmetries on fields that do not depend on $x0, x5, x6, x7$ (Sec. XVII). (b) The number of Killing vectors.

rates this is the known incompleteness of the flat slicing of de Sitter space. Section V shows the quantum form of the blueshift. (On the 4+4 spacetime the sign in Eq. (13) is the opposite, and such a particle is turned back.)

III. THE DIRAC FIELD WITH MASS TERM AND SELF-INTERACTION

A. Matrices and spin connection

At each point of a curved spacetime the tangent space is flat. The Dirac matrices of the orthonormal frame are therefore eight constant matrices γ^A, $A = 0, \ldots, 7$. They are real 16×16 matrices with entries 0 and ± 1, printed in full in Appendix B. The matrix of the time, γ^4, is symmetric and squares to $+1$; the seven matrices of the space-like coordinates are antisymmetric and square to -1; different matrices anticommute:

$$\gamma^A \gamma^B + \gamma^B \gamma^A = -2\,\eta^{AB}_{(7+1)}. \tag{15}$$

The minus sign is a consequence of the sign convention of the metric (space-like components positive) together with matrices whose time matrix squares to $+1$; it accompanies the matrices through all formulas below. All of these properties were verified by exact integer arithmetic for this work. The matrices with a coordinate index are $\gamma^\mu = e_A{}^\mu \gamma^A$: five of them are the constant matrices, and the three along the deflating coordinates are the constant matrices divided by the scale factors, $\gamma^{A=t}/b_t$. They satisfy $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = -2\,g^{\mu\nu}_{(7+1)}$ with the inverse metric $g^{\mu\nu}_{(7+1)}$.

A spinor is defined relative to a frame, and its derivative must account for the turning of the frame from point to point [8]. The covariant derivative of a spinor is $\nabla_\mu\psi = (\partial_\mu + \Omega_\mu)\psi$ with the spin connection

$$\begin{aligned} \Omega_\mu &= -\tfrac{1}{4}\,\omega_{\mu AB}\,\gamma^A\gamma^B, \\ \omega_{\mu AB} &= \eta^{(7+1)}_{AC}\,e^C{}_\nu\big(\partial_\mu e_B{}^\nu + \Gamma^\nu{}_{\mu\lambda} e_B{}^\lambda\big), \end{aligned} \tag{16}$$

whose coefficients $\omega_{\mu AB} = -\omega_{\mu BA}$ follow from the metric ($e^C{}_\nu$ is the inverse of the frame). The minus sign is that of Eq. (15): the matrices $-\frac{1}{4}[\gamma^A, \gamma^B]$ generate the Lorentz transformations of the spinors. For the metric (4) the only coefficients that do not vanish are $\omega_{t\,t4} = -\omega_{t\,4t} = -\mathrm{rate}_t\, b_t$, so that

$$\begin{aligned} &\Omega_t = \tfrac{1}{2}\,\mathrm{rate}_t\, b_t\, \gamma^t\gamma^4 \quad (t = 5, 6, 7), \\ &\Omega_0 = \cdots = \Omega_4 = 0, \end{aligned} \tag{17}$$

with the constant matrices of the frame on the right. Each Ω_t generates the Lorentz transformation in the plane of xt and $x4$. The spin connection is correct if and only if the Dirac matrices are covariantly constant, $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$, which holds for all 64 pairs of indices. Its curvature is the Riemann tensor, $[\nabla_\mu, \nabla_\nu]\psi = -\frac{1}{4} R_{AB\mu\nu}\gamma^A\gamma^B\psi$, a third route to Eq. (7). One combination enters the Dirac equation:

$$\gamma^\mu\Omega_\mu = -\tfrac{1}{2}\,\mathrm{gsum}\,\gamma^4. \tag{18}$$

The scale factors cancel, and only the total rate remains.

B. The Lagrangian: kinetic term, mass term, self-interaction

The field of this paper, $\psi_{1671d}(x)$, is a column of sixteen complex-valued Grassmann operators. Its Lagrangian has three parts,

$$\mathcal{L}_{1671d} = V_d\,\big[\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}} + \mathcal{L}_{\mathrm{self}}\big], \tag{19}$$

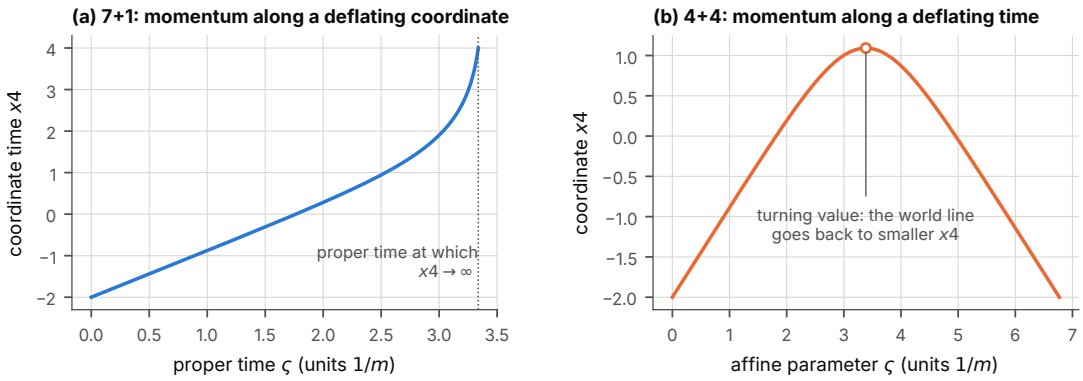


FIG. 2. A classical particle of the mass $m = 1$ with the momenta $P_1 = 0.5$ along $x1$ and $P_5 = 0.3$ along the deflating coordinate $x5$, for g55 = 1 and const5 = 0.8, started at $x4 = -2$. (a) On the 7+1 spacetime the time $x4$ grows without bound within the finite proper time 3.386. (b) On the 4+4 spacetime the path reaches the turning value $x4 = 1.0924$ and returns to smaller $x4$.

where $V_d = \sqrt{|\det g|}$ is the factor that makes $\mathcal{L}_{1671d}\, d^8x$ invariant. A bosonic field theory would call the second part the "mass times ϕ^2" term and the third the "ϕ^4" term; for a fermion field they are bilinear and quartic in ψ.

Kinetic and mass terms. With the adjoint $\bar\psi = \psi^\dagger\gamma^4$,

$$\mathcal{L}_{\rm kin} + \mathcal{L}_{\rm mass} = \bar\psi\left(i\gamma^\mu\nabla_\mu - m\right)\psi. \tag{20}$$

The mass term $-m\,\bar\psi\psi$ is real because γ^4 is symmetric, and it is a scalar under the Lorentz transformations of the frame.

Self-interaction. The most general interaction without derivatives that is quartic in the field and invariant under the Lorentz transformations of the frame is built from the products of the Dirac matrices. Let $K = (A_1 < A_2 < \cdots < A_r)$ be a strictly increasing list of r indices out of $0,\dots,7$; $r = |K|$ is its *rank*, and there are $2^8 = 256$ lists, with $1, 8, 28, 56, 70, 56, 28, 8, 1$ of the ranks $0,\dots,8$. Define

$$\Theta^K = \zeta_r\,\gamma^{A_1}\gamma^{A_2}\cdots\gamma^{A_r}, \qquad \zeta_r = \begin{cases} i, & r = 2,3,6,7,\\ 1, & \text{otherwise,}\end{cases} \tag{21}$$

($\Theta^{()} = 1$ for the empty list) and $\Theta_K = (\Theta^K)^{-1} = \mathrm{sgn}_K\,\Theta^K$, where the sign $\mathrm{sgn}_K = \pm1$ is the product of the squares of the matrices in K. The phase ζ_r makes $\gamma^4\Theta^K$ Hermitian, so that every bilinear $\bar\psi\Theta^K\psi$ is real. The *invariant of rank r* and the self-interaction are

$$I_r = \sum_{|K|=r}(\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi), \qquad \mathcal{L}_{\rm self} = -\frac{1}{2}\sum_{r=0}^{8} g_r\, I_r. \tag{22}$$

The nine real constants g_r are the couplings. $I_0 = (\bar\psi\psi)^2$ is the square of the scalar density and $I_1 = \sum_A(\bar\psi\gamma_A\psi)(\bar\psi\gamma^A\psi)$ the square of the current, where $\gamma_A = (\gamma^A)^{-1}$; with the chirality matrix χ of Sec. VI, $I_8 = -(\bar\psi\, i\chi\,\psi)^2$ and $I_7 = -\sum_A(\bar\psi\gamma_A\chi\psi)(\bar\psi\gamma^A\chi\psi)$. In the traditional names the scalar, vector, axial-vector and pseudoscalar couplings are $g_S = g_0$, $g_V = g_1$, $g_A = -g_7$, $g_P = -g_8$. A coupling g_r carries one digit as its subscript; it must not be confused with a rate (g55) or with a component of the metric ($g^{(7+1)}_{55}$). In a gravitational field each I_r is written with the frame matrices exactly as in flat spacetime, because it is invariant under the Lorentz transformations of the frame. Section IV shows that for anticommuting fields only four of the nine invariants are independent.

C. The field equation

The variation of the action with respect to $\bar\psi$ gives

$$\left(i\gamma^\mu\nabla_\mu - m\right)\psi_{1671d} = \sum_{r=0}^{8} g_r \sum_{|K|=r}\left(\bar\psi\Theta_K\psi\right)\Theta^K\psi. \tag{23}$$

(The bilinears are even elements of the Grassmann algebra, so that the derivative with respect to $\bar\psi$ acts on $\mathcal{L}_{\rm self}$ as on a product of ordinary numbers.) The left-hand side is the Dirac operator with the mass term; the right-hand side is the self-interaction, cubic in the field. Without it the equation is linear. With Eq. (18),

$$i\gamma^4\Big(\partial_4 - \frac{\mathrm{gsum}}{2}\Big)\psi + i\sum_{j=0}^{3}\gamma^j\partial_j\psi + i\sum_{t=5}^{7}\frac{\gamma^t}{b_t}\,\partial_t\psi - m\,\psi$$
$$= \sum_{r=0}^{8} g_r \sum_{|K|=r}\left(\bar\psi\Theta_K\psi\right)\Theta^K\psi, \tag{24}$$

with the constant matrices. Three things distinguish the left-hand side and the interaction from their forms on flat spacetime: the derivative along the time is shifted by gsum/2, each derivative along a deflating coordinate is divided by its scale factor, and, as the next subsection shows, the self-interaction is strengthened by the shrinking of the volume.

D. The weighted field

The shift is removed by a change of normalization. Define the *weighted field*

$$\hat{\psi} = \sqrt{V_d}\,\psi. \tag{25}$$

Because $\partial_4\sqrt{V_d} = -\frac{1}{2}\mathrm{gsum}\sqrt{V_d}$, the left-hand side of Eq. (24) applied to ψ equals $V_d^{-1/2}$ times the same expression without the term gsum/2 applied to $\hat{\psi}$. The right-hand side is cubic in the field and therefore equals $V_d^{-3/2}$ times the same expression in $\hat{\psi}$. The weighted field thus obeys Eq. (24) *without the term* gsum/2 *and with every coupling* g_r *replaced by*

$$\frac{g_r}{V_d(x4)} \;\propto\; g_r\, e^{\mathrm{gsum}\, x4}. \tag{26}$$

The quantity $\psi^\dagger\psi$ is a density per unit of proper volume and grows as the deflating coordinates shrink; $\hat{\psi}^\dagger\hat{\psi}$ is the density per unit of coordinate volume, from which this trivial growth is removed.

The central observation of this paper follows at once. *For a field that does not depend on* $x5$, $x6$, $x7$, *the equation of the weighted field is exactly the flat equation with a self-interaction whose couplings grow as* $e^{\mathrm{gsum}\, x4}$. Every result that holds on flat spacetime for such fields, in particular everything about the pairing of the fermions, holds for the weighted field at each moment with the couplings of that moment. The deflation enters those results in one way only, through the dependence of the couplings on V_d (Sec. VI): the mass term is not changed, and every quartic coupling grows at the same rate.

A field that does depend on a deflating coordinate feels the deflation directly. The rest of this section and Sec. V treat the equation without the self-interaction, which is linear and can be solved exactly; it describes a field so dilute that the cubic term is negligible, and the single-particle states from which the many-body treatment of Secs. IV–VII starts. Because the coefficients depend on $x4$ only, a solution of the linear equation can be taken proportional to $\exp(i\sum_j p_j\, xj + i\sum_t k_t\, xt)$ with seven constant wavenumbers, four along the flat coordinates xj, $j = 0, 1, 2, 3$, and three along the deflating ones; such a solution is a *mode*. Multiplying the equation of the weighted field by γ^4 gives an ordinary differential equation in the time:

$$i\,\partial_4\hat{\psi} = \mathsf{h}(x4)\,\hat{\psi}, \qquad \mathsf{h} = \mathsf{h}_\perp + k_d(x4)\,\alpha^d, \tag{27}$$

where $\mathsf{h}_\perp = m\beta + \sum_j p_j\alpha^j$ is the Hamiltonian of the modes that do not depend on the deflating coordinates ($\beta = \gamma^4$, $\alpha^A = \gamma^4\gamma^A$),

$$k_d(x4) = \Big(\sum_t \frac{k_t^2}{b_t(x4)^2}\Big)^{1/2} \tag{28}$$

is the *physical* wavenumber along the deflating coordinates, and $\alpha^d = k_d^{-1}\sum_t (k_t/b_t)\alpha^t$ is the matrix α along its direction. With $E_\perp = (m^2 + \sum_j p_j^2)^{1/2}$,

$$\mathsf{h}^2 = \big(E_\perp^2 + k_d^2\big)\,1. \tag{29}$$

The Hamiltonian is real and symmetric at every time: the evolution is unitary and $\hat{\psi}^\dagger\hat{\psi}$ is conserved. But the Hamiltonian depends on the time, through a term that anticommutes with β, and the energy of a mode with $k_t \neq 0$ grows as $e^{\mathrm{rate}_t\, x4}$. A Hamiltonian of this kind creates pairs [9].

IV. THE SELF-INTERACTION: NINE CHANNELS, FOUR COUPLINGS

This section analyzes the quartic term $\mathcal{L}_{\mathrm{self}}$ of Eq. (22) by exact algebra. Its results are used in every later section: the number of independent couplings, the couplings that act in uniform matter, the couplings that act between two particles, and the symmetries that the self-interaction keeps or breaks.

A. The Fierz matrix

The 256 matrices Θ^K are a basis of all 16×16 matrices, $\mathrm{Tr}(\Theta_K\Theta^J) = 16\,\delta_{KJ}$, so that $\sum_K (\Theta_K)_{cd}(\Theta^K)_{ab} = 16\,\delta_{ad}\delta_{bc}$. This completeness relation rearranges a product of two bilinears, as in the rearrangement of Fierz for four Dirac matrices [10]. For four columns a, b, c, d of ordinary (commuting) numbers,

$$\sum_{|K|=r} (a\,\Theta_K b)(c\,\Theta^K d) = \sum_{r'=0}^{8} \mathsf{F}_{rr'} \sum_{|J|=r'} (a\,\Theta_J d)(c\,\Theta^J b), \tag{30}$$

with the *Fierz matrix*

$$\mathsf{F}_{rr'} = \frac{1}{16}\sum_l (-1)^{rr'-l}\binom{r'}{l}\binom{8-r'}{r-l}. \tag{31}$$

The derivation takes three lines. Expanding the left-hand side of Eq. (30) in the basis gives the coefficient $\frac{1}{256}\sum_{|K|=r}\mathrm{Tr}(\Theta_K\Theta_J\Theta^K\Theta^J)$ for each list J of rank r'. Two products of Dirac matrices either commute or anticommute: $\Theta_K\Theta_J\Theta^K = (-1)^{rr'-l}\Theta_J$ if the lists K and J have l indices in common. And $\binom{r'}{l}\binom{8-r'}{r-l}$ lists K of rank r share exactly l indices with a given J of rank r'. Table I prints the matrix. It satisfies $\mathsf{F}^2 = 1$; its eigenvalues are $+1$ (five times) and -1 (four times).

B. Anticommuting fields: only four invariants are independent

The components of ψ anticommute. Setting $a = c = \bar{\psi}$ and $b = d = \psi$ in Eq. (30), the right-hand side contains

TABLE I. The Fierz matrix: the integers $16\,\mathsf{F}_{rr'}$ of Eq. (31), rows $r = 0, \dots, 8$, columns $r' = 0, \dots, 8$.

$r\backslash r'$	0	1	2	3	4	5	6	7	8
0	1	1	1	1	1	1	1	1	1
1	8	−6	4	−2	0	2	−4	6	−8
2	28	14	4	−2	−4	−2	4	14	28
3	56	−14	−4	6	0	−6	4	14	−56
4	70	0	−10	0	6	0	−10	0	70
5	56	14	−4	−6	0	6	4	−14	−56
6	28	−14	4	2	−4	2	4	−14	28
7	8	6	4	2	0	−2	−4	−6	−8
8	1	−1	1	−1	1	−1	1	−1	1

the two fields ψ in the opposite order, which costs a sign. The nine invariants therefore obey

$$I_r = -\sum_{r'} \mathsf{F}_{rr'}\, I_{r'}, \qquad \text{that is} \qquad (1+\mathsf{F})\, I = 0. \tag{32}$$

Because F has the eigenvalue $+1$ five times, these are five independent linear relations: *of the nine invariants only four are independent.* Solved for the tensor invariants,

$$\begin{aligned} I_2 &= -3I_0 - I_1 - I_7 - I_8, & I_3 &= -4I_0 - I_7 + 4I_8, \\ I_4 &= -5I_0 - 5I_8, & I_5 &= -4I_0 - I_1 + 4I_8, \\ I_6 &= -I_0 + I_1 + I_7 - 3I_8. \end{aligned} \tag{33}$$

Inserted into Eq. (22),

$$\mathcal{L}_{\text{self}} = -\tfrac{1}{2}\big(g_0^{\text{eff}}\, I_0 + g_1^{\text{eff}}\, I_1 + g_7^{\text{eff}}\, I_7 + g_8^{\text{eff}}\, I_8\big) \tag{34}$$

with the four *effective couplings*

$$\begin{aligned} g_0^{\text{eff}} &= g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6, \\ g_1^{\text{eff}} &= g_1 - g_2 - g_5 + g_6, \\ g_7^{\text{eff}} &= g_7 - g_2 - g_3 + g_6, \\ g_8^{\text{eff}} &= g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6. \end{aligned} \tag{35}$$

A tensor coupling $g_2, \dots, g_6$ is not a new parameter of the theory: it shifts the scalar, vector, axial-vector and pseudoscalar couplings. Every physical quantity computed below depends on the nine g_r only through the four combinations (35), and this is verified in each case. The theorem (32) was verified in two independent ways: by exact integer arithmetic on the 16^4 coefficients of the quartic form, antisymmetrized as anticommuting fields require, and by evaluating the nine invariants on Grassmann-valued fields with an implementation of the Grassmann algebra; in both, the rank of the nine invariants is exactly four.

C. Uniform matter: the Hartree-Fock couplings

In uniform matter at rest the expectation value of a product of two fields is the density matrix $\mathsf{n}_{ab} = \langle \psi_b^\dagger \psi_a \rangle$. For a filled Fermi sea of particles of mass m^* it is $\mathsf{n} = (n + n_S\,\beta)/16$, where $n = \langle \psi^\dagger \psi \rangle$ is the number density and $n_S = \langle \bar\psi \psi \rangle$ the scalar density, with $\beta = \gamma^4$. By Wick's theorem the expectation value of a normal-ordered product of two bilinears is a direct (Hartree) term minus an exchange (Fock) term, $\langle (\psi^\dagger M \psi)(\psi^\dagger M \psi) \rangle = [\mathrm{Tr}(M\mathsf{n})]^2 - \mathrm{Tr}(M\mathsf{n}M\mathsf{n})$. The energy density of the self-interaction, $-\langle \mathcal{L}_{\text{self}} \rangle$, becomes

$$\varepsilon_{\text{int}} = \tfrac{1}{2}\, G_V^{\text{HF}}\, n^2 + \tfrac{1}{2}\, G_S^{\text{HF}}\, n_S^2, \tag{36}$$

$$\begin{aligned} G_S^{\text{HF}} &= \tfrac{1}{16}\big(15\, g_0^{\text{eff}} - 8\, g_1^{\text{eff}} - 8\, g_7^{\text{eff}} - g_8^{\text{eff}}\big), \\ G_V^{\text{HF}} &= \tfrac{1}{16}\big(-\, g_0^{\text{eff}} + 22\, g_1^{\text{eff}} - 6\, g_7^{\text{eff}} + g_8^{\text{eff}}\big). \end{aligned} \tag{37}$$

In terms of the nine couplings, $G_S^{\text{HF}} = \frac{1}{16}(15g_0 - 8g_1 - 28g_2 - 56g_3 - 70g_4 - 56g_5 - 28g_6 - 8g_7 - g_8)$ and $G_V^{\text{HF}} = \frac{1}{16}(-g_0 + 22g_1 - 14g_2 + 14g_3 - 14g_5 + 14g_6 - 6g_7 + g_8)$. The direct terms alone would give $G_S = g_0$ and $G_V = g_1$; the exchange terms reduce the scalar coupling by one sixteenth and add to the vector coupling. Equation (36) is the reason why uniform matter is described by two numbers, a scalar coupling G_S and a vector coupling G_V, whatever the nine couplings are: Sec. VII takes these two as its parameters.

D. Two particles: the pair couplings

Two particles at rest can be in $8 \cdot 7/2 = 28$ antisymmetric states, since a particle at rest has eight internal states (the eigenvalue $+1$ of β). The self-interaction acts on a pair state Φ, an antisymmetric 16×16 matrix, as $\Phi \mapsto \sum_r g_r \sum_{|K|=r} \mathrm{sgn}_K\, (\beta\Theta^K)\, \Phi\, (\beta\Theta^K)^T$, projected on the states of positive energy. Its eigenvalue is the *pair coupling* G_{pair}, the interaction energy of the pair times the volume; a negative value means attraction. Diagonalizing this map (exactly in the program, numerically in the independent verification) gives two values:

$$G_{\text{pair}} = \begin{cases} g_0^{\text{eff}} + g_1^{\text{eff}} + 5\, g_7^{\text{eff}} & \text{for 7 pairs,} \\ g_0^{\text{eff}} + g_1^{\text{eff}} - 3\, g_7^{\text{eff}} & \text{for 21 pairs.} \end{cases} \tag{38}$$

The seven pairs are those created by the vector pair operators $\psi^T C^- \gamma^A \psi$ along the seven axes other than the time ($A \neq 4$); among them is the pair along the hidden axis $x0$, the pair of the superfluid of this paper. A scalar attraction ($g_0^{\text{eff}} < 0$) binds all 28 pairs equally; a vector repulsion ($g_1^{\text{eff}} > 0$) opposes all of them equally; the axial coupling distinguishes the seven from the twenty-one; and the pseudoscalar coupling does not act between particles at rest.

E. Symmetries

Chiral symmetry. The transformation $\psi \to e^{i\epsilon\chi}\psi$ with the chirality matrix χ (Hermitian, $\chi^2 = 1$, anticommut-

ing with every γ^A) leaves I_1 and I_7 unchanged and rotates I_0 and I_8 into each other. The self-interaction is invariant under it if and only if

$$g_0^{\text{eff}} + g_8^{\text{eff}} = 0 \tag{39}$$

The mass term breaks this symmetry in any case; it is the symmetry $U(1)_{\text{VS}}$ of the model with the Vorton-Scalar (Sec. VI), in which the mass arises from the vacuum value of a scalar field that transforms as well.

The internal symmetry of fields in ordinary space. A field that does not depend on the four extra coordinates $x0, x5, x6, x7$ has an internal symmetry $[SU(2)\times SU(2)]\times U(1)_E$ in addition to the phase symmetry (Sec. XVII). The self-interaction is invariant under the six rotations of the extra axes and under the phase for all couplings, but under $U(1)_E$ only if $g_1^{\text{eff}} = g_7^{\text{eff}} = 0$: *a vector or an axial-vector contact coupling breaks* $U(1)_E$. A scalar and a pseudoscalar coupling keep it.

V. MODES THAT DEPEND ON THE DEFLATING COORDINATES

A. Three kinds of modes

Modes fall into three kinds. A mode that does not depend on the deflating coordinates, $k_5 = k_6 = k_7 = 0$, obeys the flat equation; its solutions are the plane waves of flat spacetime, with the energies $\pm E_\perp$, eight of each sign. A mode with some $k_t \neq 0$ behaves in the same way as long as its physical wavenumber is negligible, $k_d \ll E_\perp$. But k_d grows without bound, and every such mode sooner or later reaches $k_d \gg E_\perp$. The moment at which $k_d = E_\perp$ is its *crossing*. The crossing is where the deflation acts on the field, and this section solves the equation of the weighted field across it.

The solution is exact when *one rate governs*: when only one of the k_t differs from zero, or when the three rates are equal. In both cases the matrix α^d does not depend on the time, and k_d is proportional to $e^{\text{rate}_t\, x4}$ with the rate that governs. The natural variable is then the physical wavenumber in units of the rate,

$$u_d = \frac{k_d(x4)}{\text{rate}_t}, \qquad \frac{d}{dx4} = \text{rate}_t\, u_d\, \frac{d}{du_d}. \tag{40}$$

The crossing is at $u_d = E_\perp/\text{rate}_t$, and this ratio of the energy to the rate is the one parameter on which the results depend.

B. The exact solution

In $\mathsf{h} = \mathsf{h}_\perp + k_d\,\alpha^d$ the two constant matrices satisfy $\mathsf{h}_\perp^2 = E_\perp^2$ and $(\alpha^d)^2 = 1$, and they anticommute. Let w be one of the eight eigenvectors of $\mathsf{h}_\perp$ with the eigenvalue $+E_\perp$; then $\alpha^d w$ is an eigenvector with the eigenvalue $-E_\perp$. Let Z_+ and Z_- denote two cylinder functions of the same kind (Bessel functions of the first or of the second kind, or Hankel functions) of the orders $\frac{1}{2} + iE_\perp/\text{rate}_t$ and $-\frac{1}{2} + iE_\perp/\text{rate}_t$. The recurrence relations of the cylinder functions connect two functions whose orders differ by one, and with them one verifies by substitution that

$$\hat{\psi} = \sqrt{u_d}\left[Z_+(u_d)\, w + i\, Z_-(u_d)\, \alpha^d w\right] \tag{41}$$

solves Eq. (27) for each of the eight w. Two independent kinds of cylinder function give all sixteen solutions. Equation (41) was verified with the Bessel functions of an arbitrary-precision library, to a relative residual of 10^{-15}.

C. The deflation creates particles

At early times, $u_d \to 0$, a Bessel function of the first kind behaves as the power of its argument that its order gives. The solution that has positive energy at early times, $\propto e^{-iE_\perp x4}\, w$, is therefore Eq. (41) with Z_+ the Bessel function of the first kind of the order $-\frac{1}{2} - iE_\perp/\text{rate}_t$ and Z_- minus that of the order $+\frac{1}{2} - iE_\perp/\text{rate}_t$, a pair that obeys the same recurrence relations. At late times, $u_d \to \infty$, the solution of positive energy is Eq. (41) with the Hankel functions of the second kind, which behave as e^{-iu_d}, and the solution of negative energy is the one with the Hankel functions of the first kind. The connection formula of the Bessel functions expresses the first solution through the other two, and the weight of the negative-energy part divided by that of the positive-energy part is $e^{-2\pi E_\perp/\text{rate}_t}$. A mode that starts empty therefore ends with

$$n_{\text{cr}} = \frac{1}{e^{2\pi E_\perp/\text{rate}_t} + 1} \tag{42}$$

particles in each of its eight internal states, and with as many antiparticles. This is the Fermi-Dirac distribution in the energy $E_\perp$ at the temperature

$$T_{\text{defl}} = \frac{\text{rate}_t}{2\pi}. \tag{43}$$

For equal rates it is the temperature that Gibbons and Hawking found for de Sitter space [11]; the creation of particles by a time-dependent metric is the phenomenon discovered by Parker [9, 12].

The numerical solution of Eq. (27) for all sixteen components confirms the law (Fig. 3). For $E_\perp/\text{g55}$ between 0.25 and 3 the created number agrees with Eq. (42) to a relative deviation of at most 3.2×10^{-6}, while the numbers themselves range from 0.17 to 6.5×10^{-9}; the eight internal states are created equally. With three equal rates the result depends neither on the direction of the wavenumber nor on the constants const_t. With unequal rates no exact solution is known. If the second physical wavenumber is still below $e^{-3}E_\perp$ when the first reaches $E_\perp$, the mean created number is Eq. (42) with the rate of the coordinate that becomes energetic first, to within 1.2

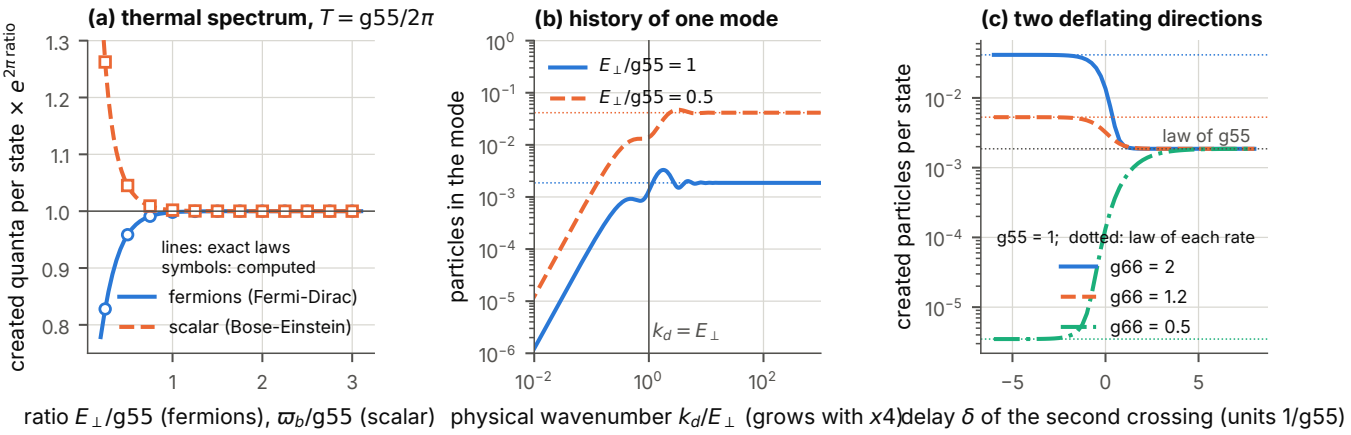


FIG. 3. Creation of particles on the deflating 7+1 spacetime. (a) Created quanta per state, multiplied by $e^{2\pi \text{ ratio}}$, against the ratio of the energy to the rate. Fermions follow the Fermi-Dirac law (42), the modes of the Vorton-Scalar the Bose-Einstein law (53), both at the temperature g55/2π. Lines: the exact laws; symbols: computed. (b) History of two modes, with $E_\perp/g55 = 1$ and 0.5: the number of particles in the mode against the physical wavenumber $k_d/E_\perp$, which grows with the time. After the crossing at $k_d = E_\perp$ the number settles at the value of Eq. (42) (dotted). (c) Two deflating directions, g55 = 1 and g66 = 2, 1.2 or 0.5: mean number of created particles per state against the delay between the two crossings; the dotted lines are Eq. (42) with each of the two rates.

per cent. If two directions become energetic together, the eight internal states are created *unequally*, in the cases computed by up to a factor 110 between the most and the least populated state, and the result is known only numerically.

D. What the result means

Each mode with $k_t \neq 0$ is populated once, at its crossing. Its particles then gain energy without bound, as $(E_\perp^2 + k_d^2)^{1/2}$, like the classical particle of Sec. II E. Modes with $k_t = 0$ stay empty. Two remarks are in order.

First, the stress of the created particles must eventually change the metric. That reaction of the fields on the metric has not been computed, and the results of this section hold for as long as it can be neglected. For any rate of deflation that the bounds of Sec. XVI allow today the temperature (43) is below 10^{-30} K, and nothing is created.

Second, the instantaneous energy of a quantum in a mode with $k_t \neq 0$ is at least $|k_t|/b_t$, which grows as $e^{\text{rate}_t\, x4}$. A process with a given energy at its disposal can excite such a mode only before the mode has crossed that energy. If the deflating coordinates are periodic, the smallest wavenumber that differs from zero is fixed, and the sector of fields that do not depend on $x5, x6, x7$ is separated from the rest by a gap that grows exponentially. This is the sense in which the deflation supports the restriction to that sector, which on flat spacetime is an assumption. If the deflating coordinates are infinite, modes of arbitrarily small k_t exist at every time, and the restriction remains an assumption.

E. The numerical method

The equations of this section are linear, with coefficients that grow exponentially, and an integrator with a fixed step fails on them. The program uses a commutator-free Magnus method of the fourth order: over each step the solution is advanced by the product of two matrix exponentials of linear combinations of the coefficient matrix at two Gauss points. Because the square of h is a multiple of the unit matrix at any one time, each exponential is evaluated in closed form, and every step is exactly unitary. The length of a step is chosen so that neither the phase accumulated in it nor the relative change of the coefficients exceeds a fixed bound. The number of particles is read off by projection on the eigenstates of the Hamiltonian corrected to the first order in the rate at which its direction changes, which removes the leading spurious oscillation of the particle number at finite time. An independent program, which uses an explicit Runge-Kutta method of the eighth order and the Bessel functions of another library, reproduces Eq. (42) to 2.2×10^{-5} (Appendix D).

VI. THE VORTON-SCALAR AND THE DEFLATION CLOCK

A. The model

The model contains, besides the fermion field, a complex scalar field, the *Vorton-Scalar* Φ_{VS}, whose vacuum value gives the fermions a mass and whose exchange binds two fermions into a pair. This subsection states the complete model on the deflating spacetime: the fermion field

with a mass term and with the self-interaction of Sec. III, and the Vorton-Scalar. The Vorton-Scalar brings three real, positive constants: its self-coupling $\lambda_{\rm VS}$, the constant $v_{\rm VS}$ that fixes the magnitude $v_{\rm VS}/\sqrt{2}$ at which the potential of the scalar is lowest, and its Yukawa coupling $y_{\rm VS}$ to the fermions. The fermion field brings the mass m_D of its Lagrangian, called the *bare Dirac mass* from here on, and the couplings g_r of its self-interaction. The chirality matrix is denoted by χ, with $\chi = i\,\gamma^0\gamma^1\cdots\gamma^7$, and the chiral projectors are $P^{L,R} = \frac{1}{2}(1 \mp \chi)$.

On a curved spacetime a Lagrangian is multiplied by the volume factor V_d, derivatives of spinors become covariant derivatives, and the flat metric is replaced by the curved one. The Lagrangian of the model is

$$\begin{aligned}\mathcal{L}^{\rm VS}_{1671d} = V_d \Big[&\bar\psi\left(i\gamma^\mu\nabla_\mu - m_D\right)\psi + \mathcal{L}_{\rm self} \\ &- g^{\mu\nu}_{(7+1)}\,\partial_\mu\Phi^*_{\rm VS}\,\partial_\nu\Phi_{\rm VS} \\ &- \lambda_{\rm VS}\Big(|\Phi_{\rm VS}|^2 - \frac{v^2_{\rm VS}}{2}\Big)^2 \\ &- y_{\rm VS}\,\bar\psi\big(\Phi_{\rm VS}\,P^R + \Phi^*_{\rm VS}\,P^L\big)\psi\Big], \end{aligned} \tag{44}$$

with $\bar\psi = \psi^\dagger\beta$, $\beta = \gamma^4$. The first line is the Lagrangian of Sec. III with the bare Dirac mass; the second is the kinetic term of the scalar (the minus sign belongs to the sign convention of the metric, in which the component of the time is negative, so that the term contains $+|\partial_4\Phi_{\rm VS}|^2$); the third is its potential; the fourth is the Yukawa coupling. The Lagrangian is invariant under the phase transformation $\psi \to e^{i\epsilon}\psi$ of the fermion number, $U(1)_F$, whatever its constants are. Without the bare Dirac mass, and with a self-interaction that satisfies the condition (39), it is also invariant under the chiral transformation $U(1)_{\rm VS}$: $\psi \to e^{i\epsilon\chi}\psi$ with $\Phi_{\rm VS} \to e^{-2i\epsilon}\Phi_{\rm VS}$. These statements hold on the deflating spacetime as on flat spacetime, because the transformations do not involve the coordinates. *The case $m_D = 0$, $g_r = 0$ is the model in which the whole mass of the fermions comes from the Vorton-Scalar and the whole interaction from its exchange;* Secs. VIII to XIV treat it first, and Secs. VII, IX and XIII then add the bare mass and the self-interaction and say what each changes.

A constant field $\Phi_{\rm VS} = v_{\rm VS}/\sqrt{2}$ with $\psi = 0$ solves the field equations on the deflating spacetime, because a constant has no derivatives: the vacuum value is the same at all times. It breaks $U(1)_{\rm VS}$. Around it the Vorton-Scalar has two modes: the *radial mode*, a real field of the mass m_s, and the *phase mode*, a real massless field, the Goldstone boson of the broken symmetry (massless if $U(1)_{\rm VS}$ is exact; Sec. XIII computes the mass that a bare Dirac mass gives it). The fermions acquire the mass $m = m_D + m_Y$, of which $m_Y = y_{\rm VS}v_{\rm VS}/\sqrt{2}$ comes from the Vorton-Scalar. Both modes couple to the fermions with the same strength $g_{\rm VS}$, the radial mode to the scalar density $\bar\psi\psi$ and the phase mode to the pseudoscalar density $\bar\psi\, i\chi\,\psi$:

$$\begin{aligned} m_Y &= \frac{y_{\rm VS}\,v_{\rm VS}}{\sqrt{2}}, \qquad & m_s &= \sqrt{2\lambda_{\rm VS}}\,v_{\rm VS}, \\ g_{\rm VS} &= \frac{m_Y}{v_{\rm VS}}, & \alpha_{\rm VS} &= \frac{g^2_{\rm VS}}{4\pi}. \end{aligned} \tag{45}$$

B. Fields that do not depend on the extra coordinates

Consider fields that do not depend on the four extra coordinates $x0, x5, x6, x7$, and introduce the weighted fields

$$\hat\psi = \sqrt{V_d}\,\psi, \qquad \hat\Phi_{\rm VS} = \sqrt{V_d}\,\Phi_{\rm VS}. \tag{46}$$

Each term of Eq. (44) is rewritten in turn. The kinetic term and the mass term of the fermions are the flat ones for $\hat\psi$ (Sec. III). The self-interaction contains four fermion fields and therefore the factor $V_d \cdot V_d^{-2}$, as in Eq. (26). The Yukawa term contains one scalar and two fermion fields and therefore the factor $V_d \cdot V_d^{-3/2}$. The potential becomes $(\lambda_{\rm VS}/V_d)\,(|\hat\Phi_{\rm VS}|^2 - V_d v^2_{\rm VS}/2)^2$. In the kinetic term of the scalar the derivative along the time acts on the weight as well, $\sqrt{V_d}\,\partial_4\Phi_{\rm VS} = \partial_4\hat\Phi_{\rm VS} + \frac{1}{2}{\rm gsum}\,\hat\Phi_{\rm VS}$, which gives a term $\frac{1}{4}{\rm gsum}^2|\hat\Phi_{\rm VS}|^2$ and a total derivative. The result is the flat Lagrangian for the weighted fields, with four replacements and one additional term:

$$\begin{aligned} &y_{\rm VS} \to \frac{y_{\rm VS}}{\sqrt{V_d(x4)}}, \qquad \lambda_{\rm VS} \to \frac{\lambda_{\rm VS}}{V_d(x4)}, \qquad g_r \to \frac{g_r}{V_d(x4)}, \\ &v_{\rm VS} \to v_{\rm VS}\sqrt{V_d(x4)}, \qquad \text{and } + \frac{{\rm gsum}^2}{4}\,|\hat\Phi_{\rm VS}|^2. \end{aligned} \tag{47}$$

The bare Dirac mass m_D is not replaced. With a constant volume of the extra coordinates the same replacements define the couplings of a four-dimensional description on flat spacetime. Here that volume is the coordinate volume of a cell times $V_d(x4)$, and it shrinks. The additional term lowers the square of the mass of each mode of the scalar by ${\rm gsum}^2/4$ in the equation of the weighted field; Section VI D shows what it does. From here on the couplings $y_{\rm VS}$, $\lambda_{\rm VS}$, $v_{\rm VS}$, $g_{\rm VS}$, $\alpha_{\rm VS}$ and g_r are those of the four-dimensional description at the time considered.

C. The deflation clock

Define the *deflation clock*

$$\tau_d = {\rm gsum}\,(x4 - x4_{\rm ref}), \tag{48}$$

the number of e-foldings by which the volume V_d has shrunk since a reference time $x4_{\rm ref}$, which is chosen in each application. By Eq. (47) every quantity of the four-dimensional description is proportional to $e^{n\tau_d}$ with a definite exponent n (Table II). Three observations organize what follows.

TABLE II. How the quantities of the four-dimensional description follow the deflation clock: each is proportional to $e^{n\tau_d}$. The last two rows hold under additional assumptions, which Secs. XVI and XVII state and examine.

quantity	meaning	exponent n
V_d	$b_5 b_6 b_7$: the factor $\sqrt{\lvert\det g\rvert}$ of the measure	-1
$y_{\rm VS}$ of the four-dimensional description	Yukawa coupling of the weighted fields	$\frac{1}{2}$
$\lambda_{\rm VS}$ of the four-dimensional description	quartic coupling of the weighted Vorton-Scalar	1
$v_{\rm VS}$ of the four-dimensional description	vacuum value of the weighted Vorton-Scalar	$-\frac{1}{2}$
m_D	bare Dirac mass (the mass term of the Lagrangian of the fermion field)	0
m_Y	part of the fermion mass that the Vorton-Scalar gives, $y_{\rm VS} v_{\rm VS}/\sqrt{2}$; the mass in vacuum is $m = m_D + m_Y$	0
m_s	mass of the radial mode $\sqrt{2\lambda_{\rm VS}}\, v_{\rm VS}$	0
g_r of the four-dimensional description	each of the nine couplings of the self-interaction of the weighted field, and with them the mean-field couplings G_S, G_V	1
$g_{\rm VS}$	coupling of both modes of the Vorton-Scalar to the fermions, $m_Y/v_{\rm VS}$	$\frac{1}{2}$
$\alpha_{\rm VS}$	$g_{\rm VS}^2/4\pi$	1
$G_{\rm VS}$	constant of the Fermi type of the neutral-current interaction, $1/(\sqrt{2}\, v_{\rm VS}^2)$	1
$D_{\rm VS}$	$\alpha_{\rm VS}\, m/m_s$: decides whether two fermions bind	1
C	$g_{\rm VS}^2 m^2/m_s^2$: decides the state of matter	1
m_ϑ	mass of the phase mode of the Vorton-Scalar when $m_D \neq 0$, $m_\vartheta^2 v_{\rm VS}^2 = \lvert\langle\bar\psi\psi\rangle_{\rm vac}\rvert\, m_D m_Y/m$	$\frac{1}{2}$
vortex energy per length	proportional to $v_{\rm VS}^2$ (four-dimensional)	-1
$G_{\rm Fermi}$ (Fermi constant of the weak interaction) if the vacuum value of the Vorton-Scalar sets the electroweak scale (assumption HV)	$1/(\sqrt{2}\, v_{\rm VS}^2)$, four-dimensional vacuum value	1
G_N (Newton's constant) if gravity is eight-dimensional (A2)	G_8 divided by the volume of $x5, x6, x7$	1

The masses do not change. The bare Dirac mass is not affected at all, and the part m_Y of the fermion mass and the mass m_s of the radial mode contain the couplings in combinations from which the volume cancels. So do the Fermi momentum and the Fermi energy of matter with a given number of particles per unit of ordinary volume.

The couplings grow. The coupling $g_{\rm VS} = m_Y/v_{\rm VS}$ grows as $e^{\tau_d/2}$, because the four-dimensional vacuum value falls. The strength of the attraction, $\alpha_{\rm VS}$, grows as e^{τ_d}: it doubles every time the volume of the deflating coordinates halves. Every coupling g_r of the self-interaction, and with them the couplings G_S, G_V of Sec. VII, grows as e^{τ_d} as well: *the contact interaction and the exchange of the Vorton-Scalar keep their ratio.*

The two numbers that decide the state of matter grow at the same rate. These are

$$D_{\rm VS} = \alpha_{\rm VS}\,\frac{m}{m_s}, \qquad C = \frac{g_{\rm VS}^2 m^2}{m_s^2}. \tag{49}$$

The first decides whether two fermions form a bound pair (Sec. X), the second whether matter keeps its fermion mass (Sec. XII). Their ratio $C/D_{\rm VS} = 4\pi\, m/m_s$ is fixed.

On flat spacetime the model is specified by the masses and the couplings. On a deflating spacetime it is specified by the masses and the values of the couplings *at one time*; the deflation then carries it through all larger couplings. The constants of the flat theory become functions of the time, and its thresholds are passed one after another (Fig. 4).

D. Modes of the Vorton-Scalar

Let $\varphi_{\rm bos}$ stand for the radial mode or the phase mode, with the mass $m_{\rm bos} = m_s$ or $m_{\rm bos} = 0$. To first order in the field, a mode with a wavenumber of the magnitude k along the flat coordinates and the wavenumbers k_t along the deflating ones obeys

$$\begin{aligned}&\frac{1}{V_d}\,\partial_4\big(V_d\,\partial_4\varphi_{\rm bos}\big)\\ &\quad+\Big(k^2+\sum_t\frac{k_t^2}{b_t^2}+m_{\rm bos}^2\Big)\varphi_{\rm bos}=0.\end{aligned} \tag{50}$$

Modes that do not depend on the deflating coordinates. With $k_t = 0$ the equation has constant coefficients, and its solutions are the exponentials

$$\varphi_{\rm bos} \propto \exp\Big\{\Big[\frac{\rm gsum}{2} \pm \sqrt{\frac{{\rm gsum}^2}{4} - k^2 - m_{\rm bos}^2}\Big]x4\Big\}. \tag{51}$$

If $k^2 + m_{\rm bos}^2 > {\rm gsum}^2/4$, the mode oscillates with the frequency $\varpi_b = (k^2 + m_{\rm bos}^2 - {\rm gsum}^2/4)^{1/2}$, and its amplitude grows as $e^{{\rm gsum}\, x4/2}$. The growth is that of the weight

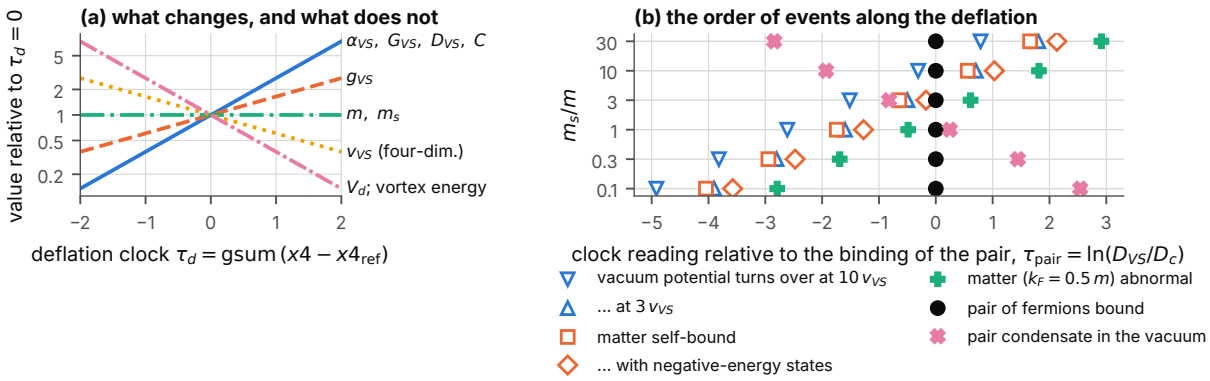


FIG. 4. The deflation clock. (a) How the quantities of the four-dimensional description change with the clock τ_d (logarithmic scale; values relative to $\tau_d = 0$). (b) The order of events along the deflation for six ratios m_s/m: the clock reading of each event relative to the moment at which two fermions form a bound pair (Table IX).

alone: the weighted field has a constant amplitude, and the number of quanta in a cell of fixed coordinates is conserved while the cell shrinks. Otherwise the mode does not oscillate. For the radial mode this never happens if $m_s > \mathrm{gsum}/2$, which holds by many orders of magnitude for any rate that observation allows (Sec. XVI). For the phase mode it happens at wavelengths longer than $4\pi/\mathrm{gsum}$. At $k = 0$ the two exponents of the phase mode are 0 and gsum: a constant phase, which costs nothing, and a phase that advances at a rate proportional to $1/V_d$, which is the conservation of the charge of $U(1)_{\mathrm{VS}}$ in a shrinking volume. Neither is an instability of the vacuum. Figure 5 shows the exponents.

The potential between two fermions. Two fermions at rest attract each other by exchanging the radial mode. The density of a fixed number of fermions grows as $1/V_d$, so that the radial mode obeys Eq. (50) with a source proportional to $e^{\mathrm{gsum}\,x4}$. The solution that follows the source is $e^{\mathrm{gsum}\,x4}/(k^2 + m_{\mathrm{bos}}^2)$ times the source, as substitution shows: the terms $\mathrm{gsum}^2 - \mathrm{gsum}\cdot\mathrm{gsum}$ cancel, and the denominator contains the *unshifted* mass. The potential between two fermions at rest is therefore, at every moment, a Yukawa potential,

$$V_s(r) = -\alpha_{\mathrm{VS}}(x4)\,\frac{e^{-m_s r}}{r}, \tag{52}$$

with the range $1/m_s$ unchanged and the strength of the moment. The numerical solution from rest approaches this form with a transient that decays as $e^{-\tau_d/2}$ [Fig. 5(c)]; after $\tau_d = 20$ the relative difference is below 4×10^{-5} in the four cases computed.

Modes that depend on a deflating coordinate. For $k_t \neq 0$ the weighted field obeys $\partial_4^2\hat{\varphi}_{\mathrm{bos}} + (\varpi_b^2 + k_d^2)\,\hat{\varphi}_{\mathrm{bos}} = 0$ with the physical wavenumber k_d of Eq. (28), and the scalar behaves as the fermion field does. The frequency grows without bound. When one rate governs, the solutions are Bessel functions of the imaginary order $i\varpi_b/\mathrm{rate}_t$ and of the argument u_d, and a mode that starts empty ends with

$$n_{\mathrm{cr}} = \frac{1}{e^{2\pi\varpi_b/\mathrm{rate}_t} - 1} \tag{53}$$

quanta: the Bose-Einstein distribution at the same temperature T_{defl} as for the fermions. The program confirms Eq. (53) in 36 cases to a relative deviation of 1.9×10^{-5} (Fig. 3).

VII. MATTER: THE KOHN-SHAM ANALYSIS

This section determines the states of uniform matter made of the fermions of the model when the self-interaction acts between them. The Vorton-Scalar is set aside here ($y_{\mathrm{VS}} = 0$, so that the mass m of the fermions in vacuum is the bare Dirac mass); Sec. XIII restores it.

A. The method

In the method of Kohn and Sham [13] a system of interacting fermions is replaced by independent fermions that move in fields which their own densities determine. The energy is written as a function of the densities; the fields are its derivatives; and the densities are computed from the occupied states in those fields, until both agree. For uniform matter of the present model the densities are two numbers (Sec. IV C): the number density $n = \langle\psi^\dagger\psi\rangle$ and the scalar density $n_S = \langle\bar{\psi}\psi\rangle$. The interaction energy is $\frac{1}{2}G_V n^2 + \frac{1}{2}G_S n_S^2$ with a scalar coupling G_S and a vector coupling G_V, and each particle obeys the Dirac equation with the Hamiltonian matrix

$$\begin{aligned} \mathsf{h}_{\mathrm{KS}} &= \sum_j \alpha^j k_j + \beta\, m^* + V_0, \\ m^* &= m + G_S\, n_S, \qquad V_0 = G_V\, n. \end{aligned} \tag{54}$$

The scalar field changes the mass of the particles in matter from m to m^*; the vector field shifts all energies by

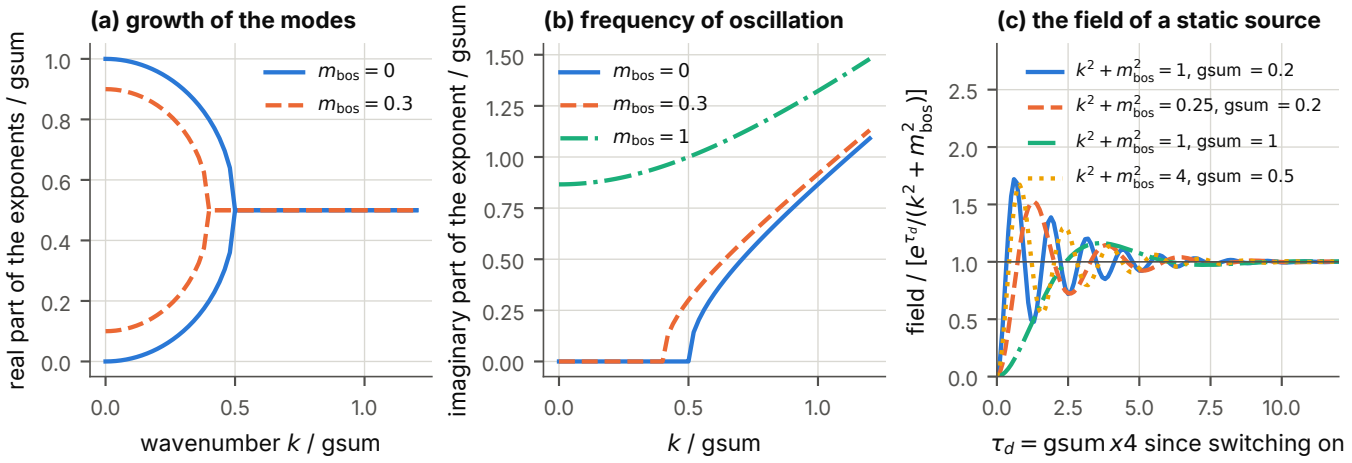


FIG. 5. Modes of the Vorton-Scalar that do not depend on the deflating coordinates. (a) Real parts of the two exponents of Eq. (51) against the wavenumber, for the phase mode and for a radial mode, in units of gsum. (b) Their imaginary part, the frequency of oscillation. (c) The field that a static source builds up from zero, divided by the form $e^{\tau_d}/(k^2+m_{\rm bos}^2)$ that leads to Eq. (52).

V_0. A negative G_S is an attraction (it lowers the mass and the energy) and a positive G_V a repulsion; this is the structure of the mean-field theory of dense matter with a scalar and a vector field [14]. With the couplings of Eq. (37) the exchange terms are contained exactly, because for a contact interaction they are a function of the same two densities.

A particle of the momentum $\vec{k}$ has the energy $E^*(k)+V_0$ with $E^*(k)=(k^2+m^{*2})^{1/2}$. The occupied states are those with $|\vec{k}|<k_F$, eight for each momentum ($\nu=8$), and averages over them are written $\langle f\rangle = \nu\int d^Dk\, f/(2\pi)^D$, where D is the number of space dimensions in which the particles have momentum. Then

$$n=\langle 1\rangle=\frac{\nu\, V_D\, k_F^D}{(2\pi)^D}, \qquad n_S=\left\langle\frac{m^*}{E^*}\right\rangle, \tag{55}$$

with the volume $V_D=\pi^{D/2}/\Gamma_{\rm E}(D/2+1)$ of the unit ball ($\Gamma_{\rm E}$ is Euler's gamma function). The second of Eqs. (54) with the second of Eqs. (55) is the *gap equation* for m^*. The energy density is

$$\varepsilon=\langle E^*\rangle+\tfrac{1}{2}G_V\, n^2-\frac{(m^*-m)^2}{2G_S}, \tag{56}$$

which is the kinetic and mass energy of the particles plus the interaction energy [the first term counts the mass as m^* instead of m, which the last term corrects]; it is stationary with respect to m^* exactly when the gap equation holds. The pressure along the coordinate xj is

$$p_j=\left\langle\frac{k_j^2}{E^*}\right\rangle+p_{\rm int}, \qquad p_{\rm int}=\tfrac{1}{2}G_V\, n^2+\frac{(m^*-m)^2}{2G_S}. \tag{57}$$

The first term is the momentum that the particles carry across a surface; the *interaction pressure* $p_{\rm int}$ is the same in all directions. *Along a coordinate on which the fields do not depend the particles have no momentum, and the pressure is* $p_{\rm int}$ *alone*; this remark is used for the deflating coordinates below. The chemical potential is $\mu=E^*(k_F)+V_0$, and for an isotropic state $p=\mu n-\varepsilon$. Matter is *self-bound* if its energy per particle, ε/n, is below the mass m at some density; the density at which ε/n is smallest is the *saturation* density, where the pressure vanishes and $\mu=\varepsilon/n$ [15].

Two limitations are stated here once. The filled states of negative energy are not counted (the *no-sea approximation*); Sec. XIII counts them. And the state is the one of lowest energy among the solutions of the gap equation: without the filled negative-energy states the energy as a function of an unrestricted trial mass has no lowest value, because a large negative trial mass places particles in states of negative rest energy.

B. Flat spacetime: the reference calculation

On the flat 7+1 spacetime the particles have momentum along all seven space coordinates, $D=7$, and a coupling has the dimension of a length to the sixth power. The reference example has $G_S=-10^5$ and $G_V=+10^4$ in units $m=1$. Table III gives its saturation point, computed by the program of this paper, together with the values of a first calculation with a notebook in the Wolfram Language; an independent script reproduces the same numbers a third time. Figure 6 shows the equation of state.

The matter is self-bound: at $k_F=0.7566\,m$ the energy per particle is $0.0283\,m$ below the mass, and the mass of the particles in it is $m^*=0.3481\,m$. The identity $p=\mu n-\varepsilon$ holds along the whole equation of state to 10^{-15} of ε, and at saturation $\mu=\varepsilon/n$.

One remark concerns the relation between these two

TABLE III. Uniform matter on the flat 7+1 spacetime at saturation, for $G_S = -10^5$, $G_V = +10^4$ (units $m = 1$, seven space dimensions): the program of this paper and the first calculation with a notebook.

quantity	this program	notebook
mass of the particles in matter, m^*/m	0.3480973401	0.3480973401
number density n (units m^7)	$1.388100213 \times 10^{-5}$	$1.388100213 \times 10^{-5}$
energy density ε (units m^8)	$1.348787233 \times 10^{-5}$	$1.348787233 \times 10^{-5}$
energy per particle minus the mass, $\varepsilon/n - m$ (units m)	-0.02832142796	-0.02832142796
Fermi momentum at saturation, k_F/m	0.7566361507	0.7566361507

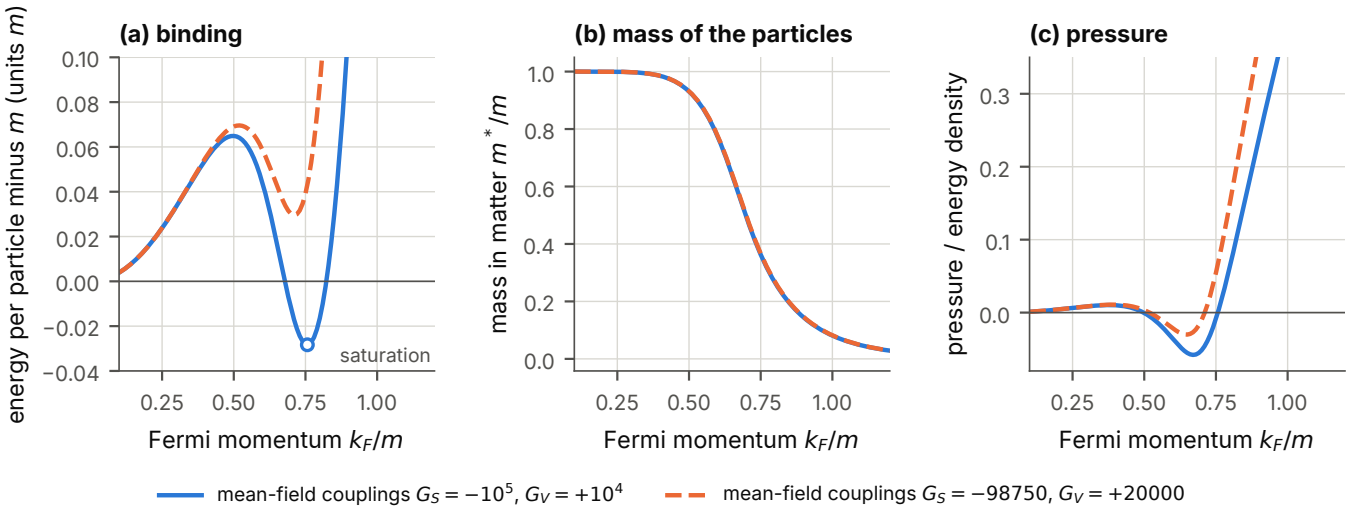


FIG. 6. Uniform matter on the flat 7+1 spacetime (seven space dimensions; units $m = 1$) for two pairs of mean-field couplings. (a) Energy per particle minus m; the circle marks the saturation point of Table III. (b) Mass of the particles in matter. (c) Pressure divided by the energy density.

numbers and the couplings of the Lagrangian. If the Lagrangian has $g_0 = -10^5$, $g_1 = +10^4$ and no other coupling, Eq. (37) gives $G_S^{\rm HF} = -98750$ and $G_V^{\rm HF} = +20000$: the exchange terms double the vector repulsion. For that pair the energy per particle is above m at all densities [dashed curves of Fig. 6; its local minimum lies $0.030\,m$ above m, at $k_F = 0.71\,m$]: *uniform matter is not self-bound.* The reference example is therefore to be read as $G_S = -10^5$, $G_V = +10^4$ for the mean-field couplings themselves, which corresponds to $g_0 = -105341.6$, $g_1 = +2484.5$ if no other coupling of the Lagrangian differs from zero.

C. Matter with momentum along the deflating coordinates

Now let $x5, x6, x7$ deflate, and consider matter whose particles have momentum along all seven space coordinates. The linear Dirac equation conserves the wavenumbers k_t of a particle (Sec. III); its proper momentum along xt is k_t/b_t and grows as the coordinate shrinks. If the particles do not collide, the occupied states remain the ball $|\vec{k}| < k_F$ of wavenumbers that they filled at a reference time, and the occupied proper momenta form an *ellipsoid* that is stretched along $x5, x6, x7$ by the factors $b_t^{\rm ref}/b_t = e^{{\rm rate}_t (x4 - x4_{\rm ref})}$. The number density grows as the volume shrinks,

$$n = n_{\rm ref}\, e^{\tau_d}, \tag{58}$$

and the equations of Sec. VII A hold with the averages taken over the ellipsoid; the pressure along a deflating coordinate is $p_t = \langle (k_t/b_t)^2/E^* \rangle + p_{\rm int}$. The energy density obeys

$$\frac{d\varepsilon}{dx4} = \sum_t {\rm rate}_t\, (\varepsilon + p_t), \tag{59}$$

the statement that the covariant divergence of the energy-momentum tensor vanishes; the program confirms it to 6×10^{-9} and the independent script to 2×10^{-8}.

Figure 7 shows the history of the saturated matter of Table III, taken as the state at the reference time, for three equal rates and for rates in the ratio $1 : 2 : 3$.

The mass of the particles falls. It reaches half its value at the reference time at $\tau_d = 0.983$ (equal rates) or 0.990 (rates $1 : 2 : 3$).

Matter is self-bound only for a limited time. The energy per particle is below m only for $-0.675 < \tau_d < 0.537$ (equal rates; $-0.658 < \tau_d < 0.523$ for $1 : 2 : 3$): about

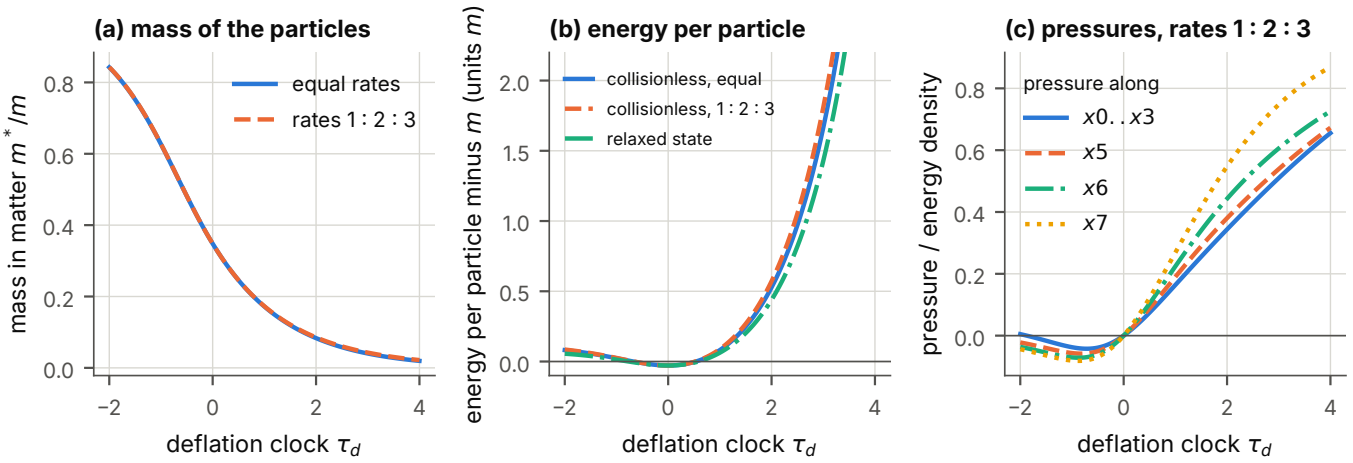


FIG. 7. Matter of the 7+1 model with momentum along all seven space coordinates while $x5, x6, x7$ deflate, without collisions. At the clock reading $\tau_d = 0$ it is the saturated matter of Table III. (a) Mass of the particles in matter. (b) Energy per particle minus m: for the collisionless state, and for the state of lowest energy at the same density ("relaxed"). (c) Pressures along the flat coordinates and along the three deflating ones, divided by the energy density, for rates in the ratio 1 : 2 : 3.

one e-folding of the deflating volume. Before that the matter is too dilute to bind, afterwards the vector repulsion $\frac{1}{2}G_V n^2$ dominates; at $\tau_d = 4$ it is 69.4 per cent of the energy density, which then grows as $e^{2\tau_d}$.

The pressure becomes anisotropic, larger along the coordinates that deflate faster [Fig. 7(c)].

Collisions would release heat. The collisionless state has more energy than the state of lowest energy at the same density, a ball of occupied states; the difference reaches $0.87\,m$ per particle at $\tau_d = 4$. Whether collisions are frequent enough to release it is not examined here.

D. Fields that do not depend on the deflating coordinates

Consider now the sector of fields that do not depend on $x5, x6, x7$. Its particles have momentum along $x0, \ldots, x3$, so that $D = 4$, or, if the fields do not depend on the hidden coordinate $x0$ either, along ordinary space, $D = 3$. By Eq. (47) the equations of Sec. VII A hold for the weighted field at each moment with the couplings of that moment,

$$G_S(x4) = G_S^{\rm ref}\, e^{\tau_d}, \qquad G_V(x4) = G_V^{\rm ref}\, e^{\tau_d}, \tag{60}$$

while the mass m and the density of particles per unit of the volume of $x0, \ldots, x3$ do not change. The ratio $G_V/|G_S|$ is fixed. The deflation therefore carries the matter through a one-parameter sequence of interactions of growing strength, and *a gas becomes self-bound matter at a definite moment*: when the scalar attraction reaches the threshold of Table IV.

If the coupling at a reference time is $G_S^{\rm ref}$, the binding begins at the clock reading $\tau_d = \ln(|G_S|_{\rm threshold}/|G_S^{\rm ref}|)$. Figure 8 follows the saturated matter from that moment on: its energy per particle falls and the mass of its particles falls, more slowly if a vector repulsion is present. The thresholds were computed twice, by the program and by the independent script, and agree to 10^{-9}.

TABLE IV. The smallest scalar attraction at which uniform matter is self-bound, for $D = 4$ and $D = 3$ space dimensions with momentum and three ratios of the vector repulsion to the scalar attraction, with the Fermi momentum of the bound state at the threshold (units $m = 1$). The ratio 0.2025 is that of the Hartree-Fock couplings for $g_1 = -0.1\, g_0$. Last column: the critical coupling of the vacuum for a cutoff equal to m (Sec. XIII).

D	$G_V/\lvert G_S\rvert$	$\lvert G_S\rvert\, m^{D-1}$	k_F/m	$\lvert G_c\rvert\, m^{D-1}$
4	0	62.0767	0.8879	29.6088
4	0.1000	85.6504	0.7707	29.6088
4	0.2025	119.5723	0.6792	29.6088
3	0	8.0671	0.7932	4.9348
3	0.1000	9.8364	0.7092	4.9348
3	0.2025	12.2272	0.6351	4.9348

Matter of a *fixed* density is carried through the same sequence (Fig. 9). Its pressure along the flat coordinates becomes negative at a moment that depends on the density (Table V): a uniform gas is then unstable against breaking up into drops of saturated matter. Along the deflating coordinates the pressure is the interaction pressure $p_{\rm int}$ of Eq. (57) alone, which is negative as long as the scalar attraction dominates: *the matter pulls on the deflating coordinates.* Section XV uses this.

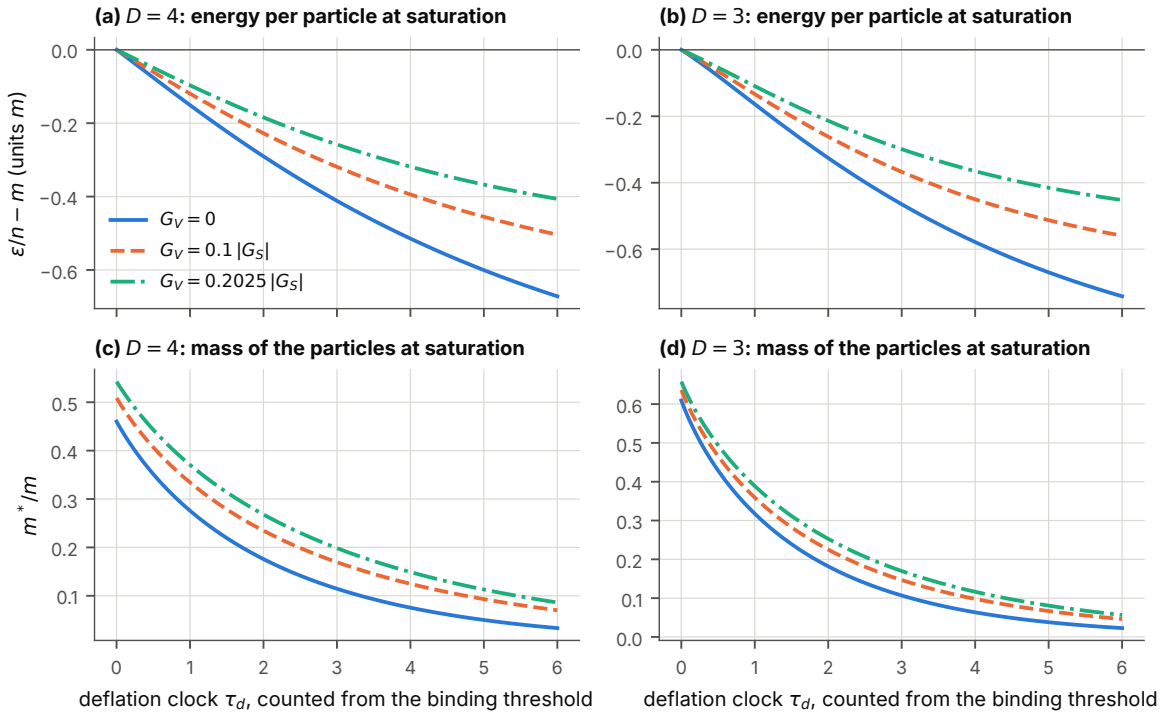


FIG. 8. Self-bound matter in the sector of fields that do not depend on $x5, x6, x7$, along the deflation clock, counted from the moment at which matter first becomes self-bound. Upper row: energy per particle of the saturated state minus m. Lower row: mass of its particles. Left: $D = 4$; right: $D = 3$. Three ratios of the vector to the scalar coupling.

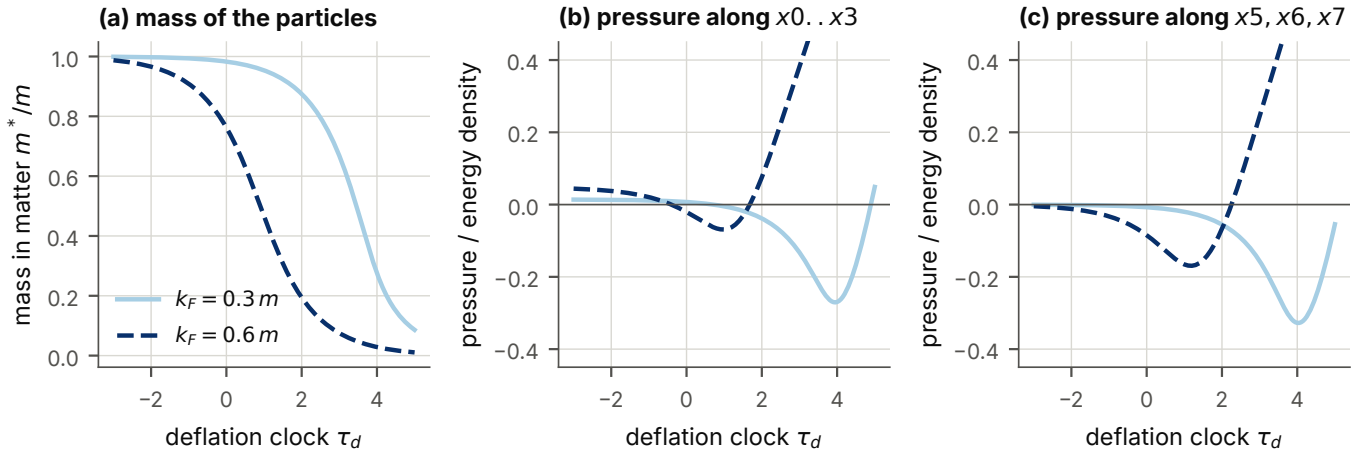


FIG. 9. A gas of fixed density in the sector of fields that do not depend on $x5, x6, x7$ ($D = 4$, $G_V = 0.1\,|G_S|$), along the deflation clock counted from the binding threshold. (a) Mass of the particles in matter. (b) Pressure along $x0, \ldots, x3$ divided by the energy density. (c) Pressure along $x5, x6, x7$, the interaction pressure alone, divided by the energy density.

VIII. PAIRING AND THE GAP ALONG THE CLOCK

A. The pair channels

A pair of fermions is created by an operator $\psi^T X \psi$ with a 16×16 matrix X; the choice of X is the *channel* of the pair. For each channel one computes the *exchange amplitudes* F_S and F_P of the radial mode and of the phase mode: the matrix elements of the exchange of one boson between two fermions of opposite momenta in the internal state of the channel. The radial mode attracts in every channel, and the phase mode attracts in the four *vector pairs along the extra axes*,

$$\psi^T C^- \gamma^a \psi, \qquad a \in \{0, 5, 6, 7\}, \tag{61}$$

and repels in their partners that are built with two Dirac matrices. Here $C^- = \gamma^4\gamma^0\gamma^1\cdots\gamma^7$ is the matrix that

TABLE V. A gas of fixed Fermi momentum in the sector of fields that do not depend on $x5, x6, x7$: the coupling at which its pressure along the flat coordinates vanishes, and the clock reading of that moment counted from the binding threshold of Table IV (negative: before the threshold is reached, at this higher density).

D	$G_V/\lvert G_S\rvert$	k_F/m	$\lvert G_S\rvert\, m^{D-1}$	τ_d
4	0	0.3	154.957	0.915
4	0	0.6	47.768	−0.262
4	0.1000	0.3	174.139	0.710
4	0.1000	0.6	57.191	−0.404
4	0.2025	0.3	199.477	0.512
4	0.2025	0.6	73.117	−0.492
3	0	0.3	10.485	0.262
3	0	0.6	6.618	−0.198
3	0.1000	0.3	11.784	0.181
3	0.1000	0.6	8.019	−0.204
3	0.2025	0.3	13.507	0.100
3	0.2025	0.6	10.631	−0.140

turns each Dirac matrix into minus its transpose, and the index a labels the four extra axes. The pair along the hidden axis, $a = 0$, has spin 0 in ordinary spacetime; it is the *hidden-axis singlet* of hypothesis H1.

The amplitudes are computed from the Dirac matrices of the frame, which the deflation does not change, and each is proportional to the square of the coupling. For fields that do not depend on $x5, x6, x7$ the amplitude of a channel at the time $x4$ is therefore

$$F_{S,P}(x4) = g_{\rm VS}(x4)^2 \times \big(F_{S,P} \text{ of the flat spacetime at } g_{\rm VS} = 1\big). \tag{62}$$

The antisymmetric matrices X fall into classes under the symmetries of the theory, with 15 representative channels. The program computes the amplitudes of these 15 channels from the matrices of the deflating spacetime, at three momenta and 13 scattering angles; the numbers agree with those computed with the matrices of flat spacetime to every digit, and the amplitudes of the hidden-axis channel reproduce the closed forms of Eq. (66) to 1.1×10^{-15}. The conclusion about the channels therefore holds at every moment (Fig. 10): the most attractive channel is the vector pair along an extra axis.

B. The gap grows much faster than the coupling

In matter of a given density the attraction opens a gap Δ_F at the Fermi momentum k_F: the energy that it costs to break a pair. The gap equation [Eq. (70) below, where it is written out with all its terms], which generalizes that of Bardeen, Cooper and Schrieffer [16], holds at each moment with the couplings of that moment. Its solution as a function of the clock is the *adiabatic history* of the paired state: the state that matter would have if it were always in equilibrium. (Section XI examines when it can be.) The strength of the pairing is measured by the dimensionless number $\lambda = N(0)\, W(k_F, k_F)$, the density of states at the Fermi surface times the kernel of the gap equation there; in the vector channel it is the sum $\lambda_S + \lambda_P$ of the contributions of the two modes. Since the kernel is proportional to $g_{\rm VS}^2$,

$$\lambda \;\propto\; e^{\tau_d}. \tag{63}$$

The program solves the gap equation at 34 moments for each of four sets of the mass m_s and the Fermi momentum (Table VI, Fig. 11). There are three results.

The order of the channels does not change. At every moment the gap of the vector channel is at least as large as the gap that the radial mode alone would give, and that at least as large as the gap of the channel in which the phase mode repels. The gap never decreases.

The critical temperature follows the gap. Wherever the gap is below 5 per cent of the Fermi kinetic energy E_F, the critical temperature is $T_c = 0.567\,\Delta_F$, the ratio of weak coupling, to within 0.002.

The law of growth. In weak coupling the gap is proportional to $e^{-1/\lambda}$. With $\lambda \propto e^{\tau_d}$,

$$\frac{d\ln\Delta_F}{d\tau_d} = \frac{1}{\lambda}. \tag{64}$$

In the computed histories the product $\lambda\, d\ln\Delta_F/d\tau_d$ lies between 1.00 and 1.03 wherever the gap is between 10^{-6} and 0.02 of E_F.

Equation (64) is the central quantitative statement about the superfluid on a deflating spacetime. While the coupling grows by one e-folding, the gap of a weakly paired state with $\lambda = 0.1$ grows by ten e-foldings, a factor of twenty thousand. A paired state that is unobservably weak at one time is strong a few e-foldings of the volume later.

IX. PAIRING WITH THE SELF-INTERACTION

Section VIII treated the pairing by the exchange of the Vorton-Scalar alone. This section adds the self-interaction of the fermion field. It writes out the amplitudes and the gap equation in full, shows that the exchange of the Vorton-Scalar is itself a self-interaction of finite range, and gives the gap along the deflation clock when both act.

A. Amplitudes of the four contact channels

Two particles of the mass m^* with the momenta $\vec{p}$ and $-\vec{p}$ in ordinary space scatter into $\vec{p}\,'$ and $-\vec{p}\,'$. Let $E = (m^{*2} + p^2)^{1/2}$ and E' likewise, and let $\mathsf{h}(\vec{p}) = \sum_j \alpha^j p_j + \beta\, m^*$ be the Hamiltonian matrix of one particle, so that $\frac{1}{2}(1 + \mathsf{h}/E)$ projects on its states of positive energy. The

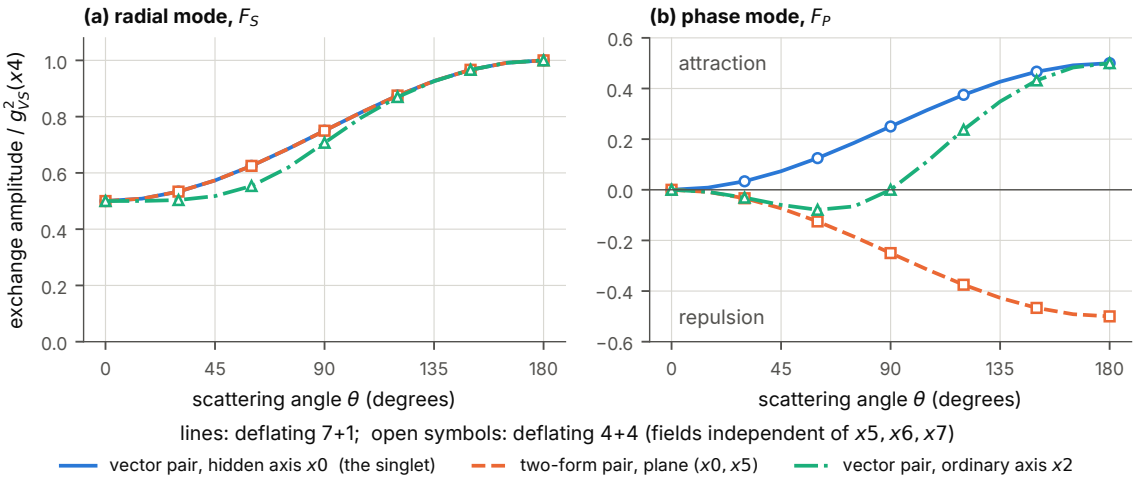


FIG. 10. Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle, for momenta of the magnitude m in the plane of $x1$ and $x2$. Lines: deflating 7+1 spacetime; open symbols: deflating 4+4 spacetime. (a) The radial mode attracts in all three channels. (b) The phase mode attracts in the vector pair along the hidden axis and repels in its partner built with two Dirac matrices. The amplitudes are those of flat spacetime: the deflation changes their overall strength and nothing else.

TABLE VI. The gap along the deflation clock for two of the four sets computed. $g_{\rm VS}$ and λ: coupling and pairing strength of the moment. E_F: Fermi kinetic energy. T_c: critical temperature. Rate of growth: $d\ln\Delta_F/d\tau_d$; the last column, its product with λ, is given where the gap is between 10^{-6} and 0.02 of E_F.

m_s/m, k_F/m	τ_d	$g_{\rm VS}$	λ	Δ_F/E_F	T_c/Δ_F	rate of growth	$\lambda\times$ rate
0.10, 0.50	−0.63	0.730	0.062	1.17×10^{-7}	0.567	—	—
0.10, 0.50	−0.13	0.939	0.102	7.40×10^{-5}	0.567	9.88	1.01
0.10, 0.50	0.38	1.208	0.169	3.70×10^{-3}	0.567	6.01	1.01
0.10, 0.50	0.88	1.554	0.279	0.0407	0.565	3.74	—
0.10, 0.50	1.39	2.000	0.463	0.189	0.555	2.50	—
0.10, 0.50	1.89	2.573	0.766	0.563	0.528	1.96	—
0.10, 0.50	2.39	3.311	1.268	1.49	0.480	1.96	—
1.0, 0.50	0.88	1.554	0.056	1.02×10^{-7}	0.567	—	—
1.0, 0.50	1.13	1.763	0.072	5.62×10^{-6}	0.567	14.02	1.01
1.0, 0.50	1.39	2.000	0.092	1.27×10^{-4}	0.567	10.93	1.01
1.0, 0.50	1.64	2.269	0.119	1.45×10^{-3}	0.567	8.54	1.02
1.0, 0.50	1.89	2.573	0.153	9.76×10^{-3}	0.567	6.71	1.03
1.0, 0.50	2.14	2.919	0.197	0.0439	0.566	5.33	—
1.0, 0.50	2.39	3.311	0.253	0.147	0.563	4.37	—

pair along the hidden axis has the wave function

$$\Phi(\vec p)=\tfrac12\Big(1+\frac{\mathsf h(\vec p)}{E}\Big)\,X^*\,\tfrac12\Big(1+\frac{\mathsf h(-\vec p)}{E}\Big)^T,\qquad X=C^-\gamma^0, \tag{65}$$

a 16×16 matrix. A vertex $\psi^\dagger M\psi$ acting on both particles gives the amplitude $\mathrm{Tr}[\Phi(\vec{p}')^\dagger M\Phi(\vec p)M^T]$ divided by the norms of the two wave functions. Summed over the lists K of one rank, with $M=\beta\Theta^K$ and the signs sgn_K, this defines the amplitude F_r of the rank r. For the four independent channels of Sec. IV B the result is

$$\begin{aligned}F_0&=\frac{EE'+m^{*2}-\vec p\cdot\vec{p}'}{2EE'},\\ F_1&=\frac{3EE'-2m^{*2}-\vec p\cdot\vec{p}'}{EE'},\\ F_7&=\frac{3EE'+2m^{*2}-\vec p\cdot\vec{p}'}{EE'},\\ F_8&=-\frac{EE'-m^{*2}-\vec p\cdot\vec{p}'}{2EE'}.\end{aligned} \tag{66}$$

F_0 is the amplitude F_S of the exchange of the radial mode of the Vorton-Scalar, which couples to the scalar density, and $F_8=-F_P$ is minus the amplitude of the exchange of its phase mode, which couples to the pseudoscalar density. At rest $(F_0,F_1,F_7,F_8)=(1,1,5,0)$,

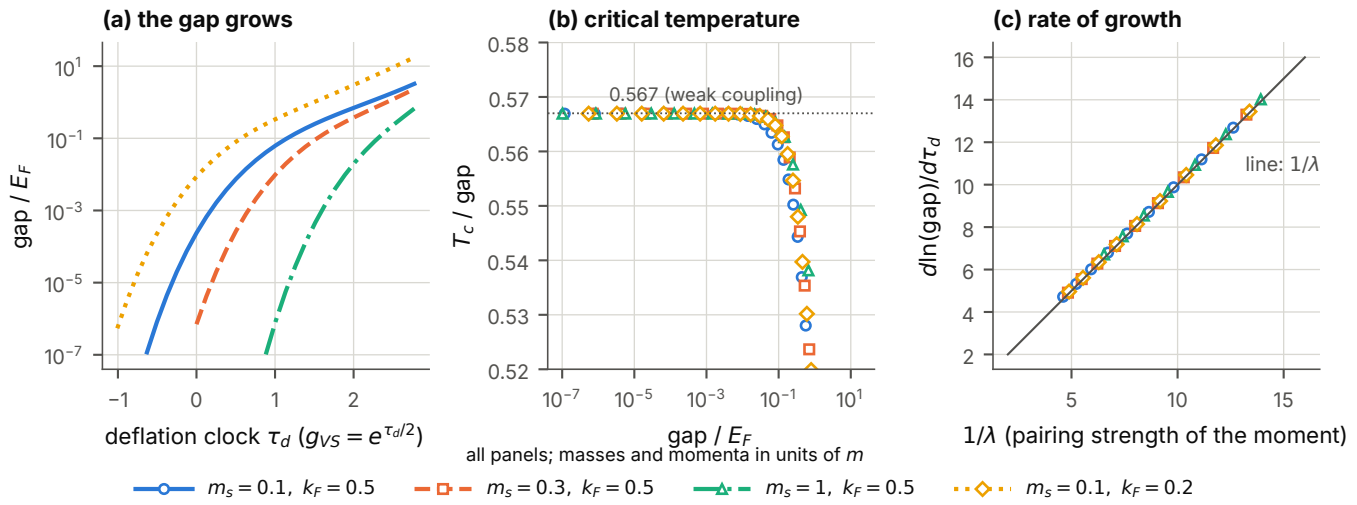


FIG. 11. The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (in units of m). (a) The gap of the vector channel in units of the Fermi kinetic energy. (b) The ratio of the critical temperature to the gap. (c) The rate of growth of the gap against $1/\lambda$; the line is Eq. (64).

the pair coupling (38). The closed forms were verified against the direct evaluation with the 256 matrices at random momenta, by the program to 1.8×10^{-15} and by the independent script with projectors built from numerical eigenvectors; the 4+4 spacetime in its quantized-sector form gives the same four functions. They hold for momenta in ordinary space; a momentum along the hidden axis itself changes them, because the pair operator points along that axis.

The relations (33) between the nine invariants hold for the amplitudes at rest, and at general momenta for the part of each amplitude that is even under $\vec{p}' \to -\vec{p}'$, which is the sum of the direct and of the exchange scattering of two identical particles; they do not hold for the odd part, which does not contribute to a contact interaction in a pair state without orbital angular momentum. For a strictly local interaction the four couplings (35) therefore suffice. When the interaction is given a range, as below, the model is *defined* by giving that range to the four channels.

B. The pairing potential and the gap equation

The pairing potential of the contact interaction in the hidden-axis channel is

$$V_{\rm ct}(\vec{p}',\vec{p}) = \left(g_0^{\rm eff} F_0 + g_1^{\rm eff} F_1 + g_7^{\rm eff} F_7 + g_8^{\rm eff} F_8\right) \frac{m_{\rm ct}^2}{q^2 + m_{\rm ct}^2}, \tag{67}$$

attractive where it is negative, with $q = |\vec{p}' - \vec{p}|$. A contact interaction in a gap equation needs a regularization at large momentum transfer. The last factor gives each coupling the range $1/m_{\rm ct}$ of an exchanged particle of the mass $m_{\rm ct}$, the *range parameter*; it is the form that the exchange of the Vorton-Scalar itself has:

$$V_{\rm VS}(\vec{p}',\vec{p}) = -g_{\rm VS}^2 \Big[\frac{F_S}{q^2 + m_s^2} + \frac{F_P}{q^2 + m_\vartheta^2}\Big], \tag{68}$$

with the mass m_s of the radial mode and the mass m_ϑ of the phase mode, which vanishes when the chiral symmetry is exact. Comparing Eqs. (67) and (68): *the exchange of the Vorton-Scalar is a self-interaction of finite range*, with the momentum-dependent couplings

$$g_0^{\rm eff}(q) = -\frac{g_{\rm VS}^2}{q^2 + m_s^2}, \qquad g_8^{\rm eff}(q) = +\frac{g_{\rm VS}^2}{q^2 + m_\vartheta^2}. \tag{69}$$

At momenta far below m_s the radial mode acts as the scalar contact coupling $g_0 = -g_{\rm VS}^2/m_s^2$. In the phase with unbroken symmetry the two masses are equal and $g_0^{\rm eff}(q) + g_8^{\rm eff}(q) = 0$, the condition (39) of chiral invariance. The program confirms that the scalar contact coupling $-g_{\rm VS}^2/m_s^2$ with $m_{\rm ct} = m_s$ reproduces the kernel of the exchange of the radial mode to 3×10^{-16}.

The gap function $\Delta(k)$ of the paired state obeys, at the temperature T,

$$\Delta(k) = \frac{1}{4\pi^2} \int_0^\infty dk'\, k'^2\, W(k,k')\, \frac{\Delta(k')}{E_{k'}} \tanh \frac{E_{k'}}{2T},$$
$$W(k,k') = \frac{1}{2} \int_{-1}^{1} d\cos\theta \,\big[- V_{\rm VS} - V_{\rm ct}\big], \tag{70}$$

with $\xi_k = (k^2 + m^2)^{1/2} - \mu$, the quasi-particle energy $E_k = (\xi_k^2 + \Delta(k)^2)^{1/2}$, the chemical potential $\mu = (k_F^2 + m^2)^{1/2}$, and the angle θ between $\vec{p}$ and $\vec{p}'$ ($k = |\vec{p}|$, $k' = |\vec{p}'|$); $\Delta_F = \Delta(k_F)$. It is the condition that the CJT effective action be stationary with respect to the pair part of the propagator, in the truncation of Sec. XII, and equally the form that the Kohn-Sham equations take with a pairing field. The angular integral is elementary, because each amplitude is linear in $\cos\theta$.

C. The gap along the clock with contact couplings

All couplings of the four-dimensional description grow as e^{τ_d} (Sec. VI), so that the ratio of a contact coupling to $g_{\rm VS}^2$ is fixed. Table VII and Fig. 12 give the gap for seven choices: the exchange alone; the exchange with a scalar contact attraction $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$; with a scalar attraction and a vector repulsion one tenth as large, $g_1^{\rm eff} = +0.1\, g_{\rm VS}^2/m^2$; with a vector repulsion alone, $g_1^{\rm eff} = +0.3\, g_{\rm VS}^2/m^2$; with an axial coupling $g_7^{\rm eff} = -0.1\, g_{\rm VS}^2/m^2$; the contact couplings without the Vorton-Scalar; and the exchange alone with a phase mode of the mass $m_\vartheta = 0.3\, m$. The range parameter is $m_{\rm ct} = m$.

Three statements summarize the result; each was verified at every clock reading at which the gap exceeds 10^{-6} of the Fermi kinetic energy.

A scalar contact attraction raises the gap, strongly so where the exchange alone is weak: for $m_s = m$ at the clock reading of Table VII by the factor 12.0.

A vector contact repulsion lowers it. A vector coupling of three tenths of $g_{\rm VS}^2/m^2$ removes most of the gap of a weakly paired state (Table VII, $m_s = m$) and reduces the gap of a strongly paired one.

A mass of the phase mode lowers it, because it weakens the attraction of that mode at small momentum transfer.

Without contact couplings the program reproduces the gaps of Sec. VIII to 7×10^{-16}. For small gaps the critical temperature remains 0.567 times the gap. An independent solution of Eq. (70) on another grid of momenta, by plain iteration, gives the gaps of Table VII for the first three rows to 1.1×10^{-3} (relative). The law of growth (64) holds with contact couplings as it does without, because every part of the kernel grows as e^{τ_d}.

X. THE BOUND PAIR AND THE ORDER OF EVENTS

A. The moment of binding

Two fermions in the Yukawa potential (52) form a bound state if and only if the number $D_{\rm VS}$ of Eq. (49) exceeds a critical value, $D_c = 1.67981$. On a deflating spacetime $D_{\rm VS} \propto e^{\tau_d}$, so that the pair becomes bound at a definite time. Measure the clock from that moment:

$$\tau_{\rm pair} = \ln \frac{D_{\rm VS}}{D_c} = {\rm gsum}\left(x4 - x4_{\rm binding}\right). \qquad (71)$$

Table VIII and Fig. 13 show the scattering length $a_{\rm sc}$ of two fermions and the binding energy E_b of the pair as the clock passes the binding.

The scattering length. Just before the binding the scattering length is large and negative, just after it large and positive: $-134.0/m_s$ at $0.99\, D_c$ and $+135.9/m_s$ at $1.01\, D_c$. At the moment of binding it is infinite: the two fermions are at a scattering resonance. *Every pair of fermions of the model passes through this resonance once, at the same time.* Section XVI examines what that means if the fermions are the dark matter.

The binding energy. Just after the binding the binding energy starts quadratically,

$$E_b \;\propto\; \left(e^{\tau_{\rm pair}} - 1\right)^2 \frac{m_s^2}{m}, \qquad (72)$$

with the fitted exponent 2.01 between 2 and 8 per cent above D_c. Because E_b vanishes at the threshold, the time $1/E_b$ that the pair needs to adjust is infinite there. However slow the deflation is, the pair cannot follow it adiabatically at first.

The same threshold from the gap equation. In vacuum the linearized gap equation is the equation of a bound state of two fermions, with a relativistic kernel. It reaches the eigenvalue 1 at $D_{\rm VS} = 1.7347$, 1.6902 and 1.6810 for $m_s/m = 0.3$, 0.1 and 0.03, approaching the value D_c of the Schrödinger equation as the radial mode becomes light and the binding non-relativistic.

B. The order of events

The other thresholds of the model are thresholds in the number C of Eq. (49), which grows at the same rate as $D_{\rm VS}$. Because $C/D_{\rm VS} = 4\pi m/m_s$ is fixed, the order in which the events happen depends on the ratio m_s/m alone. Table IX gives the clock reading $\tau_{\rm pair}$ of each event; a negative number means that the event precedes the binding of the pair. The events other than the binding are derived in Sec. XII.

For a *light radial mode*, $m_s/m \le 3$ in the table, the pair becomes bound only *after* the one-loop potential of the Vorton-Scalar has fallen below its vacuum value at a field of three times the vacuum value: the deflation endangers the vacuum of the model before it binds the pair. The condition for the opposite order is $4\pi D_c\, m/m_s < 4.27$, that is, $m_s/m > 4.9$. For a *heavy radial mode*, $m_s/m = 10$ or 30, the pair becomes bound first, and matter becomes self-bound and loses its fermion mass later. Nothing in the model decides between the two orders; the table states what follows from each.

XI. CAN THE PAIRED STATE FOLLOW THE DEFLATION?

A. The question, and the parameter that decides it

Section VIII gave the gap that the paired state would have in equilibrium at each moment. But a state can change at most at the rate of its own energy scale, which for a paired state is the gap. By Eq. (64) the equilibrium gap, written $\Delta_{\rm eq}$ in this section, changes at the rate ${\rm gsum}/\lambda$. The paired state can follow only if that rate is small compared with the gap itself:

$$\Xi_d = \frac{\rm gsum}{\lambda\, \Delta_{\rm eq}} \;\ll\; 1. \qquad (73)$$

TABLE VII. The gap at the Fermi surface in units of the Fermi kinetic energy, with contact couplings, at the clock reading 2.04 [$g_{\rm VS} = 2.771$], for the Fermi momentum $k_F = 0.5\,m$ and three masses of the radial mode. Contact couplings in units of $g_{\rm VS}^2/m^2$; $m_{\rm ct} = m$.

interaction	$m_s = 0.1\,m$	$m_s = 0.3\,m$	$m_s = m$
exchange of the Vorton-Scalar alone	0.7502	0.4117	0.0246
with a scalar contact attraction, $g_0^{\rm eff} = -1$	1.2882	0.9023	0.2968
with $g_0^{\rm eff} = -1$ and $g_1^{\rm eff} = +0.1$	1.1102	0.7403	0.1961
with a vector contact repulsion, $g_1^{\rm eff} = +0.3$	0.5438	0.2370	0.0006
with an axial-vector contact coupling, $g_7^{\rm eff} = -0.1$	1.0108	0.6454	0.1287
contact couplings alone, $g_0^{\rm eff} = -1$, $g_1^{\rm eff} = +0.1$	0.0006	0.0006	0.0006
exchange alone, phase mode of the mass $0.3\,m$	0.7340	0.3966	0.0197

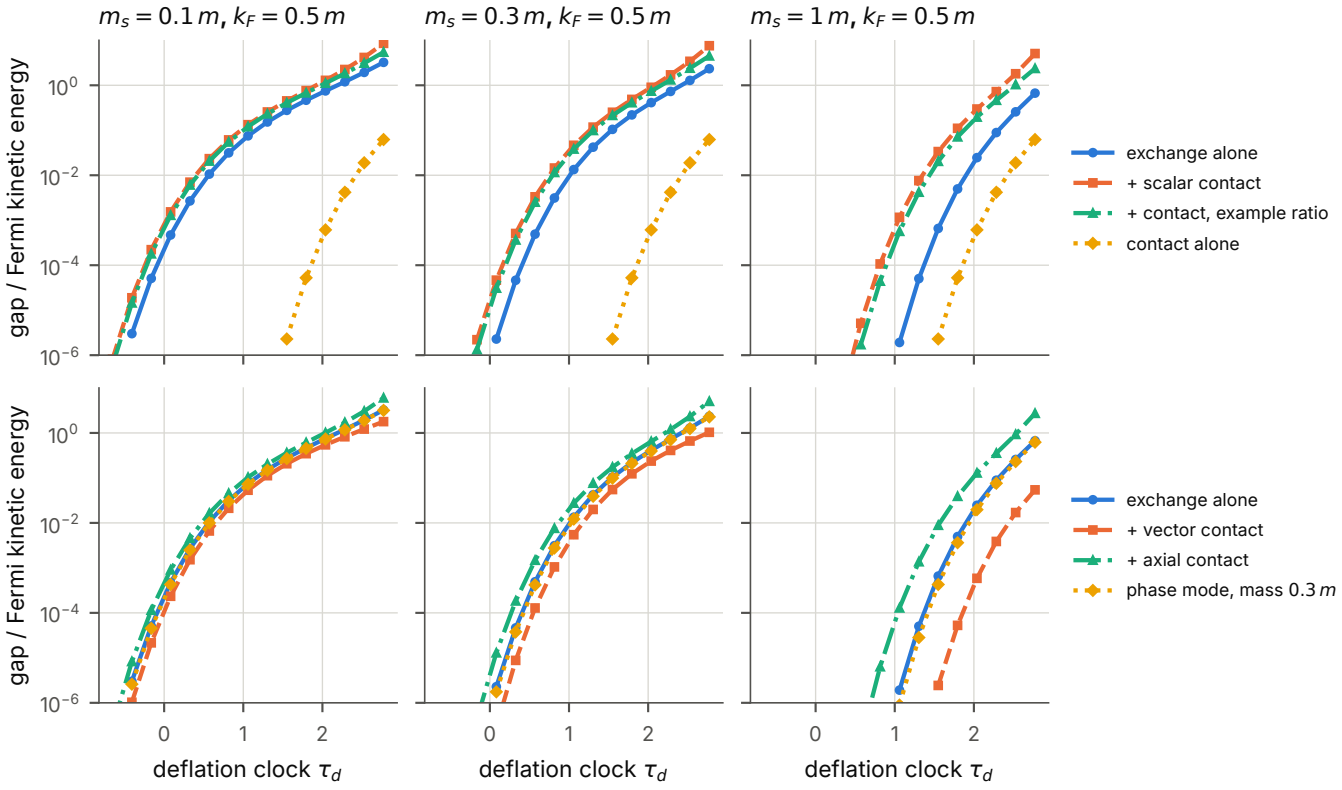


FIG. 12. The gap of the hidden-axis pair along the deflation clock with contact couplings, for the Fermi momentum $0.5\,m$ and three masses m_s of the radial mode of the Vorton-Scalar. Upper row: the exchange of the Vorton-Scalar alone; with a scalar contact attraction; with a scalar attraction and a vector repulsion one tenth as large; the contact couplings alone. Lower row: the exchange alone; with a vector contact repulsion; with an axial contact coupling; with a phase mode of the mass $0.3\,m$. Couplings and range parameter as in Table VII.

TABLE VIII. Two fermions as the clock passes the moment of binding: the scattering length in units of $1/m_s$, the number of bound states, and the binding energy in units of m_s^2/m.

$\tau_{\rm pair}$	$D_{\rm VS}$	$a_{\rm sc}\,m_s$	bound states	$E_b\,m/m_s^2$
−0.105	1.5118	−12.50	0	0
−0.095	1.5279	−13.93	0	0
−0.010	1.6630	−134.0	0	0
0.010	1.6966	135.9	1	5.50×10^{-5}
0.486	2.7314	2.698	1	0.232
0.994	4.5409	−0.1288	1	1.82

Early in the deflation the gap is exponentially small, and the condition fails however small the rate is. A system that is carried through a change faster than it can respond falls out of equilibrium, as in the formation of defects at a phase transition [17, 18]. The question is what the paired state does then, and when it catches up.

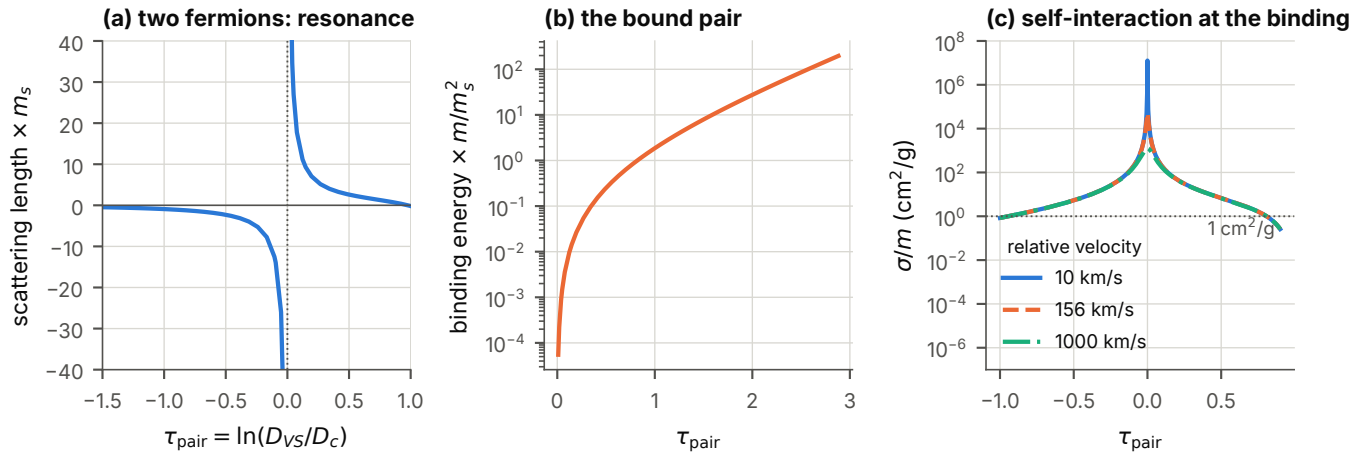


FIG. 13. The bound pair along the clock. (a) Scattering length of two fermions in units of $1/m_s$ against $\tau_{\rm pair}$: the resonance at the moment of binding. (b) Binding energy of the pair. (c) Cross-section per mass of two fermions at three relative velocities while the clock passes the binding, for the example particle of Sec. XVII.

TABLE IX. The order of events along the deflation: the clock reading $\tau_{\rm pair}$ of each event, relative to the moment at which two fermions form a bound pair, for six ratios of the mass m_s of the radial mode to the fermion mass m.

event	threshold	0.1	0.3	1	3	10	30
the one-loop potential of the Vorton-Scalar falls below its vacuum value at the field $10\,v_{\rm VS}$	$C = 1.5477$	−4.92	−3.82	−2.61	−1.51	−0.31	0.79
matter is self-bound (Fermi sea only)	$C = 3.7011$	−4.04	−2.94	−1.74	−0.64	0.56	1.66
the one-loop potential falls below its vacuum value at the field $3\,v_{\rm VS}$	$C = 4.2694$	−3.90	−2.80	−1.60	−0.50	0.70	1.80
matter is self-bound (with the negative-energy states)	$C = 5.9218$	−3.57	−2.48	−1.27	−0.17	1.03	2.13
matter of Fermi momentum $0.5\,m$ becomes abnormal (massless fermions)	$C = 12.9713$	−2.79	−1.69	−0.49	0.61	1.82	2.91
two fermions form a bound pair	$D_{\rm VS} = 1.6798$	0.00	0.00	0.00	0.00	0.00	0.00
pair condensate in the vacuum (ladder equation, cut-off $100\,m$)	$\alpha_{\rm VS} = 2.13$ to 2.93	2.54	1.44	0.25	−0.82	−1.93	−2.84

B. The model

The question is answered with the standard reduced model of a superfluid, whose dynamics without collisions is known in detail [19]. Pairs of states of opposite momenta, with the single-particle energy ξ_k measured from the Fermi energy, fill a shell $-E_{\rm sh} < \xi_k < E_{\rm sh}$ with a constant density of states, and all attract one another with the dimensionless strength λ. In the mean-field approximation each pair of states is described by its pair correlation κ, a complex number, and its occupation n_k; the order parameter is $\Delta = \lambda \int \kappa\, d\xi_k$. The equations of motion are

$$\frac{d\kappa}{dx4} = -i\left[2\,\xi_k\,\kappa - \Delta\,(1-2n_k)\right],$$
$$\frac{dn_k}{dx4} = 2\,\mathrm{Im}\big(\Delta^*\kappa\big). \qquad (74)$$

They conserve $|2\kappa|^2 + (1-2n_k)^2 = 1$ for every pair of states. In equilibrium $\kappa = \Delta/2E_k$ and $n_k = \frac{1}{2}(1-\xi_k/E_k)$ with the quasi-particle energy $E_k = (\xi_k^2+\Delta^2)^{1/2}$, and the gap equation gives

$$\Delta_{\rm eq}(\lambda) = \frac{E_{\rm sh}}{\sinh(1/\lambda)} \approx 2\,E_{\rm sh}\,e^{-1/\lambda} \quad (\lambda \ll 1). \qquad (75)$$

The model keeps what matters for the question, a gap that depends exponentially on a coupling that grows exponentially, and omits the dependence of the gap on the momentum; $E_{\rm sh}$ stands for the range of energies over which the attraction acts. The program starts the model in equilibrium at $\lambda = 0.15$, lets the coupling grow as $\lambda = 0.15\,e^{\tau_d}$ up to $\lambda = 0.5$, and integrates Eqs. (74) for 2000 energies. Doubling the number of energies and halving the step changes no digit of the results quoted. Besides $|\Delta|$ the program records the density $n_{\rm ex}$ of *broken pairs*: the number of pairs of states, per unit of $E_{\rm sh}$ times the density of states, that are not in the ground state that belongs to the order parameter of the moment. In equilibrium it is zero.

TABLE X. The paired state while the pairing strength grows from 0.15 to 0.5, for seven rates of deflation: the parameter Ξ_d at the start; the smallest ratio of the order parameter to the equilibrium gap of the moment; the clock reading at which the order parameter has returned to within 10 per cent of the equilibrium gap, and Ξ_d at that moment; the ratio at the end; and the density of broken pairs at the end.

gsum/$E_{\rm sh}$	Ξ_d at the start	smallest $\lvert\Delta\rvert/\Delta_{\rm eq}$	τ_d of the catch-up	Ξ_d there	final $\lvert\Delta\rvert/\Delta_{\rm eq}$	$n_{\rm ex}$ at the end
1×10^{-4}	0.262	0.958	0.020	0.23	1.0000	6.0×10^{-6}
3×10^{-4}	0.786	0.890	0.069	0.47	0.9998	5.1×10^{-5}
1×10^{-3}	2.62	0.737	0.234	0.52	0.9980	4.4×10^{-4}
3×10^{-3}	7.86	0.539	0.454	0.44	0.9886	2.4×10^{-3}
0.01	26.2	0.328	0.780	0.32	0.9347	0.012
0.03	78.6	0.189	1.172	0.24	0.8076	0.041
0.1	262	0.098	—	—	0.1189	0.022

C. Results

Table X and Fig. 14 show three results.

The faster the deflation, the further the order parameter falls behind. For a rate of $10^{-4}\,E_{\rm sh}$ the order parameter drops to 0.958 of the equilibrium gap of the moment before it recovers; for $10^{-1}\,E_{\rm sh}$ it drops to 0.098.

The state catches up when Ξ_d has fallen to a value between 0.23 and 0.52. In six of the seven runs the order parameter returns to within 10 per cent of the equilibrium gap at such a value. In the fastest run it has not caught up when the coupling reaches 0.5. This confirms the estimate (73) and fixes the number that "small" means.

The lag leaves broken pairs behind. Their density at the end grows with the rate, from 6.0×10^{-6} to 4.1×10^{-2} in the six slower runs. They are excitations of the superfluid that the deflation itself has made; in a system without dissipation they remain.

Because $\Delta_{\rm eq}$ grows as $e^{-1/\lambda}$ while λ grows only as e^{τ_d}, the parameter Ξ_d falls steeply with time. For any rate there is a moment at which Ξ_d passes through the window of the second result. Before it the paired state is out of equilibrium, with an order parameter below the equilibrium value; after it the adiabatic history of Sec. VIII applies. The slower the deflation, the earlier that moment and the smaller the gap at which it happens: it is the moment at which the equilibrium gap has grown to between two and four times gsum/λ.

XII. THE EFFECTIVE ACTION ALONG THE CLOCK

A. The method, and why it carries over

The results on the paired state, on matter, on temperature and on the vacuum follow from the effective action for composite operators of Cornwall, Jackiw and Tomboulis (CJT) [20]: a functional of the expectation value ϕ of the Vorton-Scalar and of the full propagators, whose two-particle-irreducible part is truncated at two loops. Its stationarity conditions are the equation for the fermion mass in matter and the gap equation. On the deflating spacetime, and for fields that do not depend on the extra coordinates, nothing in this construction changes, for two reasons established above. The weighted fermion field obeys the flat equations, so that its propagator is the flat one. And the interaction between the fermions is, at each moment, the exchange of the two modes of the scalar with the coupling of that moment and with the unshifted range (Sec. VI D).

The one new term, the shift $-\text{gsum}^2/4$ of the squares of the boson masses in Eq. (47), is smaller than every other scale of the calculation by the square of the ratio of the rate of deflation to a particle mass. For the largest rate that observation allows (Sec. XVI) and a mass of 1 keV that ratio is below 10^{-36}. The term is neglected in this section; this is the only approximation made here beyond the truncation itself. The test of the truncation on a solvable model in zero dimensions, which does not involve the spacetime, was executed again: the two-loop truncation has a solution for every coupling from 0.01 to 30 and stays within 10.7 per cent of the exact propagator.

Every result of this section is a statement about a *sequence of equilibrium states.* Whether a system can follow the sequence was examined, for the paired state, in Sec. XI.

B. The paired state

The CJT potential $V[\Delta]$ of the paired state is the effective action per unit volume, measured from the unpaired state, as a functional of the gap function $\Delta(k)$. The program evaluates it at 13 moments, for two choices of the mass m_s at the Fermi momentum $k_F = 0.5\,m$, in the vector channel (Table XI, Fig. 15).

The solution of the gap equation is a minimum at every moment. Newton's method applied to the potential converges to the gap function that the iteration of the gap equation finds, to 7.7×10^{-12}, and the smallest eigenvalue of the second variation is positive throughout. The temperature at which the second variation at $\Delta = 0$ loses its positivity equals the temperature at which the linearized gap equation has the eigenvalue 1, to 2.6×10^{-12}: two

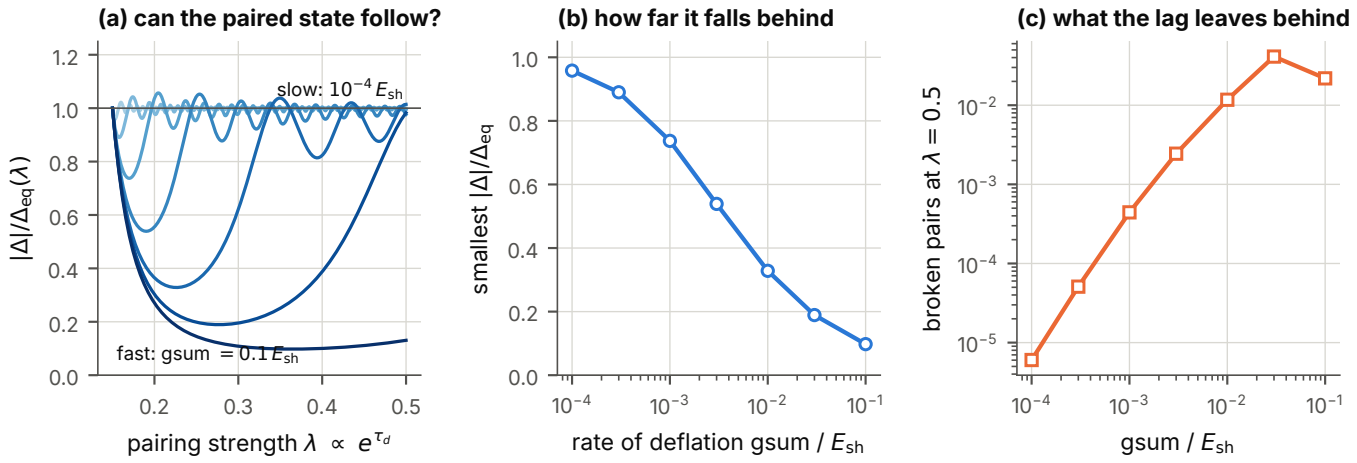


FIG. 14. The paired state while the coupling grows. (a) The order parameter divided by the equilibrium gap of the moment, against the pairing strength, for seven rates of deflation. (b) The smallest value of that ratio against the rate. (c) The density of broken pairs at the end of the run against the rate.

TABLE XI. The paired state along the clock from the CJT potential, for $k_F = 0.5\,m$. λ: pairing strength. V: the potential at the solution, the condensation energy. Ratio: V divided by its weak-coupling value $-\nu N(0)\Delta_F^2/4$, with $\nu = 8$ internal states and the density of states $N(0)$. Curvature: the smallest eigenvalue of the matrix of second derivatives of the potential. Size: the coherence length of the pair in units of $1/k_F$.

m_s/m, k_F/m	τ_d	g_{VS}	λ	Δ_F/m	V/m^4	ratio	curvature	T_c/Δ_F	size $\times k_F$
0.30, 0.50	0.00	1.80	0.216	2.81×10^{-3}	-4.46×10^{-7}	1.000	0.21	0.567	25.4
0.30, 0.50	0.21	2.00	0.266	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.26	0.566	10.3
0.30, 0.50	0.40	2.20	0.322	0.0135	-1.03×10^{-5}	0.992	0.30	0.564	5.3
0.30, 0.50	0.58	2.40	0.383	0.0230	-2.93×10^{-5}	0.981	0.34	0.559	3.1
0.30, 0.50	0.74	2.60	0.450	0.0354	-6.81×10^{-5}	0.961	0.37	0.552	2.0
0.30, 0.50	0.88	2.80	0.522	0.0511	-1.38×10^{-4}	0.932	0.38	0.542	1.4
0.30, 0.50	1.02	3.00	0.599	0.0707	-2.53×10^{-4}	0.895	0.38	0.530	1.0
1.0, 0.50	0.00	2.40	0.133	4.25×10^{-4}	-1.02×10^{-8}	1.000	0.13	0.567	167.6
1.0, 0.50	0.16	2.60	0.156	1.32×10^{-3}	-9.88×10^{-8}	1.000	0.15	0.567	53.9
1.0, 0.50	0.31	2.80	0.181	3.28×10^{-3}	-6.08×10^{-7}	0.999	0.17	0.567	21.7
1.0, 0.50	0.45	3.00	0.208	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.19	0.566	10.3
1.0, 0.50	0.58	3.20	0.237	0.0128	-9.25×10^{-6}	0.994	0.21	0.564	5.6
1.0, 0.50	0.70	3.40	0.267	0.0218	-2.65×10^{-5}	0.987	0.23	0.561	3.3

routes to the critical temperature agree.

The condensation energy deepens. It is within 15 per cent of the weak-coupling value wherever the gap is below $0.1\,E_F$, and in weak coupling its logarithmic slope along the clock is twice that of the gap, by Eq. (64):

$$\frac{d\ln|V|}{d\tau_d} = \frac{2}{\lambda}. \tag{76}$$

The pairs shrink. The coherence length of the first set falls from 25 to 1.0 in units of $1/k_F$. At its last five moments the gap exceeds $0.1\,E_F$, and at the last the pair is as small as the distance between the particles. *The deflation carries the paired state from overlapping pairs toward bound pairs.* On flat spacetime this crossover [21, 22] is reached by choosing a larger coupling; here it is reached by waiting.

The universal ratios hold while the coupling is weak. While the critical temperature rises by a factor of sixteen, from 6.3×10^{-3} to 0.10 of the Fermi energy, the ratio of the gap at zero temperature to the critical temperature stays within 0.019 of the weak-coupling value 1.7639, and the relative jump of the heat capacity at T_c within 0.048 of 1.4261. The coefficients of the Ginzburg-Landau energy agree with their weak-coupling formulas to 1.0 per cent while T_c is below $0.05\,E_F$ and differ by up to 7.9 per cent at $T_c = 0.10\,E_F$. The coherence length of the Ginzburg-Landau energy, $0.1333\,v_F/T_c$ in weak coupling with the Fermi velocity v_F, shrinks from 80 to 5.1 in units of $1/m$: the cores of the vortices of the superfluid shrink as the deflation proceeds.

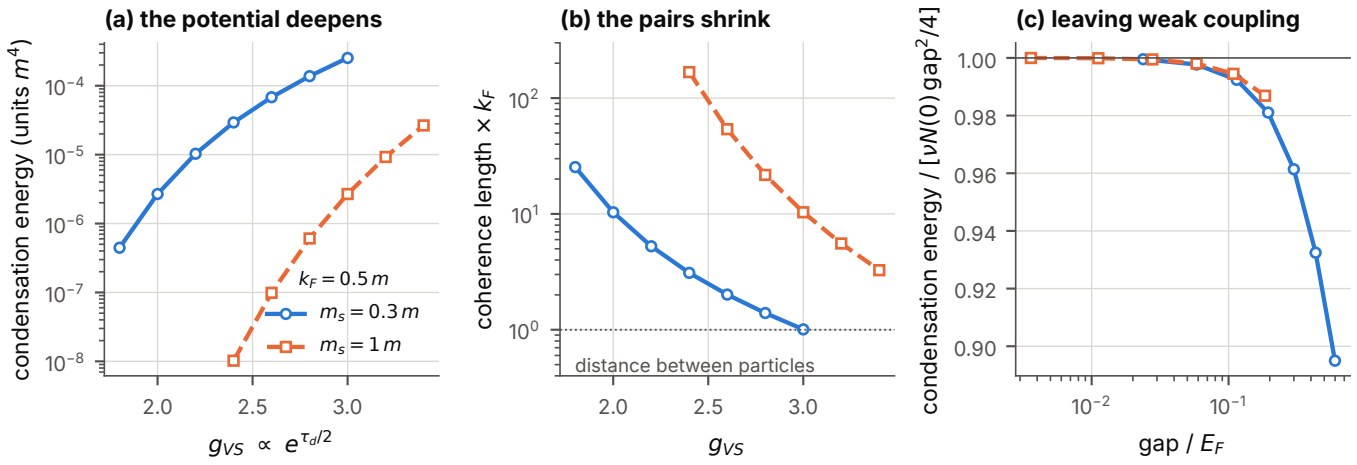


FIG. 15. The paired state along the clock, from the CJT potential, for the two sets of Table XI. (a) The condensation energy against the coupling of the moment. (b) The coherence length of the pair in units of $1/k_F$. (c) The condensation energy divided by its weak-coupling value, against the gap in units of the Fermi kinetic energy: the departure from weak coupling.

C. Matter

Matter of fixed density loses its fermion mass. In matter the expectation value of the Vorton-Scalar is reduced by the scalar density of the fermions, and with it the fermion mass in the matter, m^*. The reduction depends on the density and on the single number C, and above a critical value of C the mass vanishes: the matter is then in the *abnormal* state of Lee and Wick [23], a gas of massless fermions in a region without expectation value of the scalar. On the deflating spacetime $C \propto e^{\tau_d}$ grows at fixed density. The critical values are $C = 151.892$, 12.971, 2.556 and 0.617 for the Fermi momenta $k_F = 0.2$, 0.5, 1 and 2 in units of m; they exceed the estimate $\pi^2 m^2/4k_F^2$, which holds for dense matter, by the factors 2.46, 1.31, 1.04 and 1.00. *Every piece of matter becomes abnormal at a definite moment, the denser the earlier.*

Self-bound matter. Matter is self-bound, that is, its energy per particle at some density lies below the mass of a free fermion, if $C > 3\pi^2/8 = 3.7011$ without the filled negative-energy states and if $C > 3\pi^2/5 = 5.9218$ with them. Once the threshold is passed, the lowest energy per particle falls monotonically along the clock (Table XII), from $-0.0192\,m$ at $C = 4$ to $-0.405\,m$ at $C = 29.6$. At every one of these moments the matter of lowest energy is in the abnormal state. The exchange energies of the two modes, which the mean field omits, are 0.44 and 0.51 per cent of the mean-field energy at the first moment and 0.90 and 1.22 per cent at the last (for $m_s/m = 0.3$): the mean field remains a good approximation along the whole history. A gas of the fermions that is dilute and unbound at early times thus has, at late times, a state of lower energy: drops of dense abnormal matter, bound by up to 40 per cent of the rest mass at the last moment of the table. Whether and how fast a gas reaches that state is a question of kinetics that is not answered here.

TABLE XII. Self-bound matter along the clock, in the mean-field approximation without the negative-energy states, for $C = 4\,e^{\tau_d}$: the lowest energy per particle minus the fermion mass, and the Fermi momentum at which it is reached.

τ_d	C	$(E/N - m)/m$	k_F/m
0.00	4.00	−0.0192	0.981
0.25	5.14	−0.0787	0.921
0.50	6.59	−0.1345	0.866
0.75	8.47	−0.1869	0.813
1.00	10.87	−0.2362	0.764
1.25	13.96	−0.2825	0.718
1.50	17.93	−0.3259	0.674
1.75	23.02	−0.3668	0.633
2.00	29.56	−0.4051	0.595

D. Temperature

At high temperature the expectation value of the Vorton-Scalar vanishes and the fermions are massless. The temperature $T_{1/2}$ at which the expectation value has fallen to one half is of the order of the four-dimensional vacuum value $v_{VS} = m/g_{VS}$, which falls as $e^{-\tau_d/2}$ while the masses stay fixed. In units of the constant fermion mass, $T_{1/2}$ falls monotonically as the coupling grows, for each of the three ratios $m_s/m = 0.3$, 1 and 3 examined; for $m_s/m = 1$, from $2.33\,m$ at $g_{VS} = 0.3$ to $0.37\,m$ at $g_{VS} = 4$ [Fig. 16(d)]. *Matter of a fixed temperature melts at a definite moment of the deflation.* For $m_s/m = 0.3$ no melting temperature can be given from $C = 32$ on: there the one-loop potential of the vacuum turns over so close to the vacuum value that the Hartree potential ceases to exist before the expectation value has fallen, and the program marks these moments as not defined.

E. The vacuum

The one-loop potential of the Vorton-Scalar in vacuum contains the negative contribution of the fermion loop [24], which grows with the fourth power of the field and eventually wins: beyond some field the potential lies below its value in the vacuum. The larger C, the smaller that field: 24.7 times the vacuum value at $C = 1$, and 1.77 times at $C = 10$ [Fig. 16(c)]. As the deflation proceeds, the barrier between the vacuum of the model and the region where the one-loop potential is unbounded below moves in toward the vacuum.

The one-loop potential is not reliable at fields far from the vacuum value, and a potential that falls at large field signals that the effective theory needs completion there, not that the vacuum decays. What the result establishes is a trend: every statement of the model that rests on the vacuum of the Vorton-Scalar becomes less secure with time. The order of this event relative to the binding of the pair was given in Sec. X.

A pair condensate in the vacuum. The ladder equation with a momentum cutoff shows that the pair can condense without a medium only if $\alpha_{\rm VS}$ exceeds a critical value α_c of order one [25]; with the cutoff $100\,m$ the critical values lie between 2.13 and 2.93 for the six mass ratios of Table IX. On the deflating spacetime $\alpha_{\rm VS}$ reaches such a value at a definite moment. But at that moment the coupling is strong, the exchange of one boson is no longer a controlled approximation, and the vacuum of the model is no longer protected. A pair condensate in the vacuum is not reached by the deflation within the range of validity of the model.

XIII. THE EFFECTIVE POTENTIAL WITH THE SELF-INTERACTION

Section XII applied the effective action of Cornwall, Jackiw and Tomboulis to the model in which the Vorton-Scalar alone provides mass and interaction. This section applies the same method to the self-interaction of the fermion field: first without the Vorton-Scalar, where it must reproduce the Kohn-Sham results of Sec. VII and goes beyond them by counting the filled states of negative energy; then with the bare Dirac mass, the self-interaction and the Vorton-Scalar together.

A. The effective potential of uniform matter

The CJT effective action is a functional of the full propagator S of the fermions,

$$\Gamma[S] = -i\,\mathrm{Tr}\ln S^{-1} - i\,\mathrm{Tr}\big(S_0^{-1}S - 1\big) + \Gamma_2[S], \qquad (77)$$

where S_0 is the propagator of free particles of the mass m and Γ_2 is the sum of the vacuum diagrams that cannot be cut in two by cutting two lines, drawn with the full propagator. For the contact interaction $\mathcal{L}_{\rm self}$ the lowest order of Γ_2 consists of two diagrams, each with two loops that meet at one vertex: the direct (Hartree) and the exchange (Fock) diagram. At this order the stationarity condition $\delta\Gamma/\delta S = 0$ says that the self-energy $S_0^{-1} - S^{-1}$ does not depend on the momentum. For uniform matter at rest it is a combination of the unit matrix and of β (Sec. IV C): the full propagator is that of free particles of a mass m^* whose energies are shifted by a constant V_0. The effective action per unit volume and time then becomes an ordinary function of the two numbers m^* and V_0, the *effective potential*. At zero temperature and at the chemical potential μ,

$$\Omega_{\rm CJT}(m^*, V_0; \mu) = -\frac{(m^* - m)^2}{2\,G_S} - \frac{V_0^2}{2\,G_V} - \nu\int\frac{d^Dk}{(2\pi)^D}\,\big(\mu - V_0 - E^*\big)\,\theta\big(\mu - V_0 - E^*\big), \quad (78)$$

with $E^* = (k^2 + m^{*2})^{1/2}$, the step function θ, and the couplings G_S, G_V of Eq. (37). The integral is the grand potential of free particles of the mass m^* at the chemical potential $\mu - V_0$; the two other terms remove the interaction energy that the integral counts twice. The two stationarity conditions are

$$\begin{aligned} \frac{\partial\Omega_{\rm CJT}}{\partial V_0} = 0: &\quad V_0 = G_V\,n, \\ \frac{\partial\Omega_{\rm CJT}}{\partial m^*} = 0: &\quad m^* = m + G_S\,n_S, \end{aligned} \qquad (79)$$

the two equations (54) of the Kohn-Sham method. At the stationary point $-\Omega_{\rm CJT}$ is the pressure and $\Omega_{\rm CJT} + \mu\,n$ the energy density. *The Kohn-Sham method and the CJT effective potential at this order are two formulations of the same approximation*: the first works at a fixed density and solves the gap equation, the second works at a fixed chemical potential and searches the stationary point of a function.

The program evaluates the two with separate code (Gauss-Legendre integrals and a root search in the one, Simpson integrals and a minimization in the other) for 30 states in $D = 7$, 4 and 3 dimensions (Table XIII). Mass, density, energy density and pressure agree to 1.6×10^{-12} (sum of the four relative deviations); the independent script repeats the comparison with its own integrals and finds the same.

The effective potential says more than the gap equation in one respect. At a given chemical potential the stationary point with the lowest $\Omega_{\rm CJT}$, that is, with the highest pressure, is the state of equilibrium (the vector field V_0 is taken at its stationary value, a maximum; the minimum is over m^*). Of the 23 states of the comparison with a positive pressure, 20 are that minimum. In the others a denser state has the higher pressure at the same chemical potential: the Kohn-Sham state of that density is metastable, a dilute gas that can lower its energy by condensing into saturated matter.

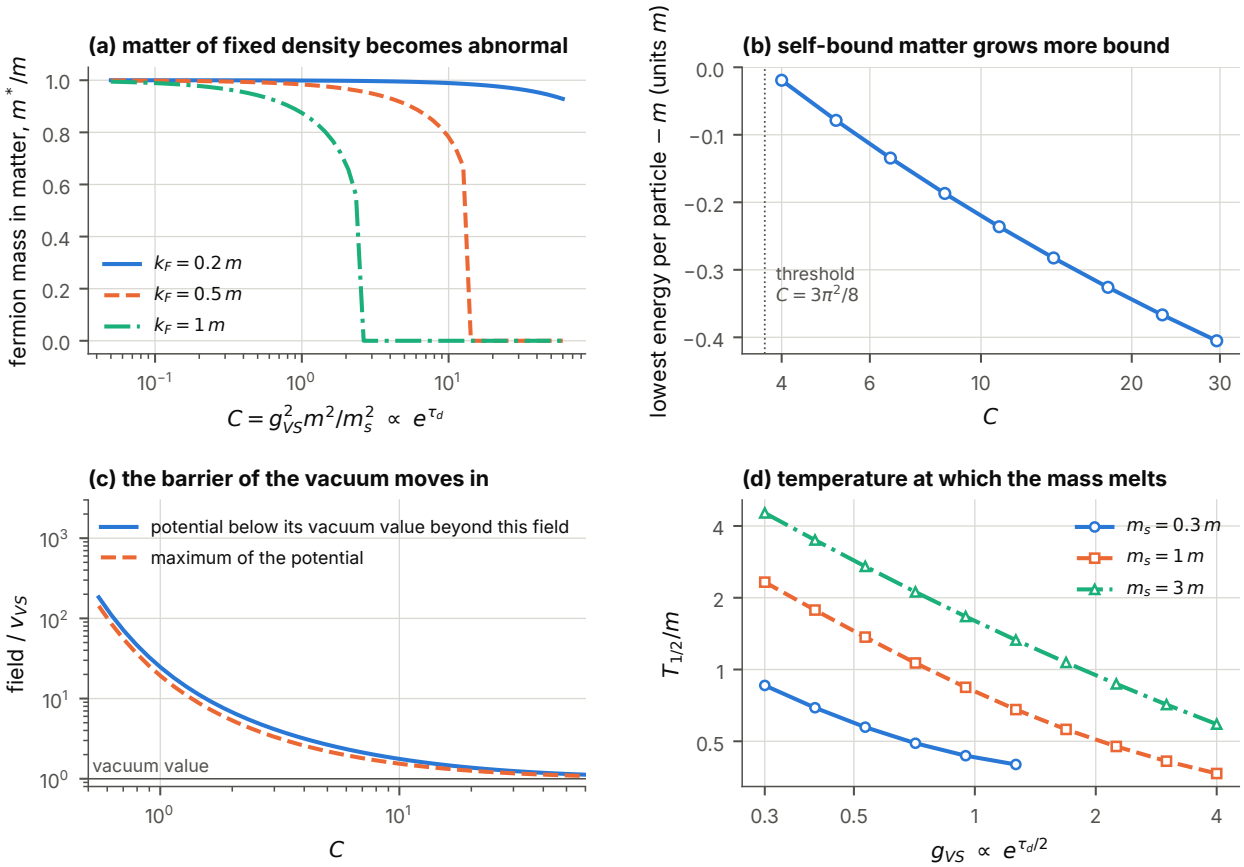


FIG. 16. Matter, vacuum and temperature along the clock. (a) The fermion mass in matter of three fixed densities against $C \propto e^{\tau_d}$: it drops to zero at the critical value. (b) The lowest energy per particle of self-bound matter, minus m. (c) The field at which the one-loop potential of the vacuum has its maximum, and the field beyond which it lies below the vacuum value, in units of the vacuum value. (d) The temperature at which the fermions have lost half of their mass, against the coupling, for three ratios m_s/m.

TABLE XIII. The Kohn-Sham method (KS) and the CJT effective potential compared, without the filled states of negative energy (units $m = 1$): the mass of the particles in matter by both methods, and $-\Omega_{\rm CJT}$ at its stationary point divided by the Kohn-Sham pressure p, for ten of the 30 states computed (the data files hold eleven digits).

D	G_S	G_V	k_F	m^* (KS)	m^* (CJT)	$-\Omega_{\rm CJT}/p$
7	−100000	10000	0.9	0.143888	0.143888	1.00000000
7	−100000	10000	1.4	0.011510	0.011510	1.00000000
7	−98750	20000	0.9	0.145495	0.145495	1.00000000
7	−98750	20000	1.4	0.011654	0.011654	1.00000000
4	−256.9512	25.6951	0.9	0.140053	0.140053	1.00000000
4	−256.9512	25.6951	1.4	0.040348	0.040348	1.00000000
3	−29.5093	2.9509	0.9	0.182319	0.182319	1.00000000
3	−29.5093	2.9509	1.4	0.079333	0.079333	1.00000000
3	−4	0	0.9	0.394848	0.394848	1.00000000
3	−4	0	1.4	0.240321	0.240321	1.00000000

B. The filled states of negative energy

The Kohn-Sham calculation counts the particles of matter only. The term $-i\,\mathrm{Tr}\ln S^{-1}$ of Eq. (77) also contains the filled states of negative energy of the vacuum, whose energy depends on the mass of the particles. Because the contact interaction is an effective interaction, their momenta are counted up to a cutoff Λ:

$$\Omega_{\rm CJT} \to \Omega_{\rm CJT} - \nu \int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D} \times \left[\sqrt{k^2 + m^{*2}} - \sqrt{k^2 + m^2}\right]. \tag{80}$$

The filled states contribute the scalar density $-\nu \int_{|\vec{k}|<\Lambda} d^D k\,(m^*/E^*)/(2\pi)^D$ to the gap equation. The mass parameter of the Lagrangian is then no longer the mass of a particle in vacuum and is called the *bare mass* $m_{\rm bare}$; the mass in vacuum, $m_{\rm vac}$, is the solution of

$$\begin{aligned} m_{\rm vac} &= m_{\rm bare} + G_S\,\langle\bar\psi\psi\rangle_{\rm vac}, \\ \langle\bar\psi\psi\rangle_{\rm vac} &= -\nu \int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D}\, \frac{m_{\rm vac}}{\sqrt{k^2 + m_{\rm vac}^2}}. \end{aligned} \tag{81}$$

TABLE XIV. The mass of the fermions in vacuum, $m_{\rm vac}/\Lambda$, with the filled states of negative energy, along the deflation clock counted from the moment at which $G_S = G_c$, for a field without bare mass and for the bare mass $0.05\,\Lambda$.

D	$m_{\rm bare}/\Lambda$	$\tau_d = -1$	$\tau_d = 0$	$\tau_d = 0.24$	$\tau_d = 0.48$
7	0	0	0	0.658	1.074
7	0.05	0.079	0.436	0.774	1.154
4	0	0	0	0.544	0.913
4	0.05	0.079	0.383	0.664	0.995
3	0	0	0	0.447	0.780
3	0.05	0.078	0.335	0.573	0.866

The scalar density of the vacuum is negative, so that an attraction ($G_S < 0$) *raises* the mass above the bare mass.

A critical coupling. For $m_{\rm bare} = 0$ Eq. (81) always has the solution $m_{\rm vac} = 0$. A second solution, of lower energy, exists if the attraction exceeds a critical value, $G_S < G_c$. Expanding the integral for small $m_{\rm vac}$, with the area $A_D = D\,V_D$ of the unit sphere,

$$G_c = -\frac{(2\pi)^D\,(D-1)}{\nu\,A_D\,\Lambda^{D-1}},$$
$$G_c\,\Lambda^{D-1} = -90\pi^4,\ -3\pi^2,\ -\frac{\pi^2}{2} \quad \text{for } D = 7,\ 4,\ 3. \tag{82}$$

A field without a mass term acquires a mass by its own self-interaction when the scalar attraction exceeds G_c. This is the mechanism of Nambu and Jona-Lasinio [26]; the minimization of the effective potential gives no mass two per cent below G_c and a mass two per cent above it in all three cases.

On the deflating spacetime the critical coupling is crossed at a definite moment. In the sector of fields that do not depend on $x5, x6, x7$ the coupling grows as e^{τ_d} while the cutoff, a momentum along the flat coordinates, does not change. A field that is massless at early times therefore acquires a mass at the clock reading $\tau_d = \ln[G_c/G_S(x4_{\rm ref})]$, and from then on the mass grows [Fig. 17(a), Table XIV]. With a small bare mass the transition is smooth. The mass reaches the cutoff about half a unit of the clock after the critical coupling is crossed; beyond that point the effective theory says nothing.

The reference example with the filled states. The coupling $G_S = -10^5\,m^{-6}$ of the reference example of Sec. VII B exceeds the critical coupling $90\pi^4/\Lambda^6$ as soon as the cutoff is larger than $\Lambda = 0.6665\,m$: *the calculation without the filled states presupposes a cutoff below this value*, which is smaller than the Fermi momentum of the saturated matter. With the filled states included, and with the bare mass adjusted so that the mass in vacuum stays the same, the saturation point moves little for the two smaller cutoffs of Table XV, and the binding grows for the largest [Fig. 17(b)]. The example is therefore meaningful only as an effective theory with a low cutoff, and its numbers depend on the cutoff at the level that the table shows.

TABLE XV. The reference example ($D = 7$, $G_S = -10^5$, $G_V = +10^4$, units $m_{\rm vac} = 1$) with the filled states of negative energy below the cutoff Λ: the bare mass that keeps the mass in vacuum, and the saturation point (Fermi momentum, mass of the particles in matter, energy per particle relative to the vacuum).

Λ	$m_{\rm bare}$	k_F	m^*	$\varepsilon/n - m_{\rm vac}$
none	1	0.7566	0.3481	−0.02832
0.3	0.9979	0.7566	0.3479	−0.02837
0.45	0.9660	0.7568	0.3438	−0.02984
0.6	0.7578	0.7565	0.3072	−0.04483

C. Bare mass, self-interaction and Vorton-Scalar together

Now let all terms of the model (44) act together, for fields in ordinary space ($D = 3$) and without the filled states. Let ϕ be the expectation value of the Vorton-Scalar, $\langle\Phi_{\rm VS}\rangle = \phi/\sqrt{2}$, and $x = \phi/v_{\rm VS}$. The mass of a fermion in a region with the expectation value ϕ is $m_D + m_Y\,x$; let $f_D = m_D/m$ be the fraction of the mass in vacuum that is a bare Dirac mass. In units $m = 1$ the energy density of matter of the Fermi momentum k_F is the stationary value of

$$\varepsilon = \frac{(1-f_D)^2\,(x^2-1)^2}{8\,C} + \langle E^*\rangle + \tfrac{1}{2}G_V\,n^2 - \frac{\left[m^* - f_D - (1-f_D)\,x\right]^2}{2\,G_S} \tag{83}$$

with respect to x and m^*: the first term is the potential of the scalar, with the number C of Eq. (49); the others are Eq. (56) with the mass $f_D + (1-f_D)x$ in place of m. For $f_D = 0$ and $G_S = G_V = 0$ this is the functional of Sec. XII, and the program reproduces its masses and energies to 2.1×10^{-8} and its binding threshold $C = 3\pi^2/8$.

At low density the Vorton-Scalar acts as a scalar contact coupling. Expanding Eq. (83) for small densities gives $m^* = m + (G_S - C/m^2)\,n_S$: the exchange of the radial mode adds $-g_{\rm VS}^2/m_s^2$ to the scalar coupling, whatever fraction of the mass is a bare mass (verified for five sets of parameters to 1.8×10^{-4} at Fermi momenta up to a twentieth of m). The scalar channel of the self-interaction and the exchange of the radial mode are the same interaction at momenta far below m_s; they differ at higher densities, where the scalar field itself changes.

The moment at which matter becomes self-bound. All three couplings C, G_S, G_V grow as e^{τ_d}, so that their ratios are fixed and the deflation carries matter through a sequence of states with growing C. Table XVI gives the smallest C at which matter is self-bound, and the shift of that moment relative to the model with the Vorton-Scalar alone [Fig. 18(a,b)]. *A bare Dirac mass makes matter bind earlier*, by 0.39 units of the clock if a quarter of the mass is a bare mass and by 0.58 if half of it is: the

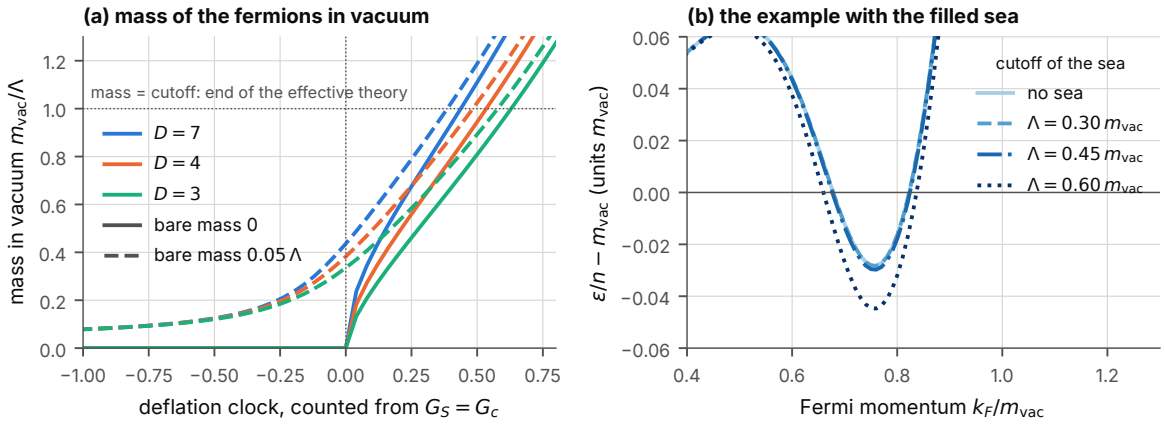


FIG. 17. The filled states of negative energy. (a) The mass of the fermions in vacuum, in units of the cutoff, along the deflation clock counted from the moment at which the scalar attraction equals the critical coupling, for $D = 7$, 4 and 3 dimensions, without bare mass and with the bare mass $0.05\,\Lambda$. (b) Energy per particle of matter relative to the vacuum, for the reference example of Sec. VII B ($D = 7$, $G_S = -10^5$, $G_V = +10^4$ in units of the mass in vacuum) without the filled states and with them for three cutoffs; the bare mass is chosen so that the mass in vacuum is the same.

matter no longer has to pay for the destruction of the whole expectation value of the scalar. *A vector repulsion delays the binding*, by 0.10 and 0.30 units for the two vector couplings of the table, without bare mass. A scalar contact attraction acts only where the particles keep a mass: without a bare mass the matter of lowest energy at the threshold consists of massless particles, whose scalar density vanishes, and the scalar coupling of the second row of the table changes nothing.

D. The mass of the phase mode

A bare Dirac mass breaks the chiral symmetry $U(1)_{\rm VS}$ explicitly, and the phase mode of the Vorton-Scalar is then no longer massless. Write $\Phi_{\rm VS} = (v_{\rm VS}/\sqrt{2})\, e^{i\vartheta_{\rm VS}}$ with a constant phase $\vartheta_{\rm VS}$. The mass term of the fermions becomes $m_D + m_Y\, e^{i\vartheta_{\rm VS}\chi}$, the mass of a particle

$$m(\vartheta_{\rm VS}) = \sqrt{m_D^2 + m_Y^2 + 2\, m_D m_Y \cos\vartheta_{\rm VS}}, \qquad (84)$$

largest at $\vartheta_{\rm VS} = 0$. The energy of the filled states of negative energy is lowest where the mass is largest, so that the vacuum selects $\vartheta_{\rm VS} = 0$, and the second derivative of the energy density with respect to the phase is $m_\vartheta^2 v_{\rm VS}^2$. With $dm/d\vartheta_{\rm VS} = 0$ and $d^2m/d\vartheta_{\rm VS}^2 = -m_D m_Y/m$ at $\vartheta_{\rm VS} = 0$, and with the derivative of the energy density with respect to the mass, which is the scalar density,

$$m_\vartheta^2\, v_{\rm VS}^2 = \Big(\big|\langle\bar\psi\psi\rangle_{\rm vac}\big| - n_S\Big)\,\frac{m_D\, m_Y}{m}. \qquad (85)$$

In vacuum ($n_S = 0$) this is the relation of Gell-Mann, Oakes and Renner [27] for the model at hand: the square of the mass of the phase mode is proportional to the bare mass and to the scalar density of the vacuum. In ordinary space $|\langle\bar\psi\psi\rangle_{\rm vac}| = (\nu m/4\pi^2)\,[\Lambda\sqrt{\Lambda^2+m^2} - m^2\,\mathrm{arsinh}(\Lambda/m)]$; the program confirms Eq. (85) against the numerical second derivative of the energy in 24 cases to 3.2×10^{-7} of the value in vacuum. Three consequences follow.

The phase mode becomes heavier along the clock. In the couplings of the model, $m_\vartheta^2 = C\, m_s^2\, f_D\,(|\langle\bar\psi\psi\rangle_{\rm vac}| - n_S)/[(1-f_D)\, m^3]$, which grows as e^{τ_d}: $m_\vartheta \propto e^{\tau_d/2}$, while all other masses stay fixed. Its value depends on the cutoff through the scalar density of the vacuum (Table XVII).

In matter the phase mode is lighter. The matter subtracts its scalar density: $m_\vartheta^2(k_F)/m_\vartheta^2(0) = 1 - n_S/|\langle\bar\psi\psi\rangle_{\rm vac}|$ [Table XVII, Fig. 18(c)].

A massive phase mode binds pairs less. The exchange of the phase mode contributes to the pairing potential (68) with the range $1/m_\vartheta$; a mass of three tenths of the fermion mass reduces the gap of the example of Table VII ($m_s = m$) from 0.0246 to 0.0197 of the Fermi kinetic energy.

XIV. ORDER PARAMETER, HIDDEN SYMMETRY AND VORTICES

A. Frame components and coordinate components

The order parameter of the superfluid is the expectation value Δ^a of the vector pair operator (61) along the extra axis $a \in \{0,5,6,7\}$: four complex numbers. On the deflating spacetime the index a is a *frame* index: Δ^a is the component along the unit vector e_a. The coordinate components along the deflating coordinates are $\Delta_t = b_t(x4)\,\Delta^t$, and they shrink with the scale factors when the frame components are constant. All statements below are about frame components.

The energy of the paired state depends on the order

TABLE XVI. Bare Dirac mass, contact couplings and Vorton-Scalar together ($D = 3$, without the filled states): the smallest C at which uniform matter is self-bound, for three vector couplings, and in parentheses the shift of the moment of binding along the deflation clock relative to the first entry (negative: earlier).

f_D	$-G_S/C$	$G_V = 0$	$G_V = 0.1\,C$	$G_V = 0.3\,C$
0	0	3.7011 (0)	4.0907 (+0.100)	5.0054 (+0.302)
0	0.5	3.7011 (0)	4.0907 (+0.100)	5.0054 (+0.302)
0	1	3.4470 (−0.071)	3.7717 (+0.019)	4.5419 (+0.205)
0.25	0	2.5112 (−0.388)	2.6869 (−0.320)	3.0810 (−0.183)
0.25	0.5	2.4285 (−0.421)	2.5931 (−0.356)	2.9598 (−0.223)
0.25	1	2.3725 (−0.445)	2.5295 (−0.381)	2.8782 (−0.251)
0.5	0	2.0643 (−0.584)	2.1763 (−0.531)	2.4244 (−0.423)
0.5	0.5	1.8878 (−0.673)	1.9829 (−0.624)	2.1914 (−0.524)
0.5	1	1.7587 (−0.744)	1.8421 (−0.698)	2.0234 (−0.604)

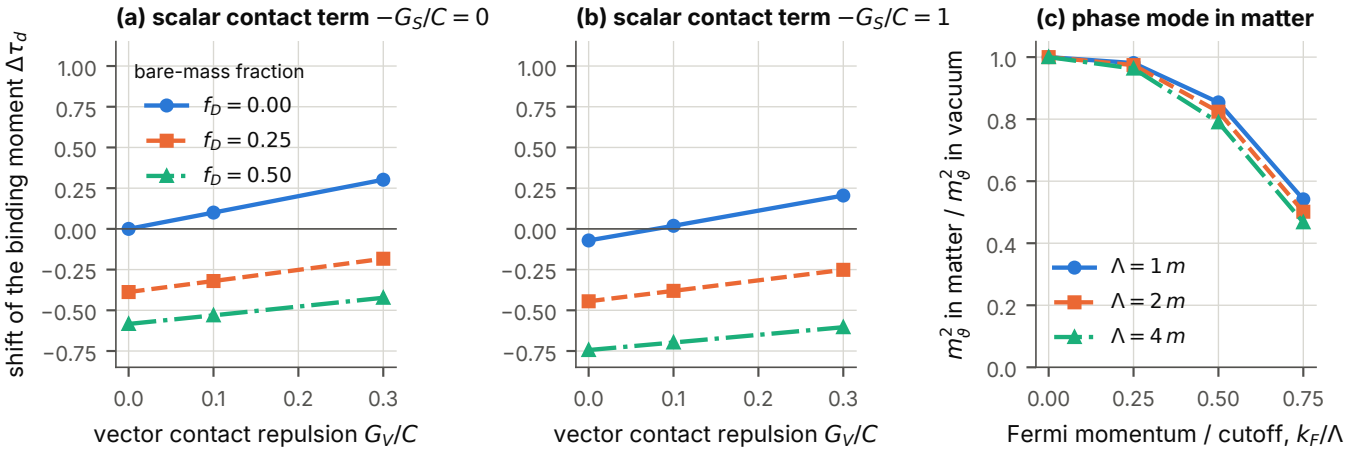


FIG. 18. Bare Dirac mass, contact couplings and Vorton-Scalar together. (a), (b) The shift of the moment at which matter becomes self-bound, in units of the deflation clock, against the vector contact coupling, for three fractions f_D of bare mass, without and with a scalar contact coupling. (c) The square of the mass of the phase mode in matter divided by its value in vacuum, against the Fermi momentum in units of the cutoff.

TABLE XVII. The phase mode of the Vorton-Scalar when a tenth of the fermion mass is a bare Dirac mass ($f_D = 0.1$; $D = 3$; units $m = 1$): scalar density of the vacuum, square of the mass of the phase mode in vacuum in units of $C\,m_s^2$, and its reduction in matter of the Fermi momenta $\Lambda/4$, $\Lambda/2$ and $3\Lambda/4$.

Λ	$\lvert\langle\bar\psi\psi\rangle_{\rm vac}\rvert$	$m_\vartheta^2/(C\,m_s^2)$	$k_F = \Lambda/4$	$\Lambda/2$	$3\Lambda/4$
1	0.1080	0.0120	0.981	0.854	0.541
2	0.6137	0.0682	0.974	0.824	0.502
4	2.9176	0.3242	0.963	0.790	0.467

parameter through the 8×8 gap matrix M on the internal states of a pair. The identity

$$\tfrac{1}{8}\,\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big] = 2\Big(\sum_a |\Delta^a|^2\Big)^2 - \Big|\sum_{a,b}\eta_{ab}\,\Delta^a\Delta^b\Big|^2, \quad (86)$$

holds on the deflating spacetime with the frame metric η_{ab} of the four axes, to 2×10^{-14}. In coordinate components the same expression contains the metric g_{tt} of the moment.

B. The ground state: no direction is preferred

Table XVIII gives the potential of the paired state for six states of the order parameter. Every *unitary* state, in which Δ^a is a real unit vector times a phase, has all eight gaps equal and the lowest potential, whichever direction the vector has. It has four flat directions, the Goldstone modes (one phase and three rotations of the direction), and four directions of positive curvature. The *half-paired* states, in which four of the eight gaps vanish, have half the condensation energy.

One might have expected the deflation to select the direction of the order parameter: the hidden axis $x0$, which does not deflate, against the three axes that do. *In the sector of fields that do not depend on the extra coordinates it does not.* The reason is the central observation of Sec. III: in this sector the weighted field obeys the flat equations, and their internal symmetry rotates the four frame axes into one another. The symmetry of the manifold of ground states is larger than the symmetry of the spacetime.

TABLE XVIII. The potential V of the paired state for six states of the order parameter, in units of $N(0)\Delta_F^2$, where Δ_F is the gap of a unitary state. Each state is taken at the magnitude of the order parameter at which its potential is stationary. d_-^2 and d_+^2: the squares of the two fourfold gaps of the state, in units of Δ_F^2. Flat and positive: the numbers of eigenvalues of the matrix of second derivatives of the potential that vanish and that are positive.

state	d_-^2	d_+^2	V	flat	positive
pair along the frame axis of $x0$ (hidden-axis singlet)	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x5$	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x7$	1.000	1.000	−2.0000	4	4
unitary, rotated between $x0$ and $x5$	1.000	1.000	−2.0000	4	4
unitary, general direction and phase	1.000	1.000	−2.0000	4	4
half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$	0.000	1.000	−1.0000	—	—

C. The internal symmetry of the sector, and the symmetries of the spacetime

For fields that depend on $x1, x2, x3, x4$ only, the equation of the weighted field contains five matrices: $\alpha^1, \alpha^2, \alpha^3$, β and $i\beta\chi$, the last through the coupling of the phase mode. The matrices that commute with all five generate the *internal symmetry* of the sector. The program computes it again for the deflating spacetime, item by item, and finds it unchanged:

- the algebra is $su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$, of dimension 8;
- the eight internal states of a fermion form the two doublets $(2,1) + (1,2)$;
- the four vector pair operators (61) form the $(2,2)$, with fermion number 2, and their one invariant quadratic form is the frame metric of the four axes, with the signs $(+,+,+,+)$ up to an overall sign;
- a condensate along one axis breaks $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$ to $SO(3)$.

Here $U(1)_E$ is a second phase symmetry of the sector, with a generator P that has the square one and distinguishes the doublets of $SU(2)_A$ from those of $SU(2)_B$.

Six of the generators are the products $\gamma^A\gamma^B$ of the six pairs taken from the four axes $x0, x5, x6, x7$. On the flat 7+1 spacetime they are the rotations of the four extra space-like coordinates, and the internal symmetry $SO(4)$ is a symmetry of the spacetime. On the deflating spacetime this is no longer so (Fig. 1). Of the six, the spacetime keeps

- none, if the three rates differ;
- one, the rotation of the two coordinates with the same rate, if two rates are equal;
- three, the rotations of $x5, x6, x7$ among themselves, if the three rates are equal.

The three rotations that mix the hidden axis $x0$ with a deflating coordinate are never symmetries of the deflating spacetime: $x0$ keeps its size and the other three shrink. The generators that are lost as symmetries of the spacetime remain exact symmetries of the sector. They are *hidden symmetries*: symmetries of the equations of a sector that are not transformations of the spacetime. On the 4+4 spacetime this is the situation from the start, with or without deflation, because the rotations that mix the hidden coordinate with the additional times are not unitary there.

D. Vortices and their zero modes

Because the vacuum value of the Vorton-Scalar breaks the phase symmetry $U(1)_{\rm VS}$, the field has line-like defects [28]: along a straight line, say the $x3$ axis, its magnitude vanishes, and around the line its phase winds by 2π times an integer. For unit winding, $\Phi_{\rm VS}$ is $v_{\rm VS}/\sqrt{2}$ times a profile that rises from 0 on the line to 1 at a distance of the order of $1/m_s$, times $e^{i\theta_{12}}$ with the angle θ_{12} around the line in the plane of $x1$ and $x2$.

Energy per unit length. It is proportional to the square of the four-dimensional vacuum value and so falls as $e^{-\tau_d}$: in the four-dimensional description vortex lines become lighter as the deflation proceeds, while their core, of the radius $1/m_s$, keeps its size.

Fermions on the line. On the line the fermion mass vanishes, and fermions can be trapped there [29]. In the field of the vortex the Hamiltonian of a fermion is

$$\sum_{j=1}^{3} \alpha^j p_j + m\,(\text{profile})\,\beta\left(\cos\theta_{12} + i\chi\,\sin\theta_{12}\right), \qquad (87)$$

with the momenta p_j along $x1, x2, x3$. A state of zero energy at zero momentum along the line is a constant spinor w_z times a function that decays away from the line. Inserting this form, the Hamiltonian annihilates the state for every angle if and only if

$$\left(i\alpha^1 + \beta\right) w_z = 0, \qquad \left(i\alpha^2 + i\beta\chi\right) w_z = 0. \qquad (88)$$

The five matrices $\alpha^1, \alpha^2, \alpha^3, \beta, i\beta\chi$ are Hermitian, have the square one and anticommute with one another. The two conditions state that w_z is an eigenvector, with the eigenvalue -1, of the two commuting matrices $i\beta\alpha^1$ and $\alpha^2\beta\chi$, and each joint eigenspace of these two matrices has the dimension $16/4 = 4$. *A vortex line of unit winding carries four zero modes of the sixteen-component field.*

For a momentum p_3 along the line the energy is the eigenvalue of $\alpha^3 p_3$. On the four zero modes α^3 equals the matrix

$$P_c = i\,\alpha^1\alpha^2\alpha^3\chi, \qquad (89)$$

the product of the five matrices, which commutes with each of them, is Hermitian and traceless and has the square one; it is, up to its sign, the generator P of $U(1)_E$. Its trace on the four zero modes vanishes. *Two zero modes move along the line in one direction at the speed of light and two in the other*, and the direction of motion is tied to the charge under $U(1)_E$. For winding -1 the two directions are exchanged.

Equations (88) contain only the matrices of the frame and the fermion mass, neither of which changes. The four zero modes therefore exist at every moment of the deflation. The count also holds on flat spacetime. (The familiar count for a four-component field is one zero mode for each unit of winding; the sixteen components are equivalent to four such fields.)

Massless fermions that run along a vortex line carry a current [30]; a closed loop of the line can be held open by that current, and such loops are the *vortons* of the literature on cosmic strings [31]. Because the energy per length of the line falls as $e^{-\tau_d}$ while the fermions on it stay massless, the balance between the tension of a loop and the current on it shifts in favor of the current as the deflation proceeds. This paper establishes the ingredients of that statement; it does not compute the equilibrium radius of a loop.

XV. GRAVITY: THE DEFLATING SPACETIME WITH MATTER AS ITS SOURCE

The hypothesis (1) prescribes the metric. This section asks what the field equations of gravity make of it when the matter of Sec. VII is their source: whether such matter can produce the deflation, and what it does to a deflation that is already under way.

A. Two scale factors and the Einstein-Lovelock equations

Let the three rates be equal, and let the four flat coordinates have a scale factor of their own:

$$ds^2 = -(dx4)^2 + \mathsf{a}(x4)^2 \sum_{j=0}^{3} (dxj)^2 + \mathsf{b}(x4)^2 \sum_{t=5}^{7} (dxt)^2, \tag{90}$$

with the rates $H_\mathsf{a} = \mathsf{a}'/\mathsf{a}$ and $H_\mathsf{b} = \mathsf{b}'/\mathsf{b}$ (a prime is a derivative with respect to $x4$). The deflating spacetime of the hypothesis is $\mathsf{a} = 1$, $\mathsf{b} = \text{const5}\, e^{-\text{g55}\, x4}$, that is, $H_\mathsf{a} = 0$ and $H_\mathsf{b} = -\text{g55}$; it is called the *undisturbed deflation* below.

For a metric of this kind the Riemann tensor is determined by one number for each pair of directions of the frame, the sectional curvature of their plane (Appendix C):

$$K_{4i} = H_i' + H_i^2, \qquad K_{ij} = H_i\, H_j \quad (i \neq j), \tag{91}$$

where i, j run over the seven directions other than the time and H_i is the rate of the direction i. The field equations of second order for the metric in eight dimensions are those of Lovelock [5], with tensors $E^{(k)\mu}{}_\nu$ of the orders $k = 1, 2, 3$ in the curvature. For the metric (90) they are diagonal, and

$$E^{(k)m}{}_m = (-1)^{k+1}\, k!\, 2^{k-1} \sum_{\substack{\text{sets of } 2k \\ \text{directions} \neq m}} \sum_{\substack{\text{divisions} \\ \text{into pairs}}} \prod_{\text{pairs}} K_{\text{pair}}, \tag{92}$$

without summation over m: the sum runs over all sets of $2k$ directions that do not contain m, and over the ways to divide such a set into k pairs (one way for $k = 1$, three for $k = 2$, fifteen for $k = 3$). In this notation and sign $E^{(1)\mu}{}_\nu = -\mathcal{G}^\mu{}_\nu$ is minus the Einstein tensor and $E^{(2)\mu}{}_\nu = \mathcal{H}^\mu{}_\nu$ the Lanczos tensor of Sec. II C. The *Einstein-Lovelock equations* are

$$\sum_{k=1}^{3} \alpha_k\, E^{(k)\mu}{}_\nu - \bar{\Lambda}\, \delta^\mu_\nu = \kappa_8\, T^\mu{}_\nu, \qquad \alpha_1 = 1, \tag{93}$$

with the couplings α_2, α_3 of the two higher terms, the cosmological term $\bar{\Lambda}$, and the gravitational coupling κ_8 of the eight-dimensional theory. The energy-momentum tensor of uniform matter has the components $T^4{}_4 = \varepsilon$ and $T^m{}_m = -p_m$, with the energy density and the pressures of Sec. VII. [In terms of Eq. (9): $\bar{\Lambda} = \Lambda_8$ and $\alpha_2 = -\alpha_{\text{GB}}$.]

Written out for four directions with the rate H_a and three with the rate H_b, the first two orders are

$$\begin{aligned}
E^{(1)4}{}_4 &= 6H_\mathsf{a}^2 + 12H_\mathsf{a}H_\mathsf{b} + 3H_\mathsf{b}^2, \\
E^{(1)\mathsf{a}}{}_\mathsf{a} &= 3H_\mathsf{a}' + 3H_\mathsf{b}' + 6H_\mathsf{a}^2 + 9H_\mathsf{a}H_\mathsf{b} + 6H_\mathsf{b}^2, \\
E^{(1)\mathsf{b}}{}_\mathsf{b} &= 4H_\mathsf{a}' + 2H_\mathsf{b}' + 10H_\mathsf{a}^2 + 8H_\mathsf{a}H_\mathsf{b} + 3H_\mathsf{b}^2,
\end{aligned} \tag{94}$$

$$\begin{aligned}
E^{(2)4}{}_4 = {} & -12H_\mathsf{a}^4 - 144H_\mathsf{a}^3H_\mathsf{b} - 216H_\mathsf{a}^2H_\mathsf{b}^2 \\
& - 48H_\mathsf{a}H_\mathsf{b}^3, \\
E^{(2)\mathsf{a}}{}_\mathsf{a} = {} & -12\big(H_\mathsf{a}^2 + 6H_\mathsf{a}H_\mathsf{b} + 3H_\mathsf{b}^2\big)H_\mathsf{a}' \\
& - 12\big(3H_\mathsf{a}^2 + 6H_\mathsf{a}H_\mathsf{b} + H_\mathsf{b}^2\big)H_\mathsf{b}' \\
& - 12H_\mathsf{a}^4 - 108H_\mathsf{a}^3H_\mathsf{b} - 180H_\mathsf{a}^2H_\mathsf{b}^2 \\
& - 108H_\mathsf{a}H_\mathsf{b}^3 - 12H_\mathsf{b}^4, \\
E^{(2)\mathsf{b}}{}_\mathsf{b} = {} & -16\big(3H_\mathsf{a}^2 + 6H_\mathsf{a}H_\mathsf{b} + H_\mathsf{b}^2\big)H_\mathsf{a}' \\
& - 16\big(3H_\mathsf{a}^2 + 2H_\mathsf{a}H_\mathsf{b}\big)H_\mathsf{b}' - 60H_\mathsf{a}^4 \\
& - 192H_\mathsf{a}^3H_\mathsf{b} - 136H_\mathsf{a}^2H_\mathsf{b}^2 - 32H_\mathsf{a}H_\mathsf{b}^3,
\end{aligned} \tag{95}$$

where the index a (b) stands for any one of the four (three) directions. Every term of the third order contains at least two factors of H_a or of its derivative, for example $E^{(3)4}{}_4 = 1080H_\mathsf{a}^4H_\mathsf{b}^2 + 1440H_\mathsf{a}^3H_\mathsf{b}^3$. The component along the time contains no derivative of a rate: it is a *constraint* on the initial data. The two other components are linear in H_a' and H_b' and are solved for them at every step of an integration. The polynomials (94) and

TABLE XIX. The reference calculation: three space coordinates with the scale factor a and four with the scale factor b, with matter of seven space dimensions as the source (units $m = 1$). EL: Einstein-Lovelock gravity with $\alpha_2 = 20$, $\alpha_3 = 500$.

gravity	quantity	this program	first calculation
Einstein	$H_\mathsf{b}(0)$	0.00914371	0.00914371
Einstein	$\mathsf{a}(20)$	1.58173868	1.58173870
Einstein	$\mathsf{b}(20)$	1.19755856	1.19755860
Einstein	$H_\mathsf{a}(20)$	0.01187800	0.01187800
Einstein	$H_\mathsf{b}(20)$	0.00741911	0.00741911
EL	$H_\mathsf{b}(0)$	0.01498070	0.01498070
EL	$\mathsf{a}(20)$	1.47735148	1.47735150
EL	$\mathsf{b}(20)$	1.27608405	1.27608410
EL	$H_\mathsf{a}(20)$	0.01017591	0.01017591
EL	$H_\mathsf{b}(20)$	0.00848876	0.00848876

(95) were obtained by the program from Eq. (92) and, independently, by computer algebra in two ways: from the Riemann tensor of the metric, and as the Euler-Lagrange equations of the reduced action for the lapse and the two scale factors.

B. A reference calculation

The program was tested on a calculation done before with other tools: gravity with three coordinates of ordinary space (scale factor a) and four further space coordinates (scale factor b), driven by the matter of the reference example of Sec. VII B at its saturation density, with $\kappa_8 = 1000$, $\bar\Lambda = 0$, $H_\mathsf{a}(0) = 0.05$ and $\mathsf{a}(0) = \mathsf{b}(0) = 1$, integrated to $x4 = 20$ in Einstein gravity and in Einstein-Lovelock gravity with $\alpha_2 = 20$, $\alpha_3 = 500$. The constraint gives $H_\mathsf{b}(0)$. The ten numbers of the first calculation are reproduced to 4.8×10^{-8} by the program and by the independent script (Table XIX, Fig. 19); the constraint is preserved along the integration to 1.3×10^{-14}.

C. The undisturbed deflation as a solution

Without matter. For $H_\mathsf{a} = 0$, $H_\mathsf{b} = -\mathrm{g55}$ and constant rates, Eqs. (94) and (95) give $E^{(1)4}{}_4 = 3\mathrm{g55}^2$, $E^{(1)\mathsf{a}}{}_\mathsf{a} = 6\mathrm{g55}^2$, $E^{(1)\mathsf{b}}{}_\mathsf{b} = 3\mathrm{g55}^2$, $E^{(2)\mathsf{a}}{}_\mathsf{a} = -12\mathrm{g55}^4$, and zero for the other components. The three equations (93) without matter hold if and only if

$$\bar\Lambda = 3\,\mathrm{g55}^2, \qquad \alpha_2 = \frac{1}{4\,\mathrm{g55}^2}, \tag{96}$$

which is Eq. (11); α_3 is arbitrary. The gravity with these two constants is called the *tuned* Einstein-Lovelock gravity below. At this solution the coefficients of $(H'_\mathsf{a}, H'_\mathsf{b})$ in the two evolution equations form the matrix $\mathrm{diag}(-6, 2)$, which can be inverted: the evolution of a perturbed solution is well defined.

In Einstein gravity. Without the Gauss-Bonnet term and without $\bar\Lambda$ the same spacetime requires a source with

$$\kappa_8\,\varepsilon = 3\,\mathrm{g55}^2, \qquad p_\mathsf{a} = -2\,\varepsilon, \qquad p_\mathsf{b} = -\varepsilon, \tag{97}$$

a tension along the flat coordinates twice as large as the energy density, and a tension along the deflating coordinates equal to it. *The matter of Sec. VII cannot be this source.* Its pressure along a coordinate on which the fields do not depend is the interaction pressure p_int of Eq. (57), and $\varepsilon + p_\mathrm{int} = \langle E^* \rangle + G_V\, n^2$, which is positive for a repulsive or vanishing vector coupling; $p_\mathsf{b} = -\varepsilon$ is not reached. Something else must hold the deflation: an *agent* with the equation of state (97), or the tuned Gauss-Bonnet term.

D. Matter on the deflating spacetime

Now let matter be present, with the initial data $\mathsf{a} = \mathsf{b} = 1$, $H_\mathsf{a} = 0$, and H_b from the constraint. Its amount is given by the *loading*, $\kappa_8\,\varepsilon(0)/(3\,\mathrm{g55}^2)$, the energy density at the start in units of the energy density that would make three coordinates deflate at the rate g55 in Einstein gravity. The rate is $\mathrm{g55} = 0.01\,m$. Two kinds of matter are compared: the saturated matter of the reference example with momenta along all seven space coordinates (Sec. VII C), and matter whose fields do not depend on $x5, x6, x7$ ($D = 4$; $G_S = -256.95\,m^{-3}$, $G_V = 0.1\,|G_S|$, at its saturation density), whose pressure along the deflating coordinates is p_int. The integration runs until the clock of the undisturbed deflation, $\tau_d^0 = 3\,\mathrm{g55}\,x4$, reads 6, or until a rate exceeds $50\,\mathrm{g55}$ (Table XX, Fig. 20).

Matter makes the deflation faster than exponential and makes the flat coordinates expand. In the tuned gravity the smaller loading leaves the deflation close to the undisturbed one for about four units of the clock; then the rates grow, and at $\tau_d^0 = 6$ the three coordinates deflate 8.0 times faster than at the start while the four flat coordinates expand. With the larger loading the rates exceed $50\,\mathrm{g55}$ before the clock reads 4: they grow without bound in a finite time. In Einstein gravity with the matter as the only source the initial rate is $-H_\mathsf{b}(0) = \mathrm{g55} \times (\mathrm{loading})^{1/2}$, far below g55 for a small loading, and it grows as the matter is compressed. In four of the eight evolutions of the table the rates run away. The constraint is preserved along every evolution to 6.4×10^{-10}.

E. The undisturbed deflation is unstable

The behavior just described is not caused by the matter alone. Near the undisturbed deflation, with $H_\mathsf{b} = -\mathrm{g55} + \delta H_\mathsf{b}$ and to the lowest order in δH_b, H_a and the energy density, the constraint of the tuned gravity reads

$$-6\,\mathrm{g55}\,\delta H_\mathsf{b} - 48\,H_\mathsf{a}^2 = \kappa_8\,\varepsilon. \tag{98}$$

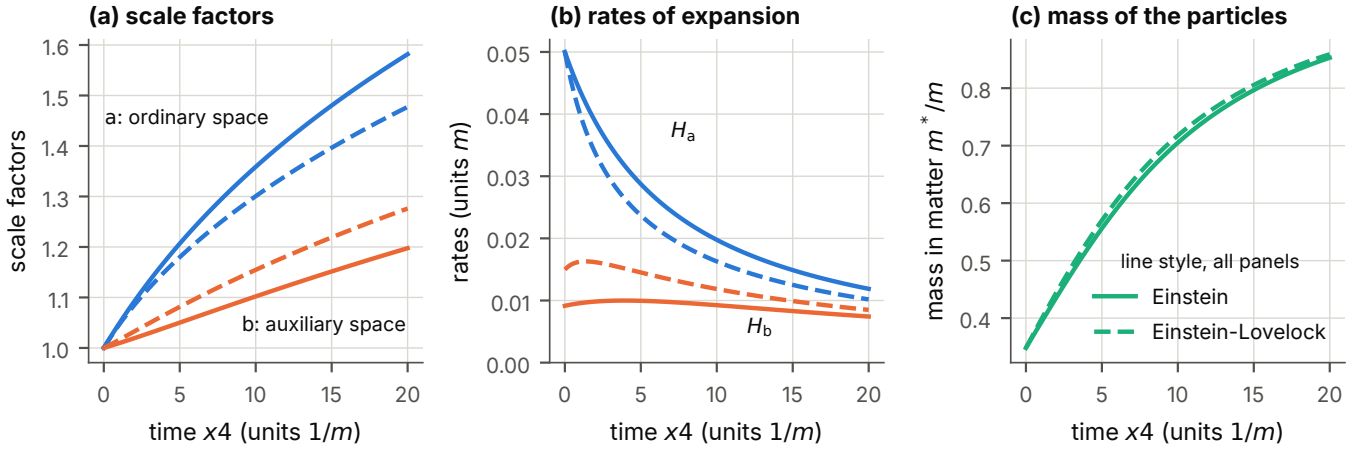


FIG. 19. The reference calculation of Table XIX: (a) the two scale factors, (b) their rates, (c) the mass of the particles in the matter, which rises as the matter is diluted. Solid lines: Einstein gravity; dashed lines: Einstein-Lovelock gravity.

TABLE XX. The deflating spacetime with matter as a source of gravity. Tuned EL: Einstein-Lovelock gravity with the constants (96), in which the undisturbed deflation is a solution without matter; Einstein: Einstein gravity with the matter as the only source. The last five columns give the state at the end of the integration; "runaway" marks an evolution in which a rate exceeded 50 g55 before $\tau_d^0 = 6$.

matter	gravity	loading	$H_b(0)/g55$	end	τ_d^0	a	b	$H_a/g55$	$H_b/g55$
seven momenta	tuned EL	10^{-3}	-1.0005	reached	6.00	1.2508	0.04908	$+1.4052$	-7.991
seven momenta	tuned EL	10^{-1}	-1.0488	runaway	3.58	1.5647	0.02320	$+8.9453$	-50.035
seven momenta	Einstein	10^{-3}	-0.0316	reached	6.00	1.0011	0.9335	$+0.0012$	-0.037
seven momenta	Einstein	10^{-1}	-0.3162	runaway	4.17	1.5278	0.09199	$+11.2603$	-50.092
sector, $D = 4$	tuned EL	10^{-3}	-1.0005	reached	6.00	1.2333	0.05683	$+1.1823$	-6.616
sector, $D = 4$	tuned EL	10^{-1}	-1.0488	runaway	3.74	1.6348	0.02202	$+9.1382$	-50.073
sector, $D = 4$	Einstein	10^{-3}	-0.0316	reached	6.00	1.0016	0.9331	$+0.0017$	-0.038
sector, $D = 4$	Einstein	10^{-1}	-0.3162	runaway	4.05	1.7118	0.08801	$+13.1968$	-50.278

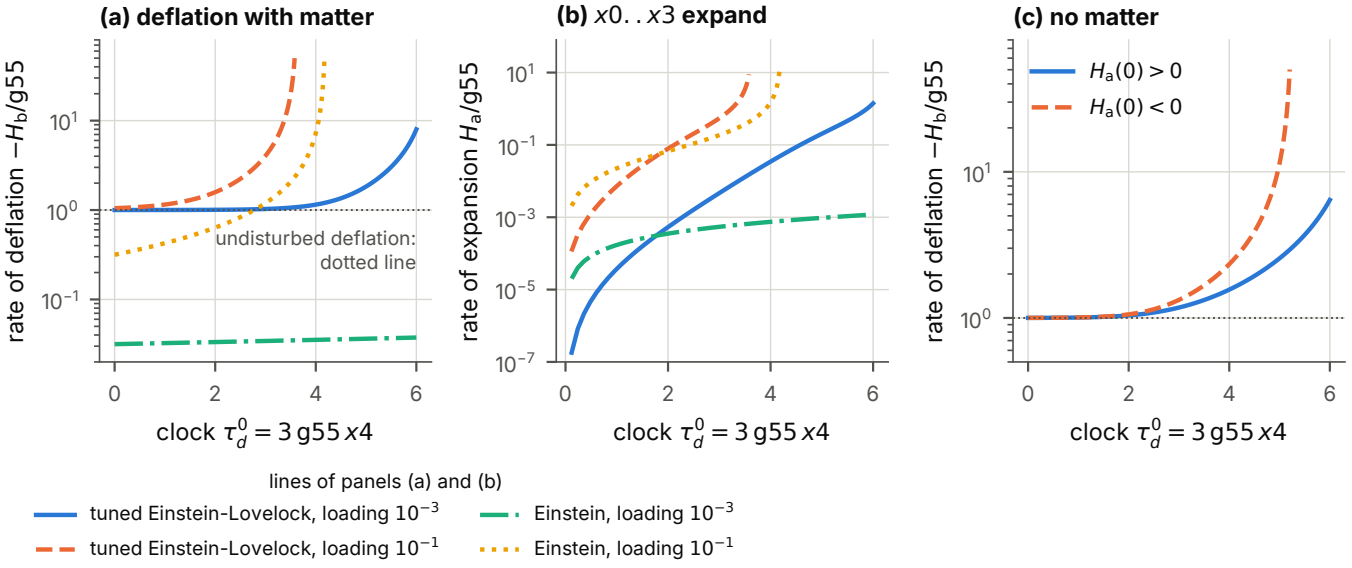


FIG. 20. The deflating spacetime with matter as a source of gravity (matter with momenta along all seven space coordinates; the runs of Table XX). (a) The rate of deflation $-H_b$ in units of g55 against the clock of the undisturbed deflation. (b) The rate H_a at which the four flat coordinates expand. (c) The tuned Einstein-Lovelock gravity without matter, started with a deflation one part in a thousand faster than the undisturbed one: the two initial data that the constraint allows.

TABLE XXI. The tuned Einstein-Lovelock gravity without matter, started at $H_\mathsf{b}(0) = -1.001\,\mathrm{g55}$ with the two values of $H_\mathsf{a}(0)$ that the constraint allows: the state at the end of the integration (at $\tau_d^0 = 6$, or where a rate exceeded $50\,\mathrm{g55}$).

$H_\mathsf{a}(0)/\mathrm{g55}$	τ_d^0 at the end	$H_\mathsf{a}/\mathrm{g55}$	$H_\mathsf{b}/\mathrm{g55}$
$+0.01147$	6.00	$+1.7334$	-6.422
-0.01088	5.20	-0.0050	-50.155

The terms linear in H_a cancel between the Einstein and the Gauss-Bonnet part, which is why H_a enters quadratically. Since $\varepsilon \geq 0$, the left-hand side must not be negative: *initial data exist only for a deflation that is at least as fast as the undisturbed one*, $\delta H_\mathsf{b} \leq 0$. For a deflation exactly at the rate g55 and without matter the constraint is $H_\mathsf{a}^2\,(H_\mathsf{a}^2 - 12\,\mathrm{g55}H_\mathsf{a} + 16\,\mathrm{g55}^2) = 0$, with the roots 0, 0 and $(6 \pm \sqrt{20})\,\mathrm{g55}$: the undisturbed deflation is a double root, on the edge of the allowed initial data.

For a deflation faster by one part in a thousand and no matter, the constraint has the two roots $H_\mathsf{a}(0) = \pm\mathrm{g55}\,(0.001/8)^{1/2}$ to the lowest order. Starting from either, the deflation does not return to the rate g55: it becomes faster and faster [Table XXI, Fig. 20(c)]. *The exactly exponential deflation is an unstable solution of the gravity in which it is a solution without matter*; every allowed perturbation, with or without matter, grows.

Three conclusions follow for the hypothesis. First, the matter of the model does not produce the deflation; an agent with the equation of state (97), or a tuned Gauss-Bonnet term, is needed. Second, with the Gauss-Bonnet term the deflation is not stable, so that the Gauss-Bonnet term alone does not explain a deflation that lasts. Third, whatever holds the deflation must also absorb the reaction of the matter, which pushes toward a faster deflation and an expansion of the flat coordinates; the sections on cosmology assume such an agent and say so (assumption A2 of Sec. XVI).

XVI. COSMOLOGY

Hypothesis H3 says that the superfluid explains the dark matter, or the Hubble tension, or both. The model offers a candidate for each. For the Hubble tension it is the phase mode of the Vorton-Scalar, a massless boson that would act as radiation in addition to photons and neutrinos (*dark radiation*) and make the early universe expand faster. For the dark matter it is the fermions themselves, as a degenerate gas or as a condensate of bound pairs. This section computes what each candidate gives when $x5, x6, x7$ deflate; the case of static extra coordinates is the case of vanishing rates, gsum = 0, and appears in every table below as the row or column without deflation. The deflation enters cosmology in several ways, and each requires an assumption beyond the hypothesis of deflation. The assumptions are stated first and are kept apart throughout, so that the reader can see which conclusion rests on which.

A. Assumptions, and the equation of the expansion

A1: ordinary space expands. The coordinates $x1, x2, x3$ carry the scale factor $a_\mathrm{sp}(x4)$ of standard cosmology, with the Hubble rate $H = d\ln a_\mathrm{sp}/dx4$; the hidden coordinate $x0$ is static; $x5, x6, x7$ deflate at constant rates. The metric of Sec. II is the case $a_\mathrm{sp} = 1$.

A2: gravity is the Einstein theory in eight dimensions, with a constant G_8. The constant of gravitation that an observer in ordinary space measures is then G_8 divided by the proper volume of the extra coordinates,

$$G_N(x4) \;\propto\; \frac{1}{V_d} \;\propto\; e^{\mathrm{gsum}\,x4}, \qquad \frac{1}{G_N}\frac{dG_N}{dx4} = \mathrm{gsum}. \tag{99}$$

Newton's constant *grows* as the extra volume shrinks: gravity was weaker in the past. That a changing volume of extra dimensions makes the constants of four dimensions change, and that nucleosynthesis bounds the change, was pointed out long ago [32]; for a review of varying constants see Ref. [33]. By Sec. II C, A2 is an assumption that the deflating spacetime itself puts in doubt: without matter the metric solves the equations of Einstein-Gauss-Bonnet gravity, not Einstein's.

For a diagonal metric whose seven directions other than the time have the rates H, H, H (ordinary space), 0 (the coordinate $x0$) and $-\mathrm{g55}, -\mathrm{g66}, -\mathrm{g77}$, the component of Einstein's equations along the time is the sum of the products of all pairs of rates, set equal to $8\pi G_8$ times the energy density in eight dimensions. Dividing the density by the extra volume turns G_8 into G_N:

$$3H^2 - 3\,\mathrm{gsum}\,H + \mathrm{gprod} = 8\pi\,G_N(x4)\,\rho, \tag{100}$$

with the density ρ of radiation, matter and the cosmological term in ordinary space. For the unperturbed universe Eq. (100) is the only one of Einstein's equations that is needed. The others determine the stresses of the unknown source that keeps the three rates constant; they enter when ordinary space is perturbed (Sec. XVI E).

The calculations take three equal rates, so that $\mathrm{gprod} = \mathrm{gsum}^2/3$, and the densities of the standard model of cosmology [34]: baryons $\omega_b = 0.02237$, cold dark matter $\omega_c = 0.1200$, and $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ without deflation. The cosmological term is chosen so that Eq. (100) holds today. Without deflation the program gives the two numbers on which this section rests: the angle θ_* under which the sound horizon at last scattering is seen, $100\,\theta_* = 1.03965$, and that sound horizon, $r_s = 144.449$ Mpc.

B. The Hubble constant inferred from the sound horizon

The microwave background fixes the angle θ_* to high precision. A model that changes the history of the

expansion must keep that angle, and the Hubble constant that it then requires is the *inferred* Hubble constant. The tension is that the standard model infers $67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [34] while the local distance network measures 73.50 ± 0.81 [35]. Table XXII gives the inferred value on the deflating spacetime under A1 and A2, at fixed θ_* and fixed densities of baryons and dark matter. The redshift z_* of last scattering and the mass fraction Y_{He} of helium, which the sound horizon needs, are computed on the deflating background (Sec. XVI G).

The inferred Hubble constant falls, with the slope $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.969$ at small rates. The reason is this. By Eq. (99) gravity was weaker in the early universe; the universe expanded more slowly at a given density; sound had more time to travel, and the sound horizon is larger. To keep the angle, the distance to last scattering must be larger, which requires a smaller Hubble constant. The deflation moves the inferred value *away* from the local measurement [Fig. 21(a)]. Weaker gravity also delays last scattering and lowers the helium fraction, because the universe expanded more slowly during recombination and during nucleosynthesis as well; both enlarge the sound horizon further. If z_* and Y_{He} are held at their values without deflation, the slope is -0.911: that simpler calculation underestimates the fall by six per cent of itself. An independent calculation of recombination confirms both numbers (Appendix D). The term gprod of Eq. (100) plays no part in this: it is of the second order in the rate, and it acts as a constant, which the cosmological term that is adjusted to today's expansion absorbs almost exactly.

C. How fast can the coordinates deflate today?

Table XXIII and Fig. 22 collect the bounds.

Under A2, lunar laser ranging measures $(dG_N/dt)/G_N$ directly, and Eq. (99) turns its limit into $\mathrm{gsum} < 2.23 \times 10^{-13}$ per year, that is, $3.24 \times 10^{-3}\, H_0$. At this rate the deflation clock has advanced by $\mathrm{gsum}\, t_0 = 3.1 \times 10^{-3}$ over the age t_0 of the universe, and the inferred Hubble constant is lower by $0.21\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Under A2, nucleosynthesis bounds the ratio of Newton's constant then to its value today; for a rate that was constant since, $\mathrm{gsum} < 0.064\, H_0$. This is weaker than the first bound but does not assume that the rate today equals the rate in the past.

Under HV, nucleosynthesis bounds the change of the Fermi constant at fixed fermion masses to 10 to 20 per cent, which gives $\mathrm{gsum} < 0.111$ to $0.234\, H_0$. HV is the assumption that the Vorton-Scalar itself plays the role of the Higgs field, so that $G_{\mathrm{Fermi}} = 1/(\sqrt{2}\, v_{\mathrm{VS}}^2)$ grows as e^{τ_d} while every mass stays constant. It is *not* the hypothesis H2, and the Vorton-Scalar, which is a singlet of every non-Abelian symmetry of the model, cannot be the Higgs field of the Standard Model, which is a doublet; the bound is given only to show what the assumption would imply.

Under the hypothesis H2, atomic clocks give by far the strongest bound, and one that needs no assumption about gravity. Section XVII derives it: $\mathrm{gsum} < 1.16 \times 10^{-6}\, H_0/n_{\mathrm{EW}}$ with an exponent n_{EW} of order one or larger.

The fine-structure constant is bounded by the same clocks to parts in 10^{18} per year. This would bound the rate, to $4.6 \times 10^{-8}\, H_0$, only if an electromagnetic field propagated along the deflating coordinates. The model contains no electromagnetic field; the row shows how strongly such an extension would be constrained.

Under A2, the microwave background would bound the rate as well. Section XVI H finds, as a forecast for mock data of the quality of the Planck satellite and not as a measurement, $\mathrm{gsum} < 0.0385\, H_0$; Fig. 22 includes it for comparison.

In the model as it stands, without H2, HV or an electromagnetic extension and without A2, no mass and no coupling of ordinary matter depends on the deflation, and none of these observations bounds the rate.

The consequence for the histories of Secs. VIII to XII is severe. Under A2 the clock has advanced by at most 0.0031 during the whole of cosmic history, if the rate was constant; under H2 by at most $1.1 \times 10^{-6}/n_{\mathrm{EW}}$. A history in which pairs become bound, matter loses its fermion mass or the gap grows by orders of magnitude within the age of the universe requires that gravity does not feel the deflation, that the superfluid is not the Higgs field, or that the rate was much larger in the past than it is today.

D. Bound pairs lose rest mass

If the dark matter consists of bound pairs of the fermions, its particles have the rest mass $2m$ minus the binding energy. The binding energy grows with the coupling, so that the pairs become lighter as the deflation proceeds, and the density of dark matter falls slightly faster than the inverse volume. A dark matter that was heavier in the past raises the inferred Hubble constant. The program computes the effect for four examples with α_{VS} between 0.4 and 1 today, where the binding energy is at most 8.2 per cent of the rest mass [Fig. 21(b)]. If gravity feels the deflation, the inferred Hubble constant changes by between -4.11 and $-0.19\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the rates that nucleosynthesis and lunar laser ranging allow: the weaker gravity of the past outweighs the loss of mass. (These examples hold z_* and Y_{He} fixed, which by Table XXII underestimates a fall by about six per cent of itself.) If Newton's constant is held fixed and the deflation acts on the dark sector alone, rates between 0.03 and $1\, H_0$ raise the inferred Hubble constant by at most $+0.48\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, for a loss of up to 7.7 per cent of the rest mass. This is the only effect of the deflation found here that raises the inferred Hubble constant, and in these examples it supplies less than a tenth of the

TABLE XXII. The Hubble constant inferred from the angular scale of the sound horizon if Newton's constant grows with the deflation (assumptions A1 and A2). G_N ratio: Newton's constant in the early universe divided by its value today, $e^{-\mathrm{gsum}\, t_0}$ with the age t_0 of the model. z_*: redshift of last scattering; Y_{He}: mass fraction of helium. Last column: the Hubble constant that results if z_* and Y_{He} are held at their values without deflation. The rows at $\mathrm{gsum}/H_0 = 0.00324$ and 0.0644 are the bounds of lunar laser ranging and of nucleosynthesis (Table XXIII).

gsum/H_0	gsum (per year)	H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	change	G_N ratio	z_*	Y_{He}	r_s (Mpc)	t_0 (Gyr)	H_0, z_* and Y_{He} held
0	0	67.36	0.00	1.0000	1089.92	0.2454	144.45	13.81	67.36
0.00100	6.89×10^{-14}	67.29	−0.07	0.9990	1089.90	0.2453	144.52	13.82	67.30
0.00300	2.07×10^{-13}	67.16	−0.20	0.9971	1089.86	0.2452	144.66	13.84	67.18
0.00324	2.23×10^{-13}	67.15	−0.21	0.9969	1089.86	0.2451	144.68	13.84	67.16
0.0100	6.89×10^{-13}	66.70	−0.66	0.9905	1089.73	0.2446	145.16	13.90	66.74
0.0300	2.07×10^{-12}	65.36	−2.00	0.9714	1089.37	0.2430	146.61	14.06	65.48
0.0644	4.43×10^{-12}	62.98	−4.38	0.9383	1088.72	0.2401	149.23	14.37	63.23
0.100	6.89×10^{-12}	60.37	−6.99	0.9037	1088.04	0.2370	152.13	14.70	60.77
0.300	2.07×10^{-11}	41.94	−25.42	0.6995	1083.78	0.2168	173.35	17.29	43.41

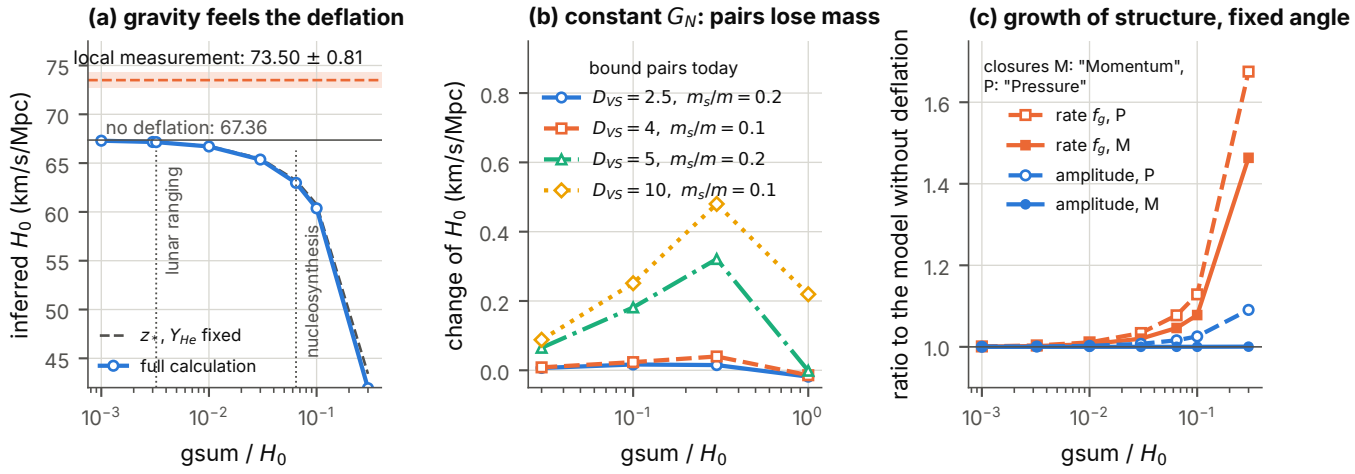


FIG. 21. The Hubble constant and the growth of structure on a deflating spacetime. (a) The Hubble constant inferred from the angular scale of the sound horizon against the rate of deflation, if gravity feels the deflation (A1, A2): with recombination computed on the deflating background, and with last scattering and the helium fraction held fixed; the bounds of Table XXIII and the local measurement are marked. (b) The change of the inferred Hubble constant if Newton's constant is held fixed and bound pairs lose rest mass. (c) Amplitude and rate of the growth of structure relative to the model without deflation, at a fixed angle of the sound horizon, for the two closures of Sec. XVI E.

6.1 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ that separate the model from the local measurement.

E. Perturbations of ordinary space: what must be assumed

The quantities of the preceding subsections belong to the unperturbed universe. The growth of structure and the spectra of the microwave background are built from small perturbations of ordinary space, and the equations of the perturbations contain more of Einstein's equations than Eq. (100). They therefore contain what the hypothesis of deflation leaves open.

The agent. Something must keep the three rates constant while ordinary space expands and matter dilutes. We call it the agent and assume nothing about its nature. Assumption A2 says that it has no energy density: otherwise its density would stand on the right of Eq. (100). It cannot be without stresses. The components of Einstein's equations along ordinary space, together with the conservation of matter in ordinary space and Eq. (99), require the agent to have, in ordinary space, the pressure

$$8\pi G_N\, p_{\mathrm{agent}} = -\frac{\mathrm{gsum}}{H}\frac{dH}{dx4} - 2\,\mathrm{gsum}\,H + 2\,\mathrm{gprod} - \frac{\mathrm{gsum}\,\mathrm{gprod}}{3H}. \tag{101}$$

In an era of matter this is $-\mathrm{gsum}\,H/2$ to the first order in the rates. The agent is a medium without energy density and with a tension, of the relative size gsum/H. When ordinary space is perturbed, such a medium will in general be perturbed too.

A3: the perturbations. We assume (i) that in a uni-

TABLE XXIII. Upper bounds on the present total rate of deflation, each under the assumption named. Uncertainties of the measured values are at 68 per cent unless stated; the bounds are at two standard deviations. The bound in units of H_0 uses $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. A2: gravity is the Einstein theory in eight dimensions. HV: the vacuum value of the Vorton-Scalar sets the electroweak scale. H2: the superfluid state is the Higgs field; the bound is given for $n_{\mathrm{EW}} = 1$ and scales as $1/n_{\mathrm{EW}}$ (Sec. XVII).

observation	measured	assumption	bound on gsum (per year)	in units of H_0
lunar laser ranging [36]	$(dG_N/dt)/G_N = (7.1 \pm 7.6) \times 10^{-14}$ per year	A2	2.230×10^{-13}	3.237×10^{-3}
nucleosynthesis, Newton's constant [37]	G_N then, divided by G_N today: $0.99^{+0.06}_{-0.05}$ (two standard deviations)	A2; constant rate	4.434×10^{-12}	0.0644
nucleosynthesis, Fermi constant [38]	G_{Fermi} at $t = 1$ s within 20 (10) per cent of today's value, at fixed fermion masses	HV; constant rate	1.615×10^{-11} (7.627×10^{-12})	0.234 (0.111)
atomic clocks, mass ratio [39]	$(d/dt)\ln(m_p/m_e) = (-8 \pm 36) \times 10^{-18}$ per year	H2, $n_{\mathrm{EW}} = 1$	8.000×10^{-17}	1.161×10^{-6}
atomic clocks, fine-structure constant [39]	$(d\alpha_{\mathrm{em}}/dt)/\alpha_{\mathrm{em}} = (1.0 \pm 1.1) \times 10^{-18}$ per year	an electromagnetic field along $x5, x6, x7$ (not part of the model)	3.200×10^{-18}	4.645×10^{-8}

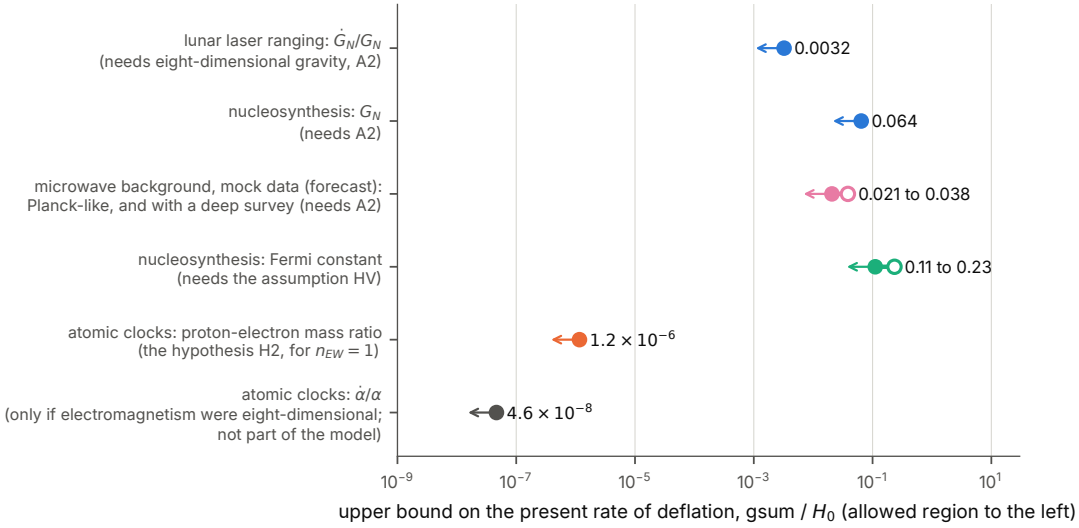


FIG. 22. Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption of Table XXIII. The allowed region is to the left of each bound. The row "microwave background" is not a measurement but the forecast of Sec. XVI H for mock data (open symbol: Planck-like; filled symbol: with a deep survey).

verse with small inhomogeneities the scale factors b_t of the deflating coordinates remain the prescribed functions of the time, the time being the proper time of observers who move with the cold matter, and that $x0$ remains static; and (ii) that the agent has no energy density in the perturbed universe either. Part (i) is the plainest reading of the hypothesis of deflation in a universe that is not exactly homogeneous; part (ii) is A2 taken to first order. Both are assumptions.

The equations. We use the conformal time τ_{cf} of ordinary space, defined by $d\tau_{\mathrm{cf}} = dx4/a_{\mathrm{sp}}$, write a prime for the derivative with respect to it, and write $H_{\mathrm{cf}} = a'_{\mathrm{sp}}/a_{\mathrm{sp}} = a_{\mathrm{sp}}H$ for the conformal Hubble rate. A scalar perturbation of ordinary space with the comoving wavenumber k is described, in the synchronous gauge, by two functions $h_{\mathrm{sy}}(\tau_{\mathrm{cf}})$ and $\eta_{\mathrm{sy}}(\tau_{\mathrm{cf}})$ of the perturbed metric; they are the h and η of Ref. [40], written here with an index because h and η have other meanings in this paper. The combination $\alpha_{\mathrm{sy}} = (h'_{\mathrm{sy}} + 6\,\eta'_{\mathrm{sy}})/2k^2$ occurs repeatedly. Matter and radiation enter through four sums over the species, each multiplied by $8\pi G_N(x4)\,a^2_{\mathrm{sp}}$ with the Newton's constant of the moment: $\mathcal{D}$, the perturbation of the energy density; $\mathcal{T}$, the divergence of the momentum density; $\mathcal{P}$, the perturbation of the pressure; and $\mathcal{A}$, the anisotropic stress. With the scale factors of the extra coordinates unperturbed, the Einstein tensor

of the eight-dimensional metric gives, to first order,

$$2k^2\eta_{\rm sy} - H_e\, h'_{\rm sy} = -\mathcal{D}, \tag{102a}$$
$$2k^2\eta'_{\rm sy} = \mathcal{T}, \tag{102b}$$
$$h''_{\rm sy} + 2H_e\, h'_{\rm sy} - 2k^2\eta_{\rm sy} = -3\,\mathcal{P}, \tag{102c}$$
$$\alpha'_{\rm sy} + 2H_e\,\alpha_{\rm sy} - \eta_{\rm sy} = -\tfrac{3}{2}\,\mathcal{A}/k^2, \tag{102d}$$

with

$$H_e = H_{\rm cf} - \tfrac{1}{2}\,a_{\rm sp}\,{\rm gsum}. \tag{103}$$

They are the energy constraint, the momentum constraint, the trace of the space-space equations and their traceless part, the equation of the shear. An independent program derives them by computer algebra for arbitrary rates and for both signatures (Appendix D). They are the four equations of general relativity [40] with two changes: Newton's constant is that of the moment, and $H_{\rm cf}$ is replaced by H_e in three of them. The second change can be read off Eq. (100). The time component of the Einstein tensor is the sum of the products of all pairs of expansion rates. If the rate of one ordinary direction is perturbed, the perturbation multiplies the sum of the rates of all *other* directions, which is $2H$ in four dimensions and $2H - {\rm gsum}$ here.

The four equations cannot all hold with matter alone. In general relativity two of the four equations (102) imply the other two, because the energy and the momentum of matter are conserved. Here they do not. Let Eq. (102a) hold, and define three *gaps*, the amounts by which the other three fail when only matter and radiation stand on their right-hand sides:

$$\begin{aligned}\mathcal{Y}_M &= 2k^2\eta'_{\rm sy} - \mathcal{T},\\ \mathcal{Y}_T &= h''_{\rm sy} + 2H_e\, h'_{\rm sy} - 2k^2\eta_{\rm sy} + 3\,\mathcal{P},\\ \mathcal{Y}_S &= \alpha'_{\rm sy} + 2H_e\,\alpha_{\rm sy} - \eta_{\rm sy} + \tfrac{3}{2}\,\mathcal{A}/k^2.\end{aligned} \tag{104}$$

The conservation of the energy and of the momentum of matter in ordinary space gives two identities between them,

$$\mathcal{Y}_M = H_e\,\mathcal{Y}_T + \mathcal{Z}, \tag{105}$$
$$2k^2\,\mathcal{Y}_S = \mathcal{Y}_T + \frac{3}{k^2}\big(\mathcal{Y}'_M + 2H_e\,\mathcal{Y}_M\big), \tag{106}$$

with

$$\begin{aligned}\mathcal{Z} &= h'_{\rm sy}\,\mathcal{B} - \tfrac{1}{2}\,a_{\rm sp}\,{\rm gsum}\,(\mathcal{D} - 3\,\mathcal{P}),\\ \mathcal{B} &= H'_e - H_e^2 + \tfrac{1}{2}(\mathcal{E} + \mathcal{E}_P).\end{aligned} \tag{107}$$

Here $\mathcal{E}$ and $\mathcal{E}_P$ are $8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe. By Eq. (100) the quantity $\mathcal{B}$ is proportional to gsum; in an era of matter it is $a_{\rm sp}\,{\rm gsum}\,H_{\rm cf}/4$. Without deflation $\mathcal{B}$ and $\mathcal{Z}$ vanish, and Eqs. (105) and (106) say that if one gap vanishes at all times, all three do: that is general relativity. With deflation $\mathcal{Z} \neq 0$, and *the three gaps cannot all vanish.* This is not a defect of the calculation. It is the statement that the agent, which has a tension, takes part in the perturbations: the gaps are, up to the factor $8\pi G_N a_{\rm sp}^2$, the perturbations of its momentum density, of its pressure and of its anisotropic stress.

Closures. The two identities relate three gaps; one more relation closes the system, and the hypothesis offers no ground for a choice. Three natural choices set one gap to zero. In the closure "Momentum", $\mathcal{Y}_M = 0$: the agent carries no momentum density, and Eqs. (102a) and (102b) hold. In the closure "Pressure", $\mathcal{Y}_T = 0$: the agent carries no perturbation of its pressure, Eqs. (102a) and (102c) hold, and Eq. (102b) is replaced by $2k^2\eta'_{\rm sy} = \mathcal{T} + \mathcal{Z}$. In the closure "Shear", $\mathcal{Y}_S = 0$: the agent carries no anisotropic stress, as a perfect fluid or a scalar field would not. Eliminating $\mathcal{Y}_T$ from the two identities then gives an equation of motion for the momentum density of the agent,

$$\mathcal{Y}'_M = -\Big(\frac{k^2}{3H_e} + 2H_e\Big)\mathcal{Y}_M + \frac{k^2}{3H_e}\,\mathcal{Z}. \tag{108}$$

Far inside the horizon, $k \gg H_{\rm cf}$, the rate $k^2/3H_e$ is enormous, and $\mathcal{Y}_M$ relaxes at once to $\mathcal{Z}$, its value in the closure "Pressure". *Inside the horizon the closure "Shear" is the closure "Pressure"*, with corrections of the relative order $(H_{\rm cf}/k)^2$. We compute with the closures "Momentum" and "Pressure" and mark every result as depending on the closure or not.

All gaps are proportional to $a_{\rm sp}\,{\rm gsum}/H_{\rm cf} = {\rm gsum}/H$. At the largest rate that nucleosynthesis allows this ratio is 2.7×10^{-6} at recombination and 6.9×10^{-2} today. The closure therefore cannot matter for anything that is decided at recombination, and it must matter for the gravitational potentials and for the growth of structure of the recent universe.

F. The growth of structure

For pressureless matter with the density contrast δ_m and the density ρ_m each closure gives one equation. In the closure "Pressure", Eqs. (102a) and (102c) give

$$\frac{d^2\delta_m}{dx4^2} + \Big(2H - \frac{\rm gsum}{2}\Big)\frac{d\delta_m}{dx4} = 4\pi\,G_N(x4)\,\rho_m\,\delta_m\,: \tag{109}$$

gravity is that of the moment, and the Hubble friction is reduced by gsum/2. In the closure "Momentum", the derivative of Eq. (102a) with Eq. (102b) gives, in conformal time,

$$\delta''_m + \frac{2H'_e + \mathcal{E}_m + \frac{4}{3}\mathcal{E}_r}{2H_e}\,\delta'_m = \frac{H_{\rm cf} - a_{\rm sp}\,{\rm gsum}}{2H_e}\,\mathcal{E}_m\,\delta_m, \tag{110}$$

where $\mathcal{E}_m$ and $\mathcal{E}_r$ are the parts of $\mathcal{E}$ from matter and from radiation. Here the source is reduced as well, by the factor $(H - {\rm gsum})/(H - {\rm gsum}/2)$. (The term with $\mathcal{E}_r$ is the momentum density of radiation that streams freely:

far inside the horizon its velocity divergence is $-h'_{\rm sy}/2$ [40].) Without deflation both equations reduce to the familiar $d^2\delta_m/dx4^2 + 2H\, d\delta_m/dx4 = 4\pi G_N \rho_m \delta_m$. With deflation that familiar equation, with $G_N(x4)$ inserted, is neither of the two. For matter alone and three equal rates the growth rate $d\ln\delta_m/d\ln a_{\rm sp}$ is, to the first order in gsum$/H$, $1-\frac{9}{16}$ gsum$/H$ in the closure "Pressure" and $1-\frac{13}{16}$ gsum$/H$ in the closure "Momentum".

Two effects oppose each other [Fig. 21(c)]. At a *fixed* Hubble constant, weaker gravity in the past lowers the amplitude σ_8 of clustering today: $d\ln\sigma_8/d(\text{gsum}/H_0) = -0.443$ in the closure "Momentum" and -0.207 in the closure "Pressure". But the Hubble constant that keeps the angle of the sound horizon is lower, the cosmological term takes over later, and matter clusters for longer. With that Hubble constant the net change is +0.003 and +0.239 per unit of gsum$/H_0$: +0.001 and +0.077 per cent at the bound of lunar laser ranging, and +0.01 and +1.60 per cent at the bound of nucleosynthesis. The growth rate today, $f_g = d\ln\delta_m/d\ln a_{\rm sp}$, rises in both closures, by the factors 1.0021 and 1.0035 at the first bound and 1.0463 and 1.0773 at the second. Common to both closures is that, at a fixed angle, the amplitude does not fall and the growth rate rises. How large the changes are depends on the closure: *the growth of structure on the deflating spacetime is not predicted until the perturbations of the agent are specified.*

G. The spectra of the microwave background

The program. The temperature and polarization spectra of the microwave background and the power spectrum of matter are computed with an Einstein-Boltzmann program written for this work. It integrates the equations of motion of photons, neutrinos, baryons and cold dark matter in the synchronous gauge [40] together with the equations of the metric perturbations, and obtains the spectra by integrating the sources along the line of sight [41]. Its version for static extra coordinates has Newton's constant built into its units. For the deflating spacetime four of its parts were copied and changed in a short, listed set of places: the expansion obeys Eq. (100); every source of the metric perturbations carries the Newton's constant of the moment; the energy constraint is Eq. (102a); and the sources of the temperature anisotropy contain the gap $\mathcal{Y}_S$ of the closure. The equations of motion of photons, neutrinos, baryons and dark matter are not changed: matter that is spread uniformly over the extra coordinates moves in ordinary space as it does in four dimensions, and Thomson scattering and recombination are atomic physics, which the model does not alter. The copies are written by a script from the original files, so that they cannot differ from them in any place that is not listed, and without deflation the two versions give the same spectra to the last binary digit.

Recombination and helium. Recombination is computed on the expansion history of Eq. (100), with the three-level atom and its standard corrections [42]. The helium fraction responds as well. At nucleosynthesis Newton's constant was smaller than today by the factor G_N ratio of Table XXII; a universe that expands more slowly by that factor in H^2 is equivalent, for nucleosynthesis, to one with $\frac{43}{7}(G_N \text{ ratio} - 1)$ additional neutrino species, a negative number, and the helium fraction follows from the standard fit [43]. (In a model with extra radiation, $\frac{43}{7}$ is replaced by $\frac{43}{7} + \Delta N$, where ΔN is the density of that radiation during nucleosynthesis in units of one neutrino species.) At the largest rate that nucleosynthesis allows, and without extra radiation, this is -0.379 species, and $Y_{\rm He}$ falls from 0.24540 to 0.24008.

Spectra at a fixed angle. Table XXIV lists the model without deflation and five models with three equal rates, each with the densities, the primordial amplitude and tilt and the optical depth of the standard model and with the Hubble constant that keeps $100\,\theta_* = 1.03965$. Keeping the angle keeps the acoustic peaks in place: the first five move by at most 0.32 in the multipole ℓ. What changes is their height and the damping tail [Fig. 23], for the reasons that apply to any change of the expansion rate at recombination: the ratio of the diffusion length to the sound horizon, the width of the last-scattering surface and the decay of the potentials shortly after recombination all change, and the lower helium fraction leaves more electrons per baryon.

At the rate that lunar laser ranging allows, the temperature spectrum differs from that of the model without deflation by at most 2.7×10^{-3} of itself, 0.134 of the cosmic variance of a single multipole, and $\Delta\chi^2 = 1.12$ against Planck-like mock data (Sec. XVI H) before any parameter is refitted. Such data cannot tell this rate from no deflation. At the rate that nucleosynthesis allows, Newton's constant at last scattering was 0.9383 of today's, the spectrum changes by up to 5.6 per cent, and $\Delta\chi^2 = 468$: the microwave background is then a stronger test than nucleosynthesis.

The closure. For $30 \le \ell \le 2500$ the spectra of the two closures differ by a thousandth of the effect of the deflation itself (1.4×10^{-3} and 1.2×10^{-3} of it at the two bounds), and $\Delta\chi^2$ is the same to five digits [Fig. 24(a)]. *The spectra at these multipoles follow from A1, A2 and A3, whatever the closure.* For $\ell < 30$, where the potentials of the recent universe act through the late integrated Sachs-Wolfe effect, the closures differ by more than the effect of the deflation in either of them, by a factor 2.1. The largest angular scales are not predicted; their cosmic variance is large, and at the rate of nucleosynthesis the closures differ by at most a third of it. The amplitude of matter is 0.9762 of its value without deflation in the closure "Momentum" and 0.9918 in the closure "Pressure" at the rate of nucleosynthesis, and 0.99892 and 0.99969 at the rate of lunar laser ranging.

Figure 24(b) shows what the agent must supply. In units of $a_{\rm sp}\,\text{gsum}/H_{\rm cf}$ times the terms of each equation, the closure "Momentum" lacks 0.54 to 0.98 in Eq. (102c) and 0.31 to 0.49 in Eq. (102d) after recombination; the

TABLE XXIV. Spectra of the microwave background on the deflating spacetime at a fixed angle of the sound horizon (assumptions A1, A2, A3; three equal rates). H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. $G_N(z_*)/G_N$: Newton's constant at last scattering divided by its value today. σ_8: amplitude of clustering for the same primordial amplitude, in the closures "Momentum" (M) and "Pressure" (P). Largest change: of the temperature spectrum D_ℓ^{TT} for $30 \le \ell \le 2500$ relative to the model without deflation. Ideal measurement: the root of the sum over those multipoles of the squared change divided by the variance $2/(2\ell+1)$ of one multipole, that is, the number of standard deviations by which a measurement of the temperature on the whole sky without noise would tell the model from no deflation if nothing else were unknown. $\Delta\chi^2$: against Planck-like mock data with all other parameters fixed.

gsum/H_0	H_0	$G_N(z_*)/G_N$	σ_8 (M)	σ_8 (P)	largest change	ideal measurement	$\Delta\chi^2$
0	67.360	1.0000	0.8228	0.8228	0	0.0	0.00
0.00324	67.149	0.9969	0.8219	0.8225	2.7×10^{-3}	3.1	1.12
0.0100	66.704	0.9905	0.8200	0.8219	8.3×10^{-3}	9.7	10.79
0.0300	65.365	0.9714	0.8142	0.8201	0.025	29.5	98.85
0.0644	62.976	0.9383	0.8032	0.8160	0.056	64.8	468.30
0.100	60.368	0.9037	0.7903	0.8104	0.090	103.5	1166.70

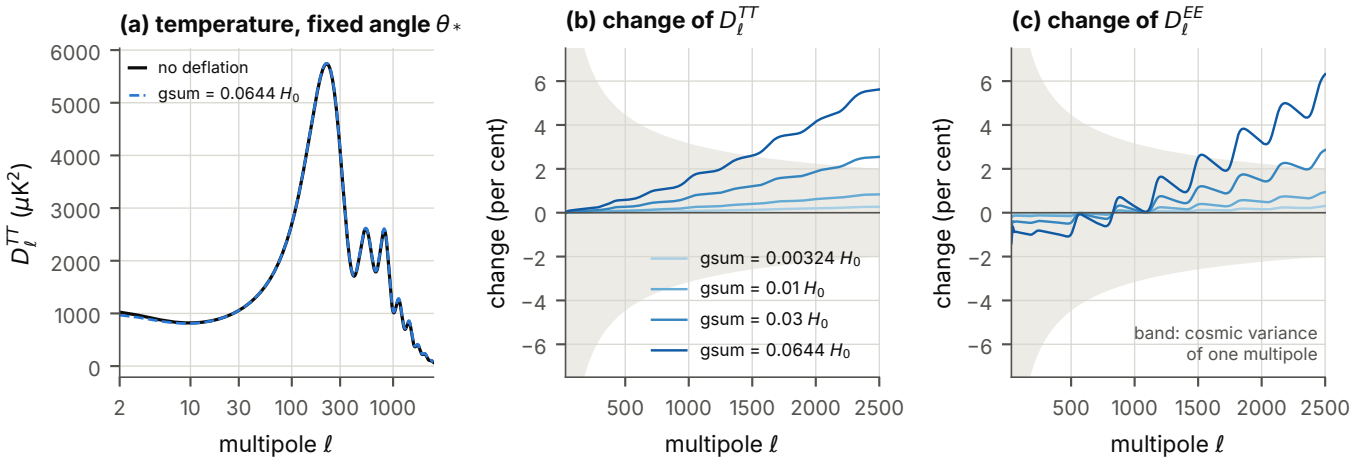


FIG. 23. The spectra of the microwave background at a fixed angle of the sound horizon. (a) The temperature spectrum without deflation and at the largest rate that nucleosynthesis allows. (b), (c) The change of the temperature and polarization spectra relative to the model without deflation for four rates (darker: faster); the band is the cosmic variance of a single multipole. Closure "Momentum"; at these multipoles the closure "Pressure" gives the same curves.

closure "Pressure" lacks 0.62 to 0.94 in Eq. (102b), and in Eq. (102d) an amount that falls as $(H_{\mathrm{cf}}/k)^2$ inside the horizon, as Eq. (108) requires. The two identities (105) and (106) hold along the numerical solutions to 8.6×10^{-4}, the accuracy of the numerical derivative with which they are tested. The complete calculation, with photons, neutrinos and baryons, reproduces the growth of Eqs. (109) and (110) to 1.3×10^{-4} [Fig. 24(c)]; with the equation of the other closure it would disagree by a quarter of gsum/H_0.

H. What mock data say about the rate, and about the Hubble constant

How far can the rate of deflation move the Hubble constant before the fit to the microwave background degrades? The question is answered here with mock data: the spectra of the standard model stand in for the measurements, with the noise and the sky coverage of the Planck satellite [43], and of Planck together with a deep survey of a quarter of the sky. The mock data contain no gravitational lensing, and the standard model fits them with $\Delta\chi^2 = 0$ by construction. The likelihood is the Gaussian likelihood of the temperature and polarization spectra with these noise levels; the six parameters of the standard model (the densities ω_b and ω_c, the Hubble constant, the amplitude A_s and the spectral index n_s of the primordial fluctuations, and the optical depth τ) are fitted again for every rate; the rate is a seventh parameter. Because the rates are positive, the derivative of the spectra with respect to gsum at zero rate is one-sided.

A bound, as a forecast. With the six parameters free, Planck-like data determine gsum/H_0 to 0.0196 (Table XXV), and the upper bound at 95 per cent is

$$\mathrm{gsum} < 0.0385\,H_0 = 2.65 \times 10^{-12}\ \text{per year}, \qquad (111)$$

or $0.0207\,H_0$ with the deep survey. With the six parameters *held* the error would be 0.00108: nearly all of

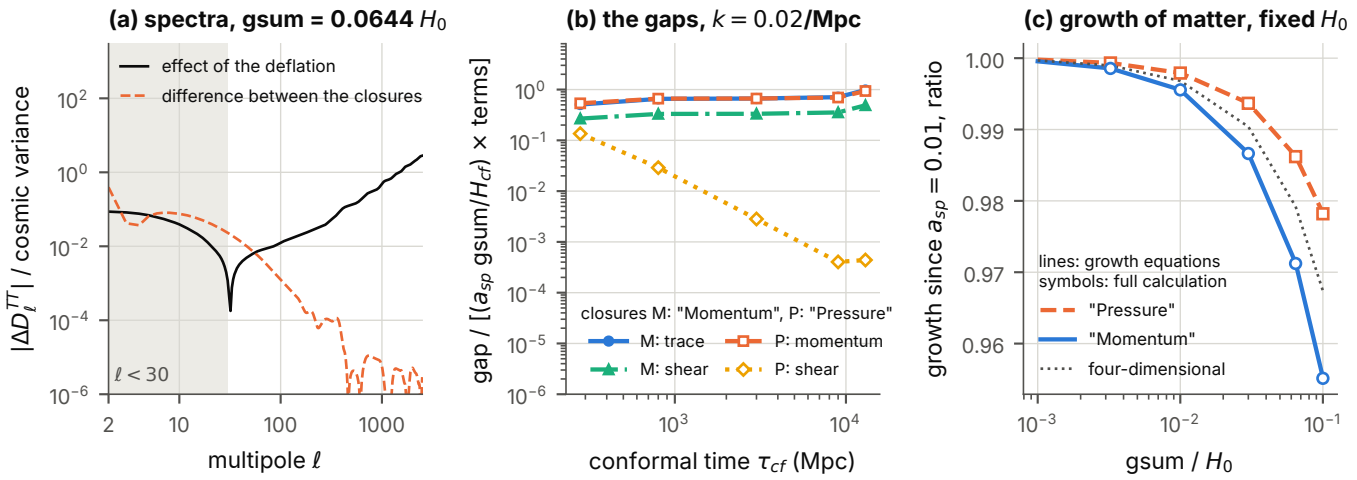


FIG. 24. The closure of the perturbation equations, at the largest rate that nucleosynthesis allows. (a) The difference between the temperature spectra of the closures "Momentum" and "Pressure", and the effect of the deflation itself, in units of the cosmic variance of a single multipole: the closure matters below $\ell = 30$ only. (b) What the agent supplies: the gaps of Eqs. (102b), (102c) and (102d) for one wavenumber, in units of $a_{\rm sp}\,{\rm gsum}/H_{\rm cf}$ times the terms of each equation. (c) The growth of the density contrast of matter since $a_{\rm sp} = 0.01$ relative to no deflation, at a fixed Hubble constant: Eqs. (109) and (110) (lines), the complete calculation (symbols), and the four-dimensional equation with the Newton's constant of the moment, which is neither closure.

TABLE XXV. Forecast for the rate of deflation from mock data of the microwave background (Fisher matrix; three equal rates; assumptions A1, A2, A3). M, P: the closures "Momentum" and "Pressure". $\sigma_{\rm rate}$: standard error of ${\rm gsum}/H_0$ with the six parameters of the standard model free. Bound: the upper bound at 95 per cent, 1.96 standard errors, for a parameter that cannot be negative and whose best value is zero. Held: the standard error if the six parameters are held fixed. Slope: the change of the best-fit Hubble constant per unit of ${\rm gsum}/H_0$, in ${\rm km\,s^{-1}\,Mpc^{-1}}$. Last two columns: the standard error of the Hubble constant in ${\rm km\,s^{-1}\,Mpc^{-1}}$ with the rate held at zero, and with the rate free.

mock data	closure	$\sigma_{\rm rate}$	bound	held	slope	error of H_0	rate free
Planck-like	M	0.01963	0.0385	0.00108	−84.35	0.561	1.748
Planck-like	P	0.01974	0.0387	0.00108	−84.40	0.561	1.758
with deep survey	M	0.01056	0.0207	0.00055	−82.11	0.372	0.943
with deep survey	P	0.01058	0.0207	0.00055	−82.13	0.372	0.945

the effect of the deflation on the spectra, above all the shift of the angle θ_*, can be absorbed by the other parameters [Fig. 25(c)]. The bound lies between those of Table XXIII: it is 11.9 times weaker than that of lunar laser ranging and 1.7 times stronger than that of nucleosynthesis. The closure changes it by half a per cent.

The Hubble constant falls in every fit. Along the degeneracy the best-fit Hubble constant changes by $-84.35\ {\rm km\,s^{-1}\,Mpc^{-1}}$ per unit of ${\rm gsum}/H_0$ for the Planck-like data and by −82.11 with the deep survey, that is, by −3.2 and $-1.7\ {\rm km\,s^{-1}\,Mpc^{-1}}$ at the respective bounds. This is steeper than the slope −65.24 at a fixed angle and fixed densities, because in the fit the density of dark matter rises and the tilt falls as well. Fits of the six parameters at fixed rates confirm the Fisher matrix [Fig. 25(a), (b)]: for rates up to $0.01\,H_0$ the minimum $\Delta\chi^2$ and the shift of the Hubble constant are 1.00 to 1.01 times its predictions, and the fits reach $\Delta\chi^2 = 3.84$ at ${\rm gsum} = 0.0376\,H_0$ and, with the deep survey, at $0.0205\,H_0$. At the rate that lunar laser ranging allows the best fit has $\Delta\chi^2 = 0.027$ and $H_0 = 67.086\ {\rm km\,s^{-1}\,Mpc^{-1}}$. At the rate that nucleosynthesis allows it has $\Delta\chi^2 = 11.6$, a deviation of 3.4 standard deviations, and $H_0 = 61.70\ {\rm km\,s^{-1}\,Mpc^{-1}}$; with the deep survey $\Delta\chi^2 = 39.8$. *The largest rate that nucleosynthesis allows would be excluded by data of the quality of Planck.*

Deflation and dark radiation together. Extra radiation is the candidate of the model for the Hubble tension. Seven one-parameter families of extra radiation were fitted to the same mock data, each with a density N in units of one neutrino species: radiation that streams freely; a fluid; the *stepped fluid of the model*, at four transition redshifts z_t; and the stepped fluid of the Wess-Zumino model [44]. The stepped fluid of the model is the massless phase mode of the Vorton-Scalar in equilibrium with the sixteen components of the fermion field: when the fermions become non-relativistic, at the redshift z_t, they annihilate and heat the phase mode, so that the temperature of the fluid rises by the factor $15^{1/3}$ relative to the

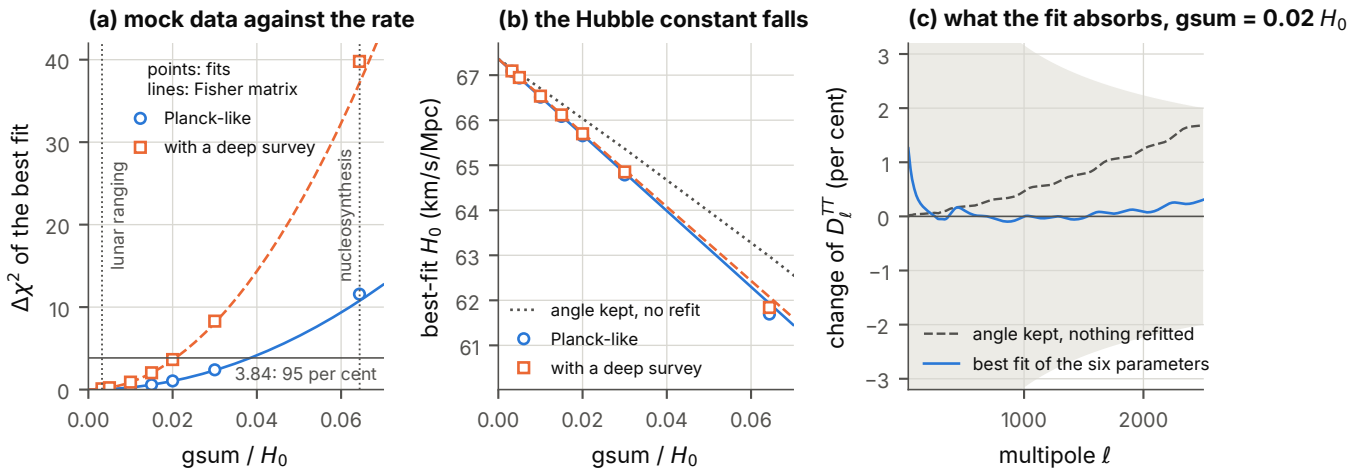


FIG. 25. Mock data of the microwave background against the rate of deflation. (a) The minimum $\Delta\chi^2$ of fits of the six parameters of the standard model at a fixed rate (symbols) and the parabola of the Fisher matrix (lines), for the two data sets; the rates that lunar laser ranging and nucleosynthesis allow are marked. (b) The best-fit Hubble constant, with the slope of the Fisher matrix and, dotted, the Hubble constant that keeps the angle when nothing else is refitted. (c) The temperature spectrum at gsum $= 0.02\,H_0$ relative to the data: with the angle kept and nothing refitted, and at the best fit, which absorbs most of the difference; the band is the cosmic variance of a single multipole.

law of dilution (N is the density after this step; the Wess-Zumino fluid has the factor $(15/7)^{1/3}$). Every family can raise the Hubble constant, and every one at a price in $\Delta\chi^2$. Does the deflation lower the price? All fits were made with static extra coordinates and on the deflating spacetime at the largest rate that lunar laser ranging allows, and the Fisher matrix was computed with N and the rate free together (Table XXVI, Fig. 26). Three results follow.

First, at that rate every best-fit Hubble constant of the static case falls by 0.274 to 0.279 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which is the shift of the model without extra radiation: the effects of the deflation and of the extra radiation add [Fig. 26(a), (b)].

Second, N and the rate are positively correlated, with correlation coefficients of 0.37 to 0.89. Extra radiation makes the early universe expand faster, the weaker gravity of the deflating spacetime makes it expand more slowly, and in the spectra the two compensate in part. Fits with extra radiation are therefore slightly *better* on the deflating spacetime than on the static one, slightly more extra radiation is allowed at a given $\Delta\chi^2$, and with both parameters free the error of each grows, for free-streaming radiation from 0.143 to 0.250 in N.

Third, the compensation does not help the Hubble constant. When the rate grows and N follows its best-fitting value, the best-fit Hubble constant still falls: by 35.3 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum$/H_0$ for free-streaming radiation, against 84.35 without extra radiation, and by 22.1 to 76.9 for the seven families. The largest Hubble constant among the models with $N \geq 0$, gsum ≥ 0 and $\Delta\chi^2 \leq 3.84$ is therefore reached at zero rate, for every family and both data sets [Fig. 26(c)]. *The deflation lowers the Hubble constant faster than it makes room for extra radiation.*

I. Dark matter

What does not change. A superfluid needs particles that are degenerate (fermions) or condensed (bound pairs) at the mass density ρ and the velocity dispersion σ_v of observed halos. A gas of fermions with $\nu = 8$ internal states is degenerate if its Fermi velocity is at least σ_v, and bosons of the mass m_B are condensed if the number of particles in a cube with the edge of their thermal wavelength, at the temperature $k_B T = m_B \sigma_v^2$, is at least 2.612. Both conditions are upper limits on the mass,

$$m_{\mathrm{deg}} = \left[\frac{6\pi^2\hbar^3\rho}{\nu\,\sigma_v^3}\right]^{1/4}, \qquad m_{B,\mathrm{max}} = \left[\frac{(2\pi)^{3/2}\hbar^3\rho}{2.612\,\sigma_v^3}\right]^{1/4}, \tag{112}$$

the first being the bound of Tremaine and Gunn [45] for this model. They contain the densities and the velocities only, and the masses of the model do not change along the deflation (Table XXVII): in the densest systems of dark matter that are observed, the cores of dwarf spheroidal galaxies, fermions are degenerate only if they are lighter than 0.276 keV and bound pairs are condensed only if they are lighter than 262 eV. The first limit conflicts with the bound of the Lyman-α forest derived below, which demands more than 1.5 keV; the second conflicts with the bound on the self-interaction of the pairs, which demands far heavier fermions. *The dark matter of galaxies is not a superfluid of these particles.* Heavy fermions or pairs can be cold dark matter of an ordinary kind. The deflation does not change this verdict.

Self-interaction, and the epoch of the resonance. The

TABLE XXVI. The seven families of dark radiation on the deflating spacetime (Planck-like mock data; assumptions A1, A2, A3). "Model fluid" is the stepped fluid of the model with the transition redshift z_t. Columns 2 to 5: the density N of extra radiation at which the best fit reaches $\Delta\chi^2 = 4$, and the Hubble constant there in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, with static extra coordinates and deflating at the largest rate that lunar laser ranging allows. Columns 6 to 10, from the Fisher matrix with N and the rate free together: the standard error of N with the rate held at zero and with the rate free; the correlation coefficient of N and the rate; the net slope, that is, the change of the best-fit Hubble constant in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum/H_0 when N follows its best-fitting value; and the largest rise of the best-fit Hubble constant, in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, among the models with $N \geq 0$, gsum ≥ 0 and $\Delta\chi^2 \leq 3.84$, which is reached at zero rate for every family.

	N at $\Delta\chi^2 = 4$		H_0 there		error of N			net	largest
family	static	deflating	static	deflating	rate zero	rate free	correlation	slope	rise
free-streaming radiation	0.29	0.31	69.74	69.62	0.143	0.250	0.82	−35.3	2.30
fluid, no step	0.27	0.29	70.20	70.12	0.135	0.249	0.84	−23.9	2.77
model fluid, $z_t = 10^4$	0.25	0.26	68.75	68.53	0.122	0.145	0.54	−65.4	1.35
model fluid, $z_t = 2 \times 10^4$	0.17	0.18	68.16	67.90	0.085	0.091	0.37	−76.9	0.78
model fluid, $z_t = 5 \times 10^4$	0.12	0.13	68.48	68.27	0.060	0.076	0.61	−67.0	1.10
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.32	0.35	70.13	70.06	0.160	0.350	0.89	−22.1	2.69
model fluid, $z_t = 10^6$	0.30	0.32	70.39	70.32	0.150	0.250	0.80	−23.0	2.95

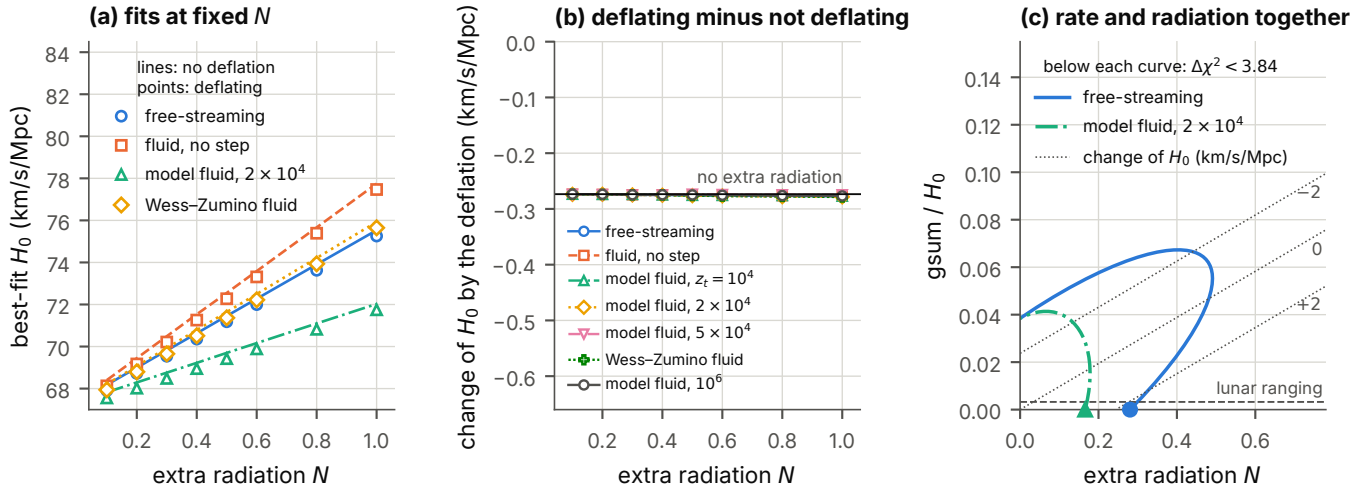


FIG. 26. The seven families of dark radiation on the deflating spacetime (Planck-like mock data). (a) The best-fit Hubble constant against the density N of extra radiation for four families: without deflation (lines) and at the largest rate that lunar laser ranging allows (symbols). (b) The difference for all seven families; the horizontal line is the shift of the model without extra radiation. (c) The rate and the extra radiation as free parameters together: the regions with $\Delta\chi^2 < 3.84$ of the Fisher matrix for two families, lines of constant change of the best-fit Hubble constant for free-streaming radiation (dotted), the point of the largest Hubble constant of each region (filled symbols, at zero rate), and the rate that lunar laser ranging allows (dashed).

TABLE XXVII. Could the fermions of the model be degenerate, or their bound pairs condensed, in observed systems of dark matter? ρ: density; σ_v: velocity dispersion; m_{deg} and $m_{B,\mathrm{max}}$: the limits of Eq. (112).

system	ρ ($M_\odot$ pc^{-3})	σ_v (km/s)	m_{deg} (keV)	$m_{B,\mathrm{max}}$ (eV)
dwarf spheroidal galaxy, core	0.100	10	0.276	262
Milky Way at the Sun	0.0105	156	0.020	19.0
cluster of galaxies, core	0.00300	1000	0.004	3.45

cross-section per mass of bound pairs must respect the bound of about $\kappa_{\mathrm{SI}} = 1$ cm^2/g from colliding clusters [46, 47]. A pair bound by the Yukawa potential (52) has a scattering length of at least the range $1/m_s$; identical bosons with that scattering length scatter at low energy with the cross-section $8\pi/m_s^2$; and the pair exists only if $D_{\mathrm{VS}} = \alpha_{\mathrm{VS}}\, m/m_s > D_c$ (Sec. X). Together, for pairs of the mass $2m$, these give a smallest fermion mass,

$$m \geq \left[\frac{4\pi\, D_c^2}{\alpha_{\mathrm{VS}}^2\, \kappa_{\mathrm{SI}}}\right]^{1/3}, \tag{113}$$

which is 918 MeV for $\alpha_{\mathrm{VS}} = 0.1$ and falls as $\alpha_{\mathrm{VS}}^{-2/3}$, that

is, as $e^{-2\tau_d/3}$ along the deflation. For a given particle the masses are constant and so is this cross-section. What the deflation changes is whether the pair *exists*: it is bound only after $D_{\rm VS}$ has grown to D_c. And at that moment the scattering length of two unbound fermions diverges (Sec. X), so that their cross-section rises to the limit that unitarity allows, which is larger the slower the particles. For the example particle, $m = 918$ MeV and $m_s = 54.7$ MeV, at the relative velocities 10, 156 and 1000 km/s [Fig. 13(c)]: one unit of the clock before the binding the cross-section per mass is 0.84 cm^2/g at all three velocities; at the binding it peaks at 1.3×10^7, 5.2×10^4 and 1.3×10^3 cm^2/g; and it exceeds 100 cm^2/g during an interval of the clock of 0.27, 0.27 and 0.26 around the binding. *A deflating universe whose dark matter consists of these fermions passes through an epoch of strong, velocity-dependent self-interaction*, strongest in the smallest halos. Whether the epoch lies in the observable past depends on the rate. Under A2 and a constant rate it does so only if $D_{\rm VS}$ today exceeds D_c by less than a third of a per cent, a coincidence that nothing in the model explains.

Superfluid dark matter and small-scale structure. What becomes of the growth of structure if the dark matter is the superfluid itself, a fluid with first sound? Two limits are computed: a degenerate gas of the fermions with eight internal states, and a condensate of bound pairs with the sound speed c_{s0} today. The pressure of such a fluid suppresses structure below a wavenumber, and the bound of the Lyman-α forest on warm dark matter [48, 49] then limits the fermion mass from below and the sound speed from above. Computed again on the deflating background under A1 to A3, at the Hubble constant that keeps the angle, the wavenumber at which the power falls to a quarter of that of cold dark matter is lower by 0.15 to 0.16 per cent at the rate that lunar laser ranging allows and by 3.0 to 3.3 per cent at the rate that nucleosynthesis allows, for fifteen models (Fig. 27). The reason is the weaker gravity of the time at which the suppression is set: the Jeans wavenumber is proportional to the square root of Newton's constant. The limits move accordingly, from $m > 1.514$ keV and $c_{s0} < 1.645 \times 10^{-11}\, c$ without deflation to $m > 1.516$ keV and $c_{s0} < 1.641 \times 10^{-11}\, c$ at the first rate and $m > 1.555$ keV and $c_{s0} < 1.565 \times 10^{-11}\, c$ at the second. The two closures give the same wavenumbers to 1.5×10^{-5}. In this calculation the equation of state of the superfluid in ordinary space is that of static extra coordinates; whether the self-interaction of a condensate would itself follow the volume of the deflating coordinates is a question about the microscopic model that the calculation does not answer.

How fast the quantities drift. With $n = d\ln(\text{quantity})/d\tau_d$, the relative change of a quantity over the age of the universe is $n\,\text{gsum}\,t_0$. The couplings themselves drift with $n = 1$ or $\frac{1}{2}$ (Table II). Quantities near a threshold drift much faster: the scattering length just below the binding, at $D_{\rm VS} = 1.5$, with $n = 9.1$; the binding energy just above it, at $D_{\rm VS} = 1.8$, with $n = 30$; the gap of a weakly paired state with $n = 1/\lambda$. These are the quantities in which a slow deflation would first be seen.

J. The quanta that the deflation creates

The temperature (43) of the quanta that the deflation creates in modes that depend on a deflating coordinate is, for three equal rates at the bound of lunar laser ranging, 2.9×10^{-33} K. The number of created particles of mass m per mode is $e^{-2\pi m/\text{rate}_t}$, with an exponent above 10^{37} for every mass above 1 keV at any rate allowed today. *No particle is created today.* Creation by deflation matters only for an epoch in which the rate was comparable with the masses, which no bound of this section addresses.

K. Verdict on hypothesis H3

The Hubble tension. The candidate of the model, the phase mode of the Vorton-Scalar as dark radiation, is untouched at the allowed rates, since the couplings drift by parts in a thousand: with static extra coordinates it raises the Hubble constant by the amounts of the last column of Table XXVI at most, a fraction of the 6.1 km s^{-1} Mpc^{-1} that are needed, and the deflation does not improve this. The deflation itself contributes two effects. The first, through Newton's constant, has the wrong sign: at a fixed angle of the sound horizon the inferred Hubble constant falls by 0.2 km s^{-1} Mpc^{-1} at the largest rate that lunar laser ranging allows, and in fits to mock data of the microwave background it falls by 84 km s^{-1} Mpc^{-1} per unit of gsum/H_0. Nor can the rate help the dark radiation: more extra radiation is allowed when the rate is free, but the largest Hubble constant that the mock data allow is reached at zero rate. The second effect, the loss of mass of bound pairs, has the right sign and is bounded, in the examples computed, to 0.5 km s^{-1} Mpc^{-1}. *The hypothesis that the model resolves the Hubble tension is not supported by the deflation, and under A2 it is mildly disfavored.*

Dark matter. The mass ranges stated above hold unchanged, and the limits on superfluid dark matter move by two tenths of a per cent at the rate that lunar laser ranging allows. The deflation adds one qualitative prediction, the epoch of the resonance, and one quantitative one, a rise of the growth rate, whose size depends on the perturbations of the agent. Neither is in conflict with observation at the allowed rates, and neither is detectable at the rate that lunar laser ranging allows. *The hypothesis that the superfluid is the dark matter is neither strengthened nor weakened by the deflation.* It acquires a history, whose observable part is empty if gravity is eight-dimensional and the rate constant.

What rests on the additional assumptions. Everything in this section that concerns Newton's constant rests on A2. The spectra, the forecast and the growth of struc-

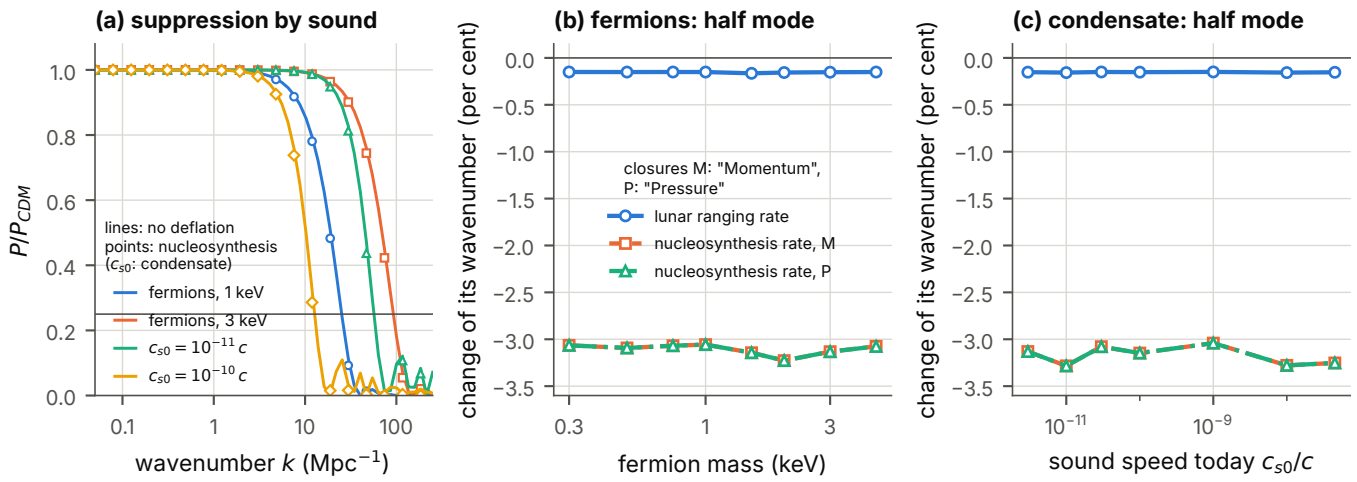


FIG. 27. Superfluid dark matter on the deflating spacetime. (a) The power spectrum of matter divided by that of cold dark matter, for a degenerate gas of the fermions of two masses and for a condensate of bound pairs with two sound speeds today: without deflation (lines) and at the rate that nucleosynthesis allows (symbols). The horizontal line marks one quarter, the value that defines the half mode. (b) The change of the wavenumber of the half mode for the degenerate gas, against the fermion mass, at the rates that lunar laser ranging and nucleosynthesis allow, for the closures "Momentum" (M) and "Pressure" (P). (c) The same for the condensate, against its sound speed today.

ture rest in addition on A3, and the growth of structure and the largest angular scales on a closure that the hypothesis of deflation does not supply. The forecasts are for mock data without gravitational lensing; no measured spectrum has been fitted.

XVII. THE HIGGS HYPOTHESIS

A. The hypothesis and the conditions

Hypothesis H2 says that the superfluid state, the condensate of the fermion pair along the hidden axis, *is the Higgs field* [50]. In the Standard Model the Higgs field is a doublet of the weak-isospin group with a vacuum value $v_{\rm EW} = 246$ GeV, which breaks the electroweak symmetry, gives the W and Z bosons and the quarks and charged leptons their masses, and leaves one scalar particle of 125 GeV. Its four real components form a vector of a group $SO(4)$, which the vacuum value breaks to $SO(3)$ [51]. From these facts follow seven conditions that a condensate must meet in order to *be* the Higgs field (Table XXVIII). With static extra coordinates the hypothesis is unproven on the 7+1 spacetime, with one structural point in its favor: the order parameter has the representation and the breaking pattern of the Higgs field under a symmetry of the spacetime, the rotations of the four extra axes. This section examines the conditions on the deflating spacetime.

Two statements must be kept apart. H2 is the hypothesis that the *pair condensate* is the Higgs field; the electroweak scale is then the scale of the condensate. HV, used in one row of Table XXIII, is the different assumption that the vacuum value of the *Vorton-Scalar* sets the electroweak scale. HV is not examined here.

B. What the deflation leaves unchanged

The pair operators, their transformation properties and the internal symmetry are properties of the Dirac matrices of the frame, which the deflation does not change, and of the equations of the fields that do not depend on the extra coordinates, which for the weighted fields are the flat equations. By Sec. XIV C the four vector pair operators form the $(2,2)$ of $SU(2)_A \times SU(2)_B$, the representation of a Higgs field, and a condensate along one axis breaks $SO(4)$ to $SO(3)$. Hence, *within the sector*, C1 holds as on flat spacetime: the pair along the hidden axis is invariant under the rotations of $x1, x2, x3$, which remain symmetries of the spacetime, and under the boosts of ordinary spacetime as they act on the fields of the sector. (The boosts are no longer symmetries of the deflating spacetime as a whole; in the sector this shows only in the dependence of the couplings on the time.) C3 and C4 hold in the sector as well, in the sense that the pattern is right under its internal symmetry.

C. What the deflation changes

On the flat 7+1 spacetime the six generators of $SO(4)$ are rotations of the extra dimensions. This is the structural point in favor of the hypothesis: a symmetry of extra dimensions is the kind of symmetry from which a gauge interaction can arise, as in the theories of Kaluza and Klein [52]. By Sec. XIV C the deflating spacetime keeps none, one or three of the six, and never one of the

TABLE XXVIII. The conditions that a condensate must meet in order to be the Higgs field, and their status on the deflating 7+1 spacetime. The last row is a condition that only a deflating spacetime imposes.

condition	status
C1: a scalar under the Lorentz group of ordinary spacetime	met in the sector of fields that do not depend on the extra coordinates
C2: non-zero in empty space	open; reached by the deflation only outside the range of validity of the model
C3: a doublet of the weak-isospin group with a neutral component	only under a hidden symmetry (three different rates), or with $SO(3)$ alone as a symmetry of the spacetime (equal rates)
C4: breaks $SO(4)$ to $SO(3)$	met in the sector; the spacetime itself has at most the $SO(3)$
C5: scale 246 GeV, with a scalar particle of 125 GeV with the observed couplings	unfavorable (estimate of Sec. XVII E)
C6: its couplings give the quarks and leptons their masses	not addressed
C7: no further light or charged particles that would have been seen	additional particles are predicted
new: the electroweak scale does not drift	requires gsum $< 1.16 \times 10^{-6}\, H_0/n_{\rm EW}$

three that mix the hidden axis with a deflating coordinate.

For equal rates the part of $SO(4)$ that remains a symmetry of the *spacetime* is $SO(3)$, the rotations of the three deflating coordinates: precisely the subgroup that a condensate along $x0$ would leave unbroken. The deflation thus breaks $SO(4)$ to $SO(3)$ *explicitly*, at the level of the geometry, in the same pattern in which the hypothesis needs the condensate to break it *spontaneously*. Three consequences follow. First, the three broken generators, which under H2 would correspond to the massive weak bosons, are not symmetries of the spacetime at all; a gauge interaction for them cannot arise from the geometry of the deflating spacetime. Second, in the sector the full $SO(4)$ nevertheless acts as an exact symmetry, but as a hidden one. This is the situation of the model on the spacetime with four space-like and four time-like dimensions, where the six generators are never rotations of the spacetime and the hypothesis has no geometric basis. *The deflation puts the* 7+1 *model in the position of the* 4+4 *model.* Third, the potential of the order parameter does not single out the hidden axis (Sec. XIV): within the sector the direction of the condensate is as free as on flat spacetime. Outside the sector the axes differ, because a variation of the order parameter along a deflating coordinate costs a gradient energy that grows as $1/b_t^2$; the hidden symmetry is exact only for fields that are strictly independent of the extra coordinates.

D. Condition C2: a condensate in empty space

With static extra coordinates C2 is open: a pair condensate forms in the vacuum only if $\alpha_{\rm VS}$ exceeds a critical value α_c of about 2 or more, depending on the cutoff (Sec. XII). On a deflating spacetime $\alpha_{\rm VS}$ grows as e^{τ_d} and reaches any such value at a definite time. Section XII showed what else has happened by then: the exchange of one boson is no longer a controlled approximation, and the field at which the one-loop potential of the Vorton-Scalar falls below its vacuum value has moved to within a factor of two of the vacuum value. *C2 can be reached by waiting, but not within the range of validity of the model.* Its status remains open.

E. Conditions C5, C6 and C7: scale, masses and spectrum

These conditions concern dynamics that the model does not determine, and the deflation does not change what can be said about them.

C5. In models in which a condensate of fermions replaces the elementary Higgs field [26, 53] the scalar bound state that accompanies the condensate has a mass of about twice the mass that the condensate gives to its constituents. With 125 GeV for the scalar the constituents would weigh about 60 GeV. By C3 they carry electroweak charges, and searches at the electron-positron collider LEP exclude new charged fermions below about 100 GeV [54]. The estimate speaks against the hypothesis; it is not a proof, because the dynamics at strong coupling is not under control.

C6. The model contains no coupling of the pair field to ordinary fermions. A pair of fermion number 2 cannot couple to a fermion-antifermion bilinear of ordinary matter without violating the fermion number or the conservation of the ordinary quantum numbers.

C7. Besides the Higgs particle the model predicts the partners that complete the complex $(2, 2)$, a Goldstone boson of the fermion number, the fermions themselves with electroweak charges, and the two modes of the Vorton-Scalar. None of these is observed.

F. The self-interaction and the hidden symmetry

The internal symmetry of the sector, on which C3 and C4 rest, was established in Sec. XIV C for the model without self-interaction. By Sec. IV E a scalar and a

pseudoscalar contact coupling keep the whole of it, and so does the exchange of the Vorton-Scalar, which is a self-interaction of these two kinds with a finite range [Eq. (69)]. A vector or an axial-vector contact coupling keeps the six generators of $SU(2)_A \times SU(2)_B$ and the phase symmetry but breaks $U(1)_E$. The representation and the breaking pattern that C3 and C4 require are therefore present in the sector for any couplings of the self-interaction; what a vector or axial-vector coupling removes is the additional symmetry $U(1)_E$.

G. A new condition: the electroweak scale must not drift

Suppose that H2 holds. The electroweak scale v_{EW} is then the scale of the pair condensate in the vacuum, as in models in which a condensate of fermions replaces the elementary Higgs field [53]. That scale is determined by the parameters of the model: the masses m and m_s, the cutoff, and the coupling α_{VS}. On the deflating spacetime the masses are constant and $\alpha_{\mathrm{VS}} \propto e^{\tau_d}$. Therefore

$$v_{\mathrm{EW}} \;\propto\; e^{\,n_{\mathrm{EW}}\,\tau_d}, \qquad n_{\mathrm{EW}} = \frac{d\ln v_{\mathrm{EW}}}{d\ln \alpha_{\mathrm{VS}}}. \tag{114}$$

The exponent cannot vanish: below the critical coupling there is no condensate, so that the scale depends on the coupling, and most strongly just above the critical coupling.

Under H2 the masses of the electron and of the quarks are proportional to v_{EW} and change at the rate n_{EW} gsum. The mass of the proton is due mostly to the strong interaction and does not follow. The ratio m_p/m_e of the proton mass to the electron mass therefore changes at the rate n_{EW} gsum. Comparisons of atomic clocks find this ratio constant to 8.0×10^{-17} per year at two standard deviations [39]. Hence

$$\mathrm{gsum} < \frac{8.0\times10^{-17}\ \text{per year}}{n_{\mathrm{EW}}} = \frac{1.16\times10^{-6}\,H_0}{n_{\mathrm{EW}}}. \tag{115}$$

The bound needs no assumption about gravity. For $n_{\mathrm{EW}} = 1$ it is 2787 times stronger than the bound that lunar laser ranging gives if gravity is eight-dimensional.

The size of the exponent. For the model itself, with massive fermions and a massive radial mode, the scale of the vacuum condensate has not been computed as a function of the coupling. It is known in closed form for the scale-invariant case of the ladder equation: massless fermions and massless bosons between an infrared scale and a cutoff [25]. There the scale of the condensate obeys

$$\varrho_c \,\ln\frac{\text{cutoff}}{\text{scale}} = \arctan\frac{1}{2\varrho_c}, \qquad \varrho_c^2 = \frac{1}{4}\Big(\frac{\alpha_{\mathrm{VS}}}{\alpha_c}-1\Big), \tag{116}$$

where ϱ_c is only an abbreviation for the square root that the second equation defines, from which

$$n_{\mathrm{EW}} = \frac{(1+4\varrho_c^2)\,\arctan(1/2\varrho_c)}{8\,\varrho_c^3} + \frac{1}{4\varrho_c^2}. \tag{117}$$

TABLE XXIX. The exponent n_{EW} with which the scale of the vacuum condensate follows the coupling in the scale-invariant ladder equation, and the bound (115) on the present rate of deflation that follows from it under hypothesis H2.

$\alpha_{\mathrm{VS}}/\alpha_c$	ln(cutoff/scale)	n_{EW}	gsum (per year)	gsum/H_0
1.05	12.082	147	5.4×10^{-19}	7.9×10^{-9}
1.10	7.998	54.0	1.5×10^{-18}	2.2×10^{-8}
1.25	4.429	15.1	5.3×10^{-18}	7.7×10^{-8}
1.50	2.702	6.05	1.3×10^{-17}	1.9×10^{-7}
2.00	1.571	2.57	3.1×10^{-17}	4.5×10^{-7}
3.00	0.870	1.15	6.9×10^{-17}	1.0×10^{-6}
5.00	0.464	0.540	1.5×10^{-16}	2.2×10^{-6}
10.0	0.215	0.230	3.5×10^{-16}	5.0×10^{-6}

The program verifies Eq. (117) against the numerical derivative to 3×10^{-8} and tabulates it with the bound (115) (Table XXIX). The exponent diverges at the critical coupling, equals $1+\pi/2 = 2.57$ at twice the critical coupling, and exceeds 1 for $\alpha_{\mathrm{VS}} < 3.27\,\alpha_c$. It falls below 1 only where the scale of the condensate approaches the cutoff, that is, where the four-dimensional description ends. Within the range in which a condensate of the kind that H2 needs can be described at all, n_{EW} is of order one or larger.

The conclusion. Under H2 the deflation clock can have advanced by at most $\mathrm{gsum}\,t_0 < 1.1\times10^{-6}/n_{\mathrm{EW}}$ during the age of the universe at a constant rate. Every effect of the deflation described in Secs. VIII to XVI is then unobservably small. Conversely, a deflation at any rate at which those effects matter excludes H2. *The hypothesis H2 and a deflation of any consequence today exclude each other.* This conclusion does not depend on the unresolved conditions C2 and C5 to C7. It depends on two properties only: that the electroweak scale is set by the dynamics of the model, which is the content of H2, and that the coupling of the model follows the volume of the deflating coordinates, which is exact (Sec. VI).

H. Verdict on hypothesis H2

On the deflating spacetime the hypothesis that the singlet superfluid is the Higgs field is unproven, with less in its favor than on the flat 7+1 spacetime, and it is incompatible with any deflation that is fast enough to matter today. Whether a mechanism exists that would make the electroweak scale insensitive to the coupling of the pair is not known; none is present in the model.

XVIII. WHAT HAS BEEN ESTABLISHED, AND WHAT HAS NOT

The sections above reported results of very different standing: identities verified in exact arithmetic, numerical results with a stated accuracy, consequences of ad-

ditional assumptions, and estimates. This section sorts them. An item is called established if a machine check stands behind it (Appendix D); statements of interpretation deserve the reader's independent scrutiny more than the numbers do.

A. Established

Geometry (Sec. II). The curvature and the Einstein tensor of the deflating spacetime in closed form, for arbitrary rates and constants. The Killing vectors, 14, 15 or 20 in number, and the completeness of the lists. That the spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant; that with three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for $\Lambda_8 = 3\,\mathrm{g}55^2$ and $\alpha_{\mathrm{GB}} = -1/(4\,\mathrm{g}55^2)$; that with unequal rates it is not; and that this solution is unstable (Sec. XV). The unbounded blueshift of a classical particle that moves along a deflating coordinate, and the incompleteness of its geodesic.

The Dirac field (Secs. III, V). The Dirac matrices in coordinate components, the spin connection, the covariant constancy of the matrices, and the curvature of the spinor derivative. The Lagrangian with the kinetic term, the mass term and the self-interaction; the field equation (24) with the self-interaction on its right-hand side; the weighted field; and the statement that fields which do not depend on $x5, x6, x7$ obey the flat equation with the couplings g_r/V_d of the moment. The exact solutions when one rate governs, and the law of creation (42) with the temperature (43). All of this is confirmed numerically by two independent methods.

The self-interaction (Sec. IV). The Fierz matrix in closed form and from the 256 matrices. That for anticommuting fields the nine invariants obey five linear relations, so that only four couplings are independent, verified by exact integer arithmetic and on Grassmann-valued fields. The four effective couplings (35), the Hartree-Fock couplings (37), the pair couplings (38), and the conditions under which the self-interaction keeps the chiral symmetry and the symmetry $U(1)_E$.

Matter with the self-interaction (Sec. VII). The Kohn-Sham states of uniform matter and their thermodynamic consistency. The reference calculation on flat spacetime, reproduced to ten digits by two programs. The history of matter with momentum along the deflating coordinates: the growth of its density, the fall of the mass of its particles, the limited time during which it is self-bound, the anisotropy of its pressure, and the conservation law (59). The thresholds of Table IV at which a gas becomes self-bound matter in the sector of fields that do not depend on the deflating coordinates, and the history of the saturated matter after that moment.

The effective potential with the self-interaction (Sec. XIII). That the Kohn-Sham method and the CJT effective potential at the order of the Hartree and Fock diagrams give the same states. The critical coupling (82) of the vacuum with the filled states of negative energy, and the moment at which the deflation crosses it. The functional (83) with bare mass, self-interaction and Vorton-Scalar; that the exchange of the radial mode acts at low density as a scalar contact coupling; the shifts of the moment of binding of Table XVI; and the mass (85) that a bare Dirac mass gives to the phase mode.

Pairing with the self-interaction (Sec. IX). The amplitudes (66) of the four contact channels in closed form; that the exchange of the Vorton-Scalar is a self-interaction of finite range; and the gaps of Table VII.

Gravity (Sec. XV). The Einstein-Lovelock tensors (94) and (95), derived in three ways. The reference evolution, reproduced by two programs. That the undisturbed deflation needs the constants (96) or a source with the equation of state (97); that the matter of the model cannot be that source; that matter accelerates the deflation and makes the flat coordinates expand; and that the undisturbed deflation is unstable in the tuned gravity.

The model along the clock (Secs. VI to XII). For fields that do not depend on the extra coordinates, the model on the deflating spacetime is the model on flat spacetime with the couplings of the moment, exactly. The masses are constant; g_{VS} grows as $e^{\tau_d/2}$; α_{VS}, G_{VS}, D_{VS} and C grow as e^{τ_d}. The modes of the Vorton-Scalar grow with the weight only, and the potential between two fermions has the unchanged range and the strength of the moment. The exchange amplitudes are those of flat spacetime times the square of the coupling of the moment. In weak coupling the gap grows according to Eq. (64), and the critical temperature follows it. Two fermions become bound at a definite time, at a scattering resonance, with a binding energy that starts quadratically; the order of the other events depends on m_s/m alone. The paired state follows the deflation if and only if Ξ_d is below about 0.2 to 0.5; before that it lags and accumulates broken pairs. The solution of the gap equation is a minimum of the effective potential at every moment examined. Matter of fixed density loses its fermion mass at a critical value of C, is self-bound for $C > 3.7011$ and is more strongly bound later; the melting temperature of the fermion mass falls; the barrier of the one-loop potential of the vacuum moves toward the vacuum value.

Order parameter and symmetry (Sec. XIV). The potential of the order parameter prefers no direction among the four frame axes. The internal symmetry of the sector, the representation of the pair operators and the breaking pattern are unchanged by the deflation. Of the six generators that belong to the pairs of the four extra axes, the deflating spacetime keeps none, one or three as symmetries of the spacetime. A vortex line of unit winding carries four fermion zero modes, two for each direction along the line, at every moment; its energy per length falls as $e^{-\tau_d}$.

Cosmology (Sec. XVI), each item under the assumption named there. The equation of the expansion (100) and the fall of the inferred Hubble constant, with recom-

bination computed on the deflating background and verified by an independent calculation. The bounds of Table XXIII, and that in the model as it stands the atomic clocks give none. The effect of the loss of mass of bound pairs. The pressure (101) of the agent. The equations (102) of the perturbations of ordinary space for unperturbed extra scale factors, the identities (105) and (106), and that the four equations cannot all hold with matter alone. The growth equations of the two closures, and that the closure "Shear" coincides with the closure "Pressure" inside the horizon. The spectra of the microwave background for $\ell \geq 30$ and their independence of the closure; the forecast for the rate from mock data; that the Hubble constant falls in every fit, with or without extra radiation; and that extra radiation and the rate of deflation compensate in part in the spectra, without any gain for the Hubble constant. The unchanged estimates of degeneracy and of condensation in halos, the limits on superfluid dark matter, the epoch of the resonance, and the drift of the quantities. The temperature of the created quanta at the allowed rates.

The Higgs hypothesis (Sec. XVII). Under H2 the electroweak scale follows the coupling with the exponent n_{EW}, and atomic clocks bound the rate of deflation by Eq. (115). The exponent is given in closed form for the scale-invariant ladder equation.

B. What the mass term and the self-interaction change

Sections VIII to XIV were derived for the model in which the whole mass of the fermions comes from the Vorton-Scalar and the whole interaction from its exchange ($m_D = 0$, $g_r = 0$). Table XXX lists, result by result, what a bare Dirac mass and the self-interaction change.

C. The two methods as checks of each other and of the other results

The Kohn-Sham method and the CJT effective potential were used not only for new results but as checks, and each result of this paper that could be computed in two ways was (Table XXXI).

D. Not established

The source of the deflation. The metric is prescribed. For equal rates it solves the equations of Einstein-Gauss-Bonnet gravity without matter, but that solution is unstable (Sec. XV); in Einstein gravity it needs a source with the equation of state (97), which the matter of the model cannot be. For unequal rates no field equation has been found that the metric solves. What holds the deflation, and how it absorbs the reaction of matter, is not known. All results on fields are results on a given background.

The cutoff. The self-interaction has couplings of negative mass dimension and defines an effective theory. The results without the filled states of negative energy presuppose a cutoff below the mass of the particles (Sec. XIII B); with the filled states the results depend on the cutoff, and the mass of the phase mode depends on it through the scalar density of the vacuum. The cutoff is a parameter of the model that nothing in the hypotheses fixes.

The couplings. The nine couplings g_r, of which four are independent, are free parameters. The examples use a scalar attraction and a vector repulsion in the ratio 10 to 1; no value is predicted. For the reference example the two numbers -10^5 and $+10^4$ are taken as the mean-field couplings themselves; taken as couplings of the Lagrangian they would not bind matter (Sec. VII B).

Matter beyond the mean field. The Kohn-Sham states and the effective potential at the order of the Hartree and Fock diagrams neglect correlations. Without the filled states the energy has no lowest value as a function of an unrestricted trial mass; the states reported are solutions of the gap equation. Temperature was not included in the calculations with the self-interaction. The matter with momentum along the deflating coordinates was treated without collisions; whether collisions are frequent enough to release the heat of Sec. VII C was not examined.

Pairing inside dense matter. The gaps of Table VII are computed for particles of the mass m. In self-bound matter the mass is $m^* < m$; the gap equation and the equation for m^* have not been solved together. The contact couplings enter the gap equation with a range $1/m_{\mathrm{ct}}$ that is a choice.

The theory of gravity and the assumption A2. If gravity is of the Einstein-Gauss-Bonnet kind, the relation (99) between Newton's constant and the volume of the deflating coordinates, and with it every result of Sec. XVI that is marked A2, would have to be derived anew.

The agent. Nothing in the hypothesis says what keeps the three rates constant while ordinary space expands. Under A2 it must have the tension (101) and no energy density; whether a physical source with these properties exists has not been examined. Nor does the hypothesis say how the agent responds when ordinary space is perturbed. That the scale factors of the deflating coordinates are not perturbed (assumption A3) is an assumption; and even with it, a closure is needed that the hypothesis does not supply. The growth of structure, the amplitude σ_8 and the spectra of the microwave background at the largest angular scales are computed for two closures and differ between them at the first order in the rate. They are the quantities that a theory of the agent would have to predict.

The reaction of the created particles on the metric. The particles created by the deflation gain energy without bound. Their stress must eventually change the metric; this has not been computed. Nor do the coordinates cover the whole future of particles that move along deflating

TABLE XXX. What a bare Dirac mass m_D and the self-interaction (couplings g_r) change in the results obtained with the Vorton-Scalar alone.

result with the Vorton-Scalar alone	with a bare Dirac mass	with the self-interaction
fields that do not depend on $x5, x6, x7$ obey the flat equations with couplings of the moment (Secs. III, VI)	unchanged; m_D itself does not change with the clock	unchanged; every g_r grows as e^{τ_d}, in a fixed ratio to $g_{\rm VS}^2$
the chiral symmetry $U(1)_{\rm VS}$ is exact and the phase mode massless (Sec. VI)	broken explicitly; the phase mode has the mass of Eq. (85), which grows as $e^{\tau_d/2}$	kept only if $g_0^{\rm eff} + g_8^{\rm eff} = 0$
the gap of the paired state and its growth (Sec. VIII)	smaller, because the massive phase mode binds less	larger with a scalar contact attraction, smaller with a vector repulsion (Table VII)
two fermions bind when $D_{\rm VS}$ reaches D_c (Sec. X)	the Yukawa potential of the radial mode is unchanged	a contact term $G_{\rm pair}$ is added to the potential; the threshold with it has not been computed
matter is self-bound for $C > 3\pi^2/8$ (Sec. XII)	earlier (Table XVI)	later with a vector repulsion; earlier with a scalar attraction where the particles keep a mass (Table XVI)
matter of fixed density loses its whole fermion mass (Sec. XII)	the particles keep the mass m_D	the mass in matter is m^* of Eq. (54)
the internal symmetry $[SU(2) \times SU(2)] \times U(1)_E$ of the sector (Sec. XIV)	unchanged	$U(1)_E$ is broken by vector and axial-vector couplings
cosmology and the bounds on the rate (Sec. XVI)	unchanged	unchanged

coordinates; what lies beyond is outside the hypothesis.

The restriction to the sector. That the fields do not depend on $x5, x6, x7$ is assumed wherever the sector is used. The deflation supports the assumption only if the deflating coordinates are periodic (Sec. V).

Unequal rates with simultaneous crossings. No closed form is known for a mode that depends on two deflating coordinates whose crossings coincide; the results are numerical.

The hidden coordinate. That the fields do not depend on $x0$ is assumed. Nothing in the deflation explains it.

The end of the history. The couplings grow without bound. The four-dimensional description with the exchange of one boson holds only while $\alpha_{\rm VS}$ is not large, and the eight-dimensional theory itself is an effective theory, with couplings of negative mass dimension, that must fail when the proper size of the deflating coordinates reaches its scale of validity. What happens then is outside the model. Every history of Secs. VIII to XII ends in a regime that the two-loop truncation does not control; the numbers at the last moments of each table indicate the trend and are not predictions.

The energy. The energy of matter is not conserved while the couplings change. Where it goes, into whatever drives the deflation, is not described.

Dynamics. Sections VIII, X and XII give sequences of equilibrium states. Only for the paired state has the ability to follow been examined, in a model without dissipation; with dissipation the broken pairs would relax, and their lifetime is not known. The time-dependent two-body problem at the moment of binding, which decides how much of the population ends in the bound state, has not been solved. How a gas reaches the self-bound state, and whether the abnormal state forms by nucleation, are open. The equilibrium of a loop of vortex line with its current, and its stability, have not been computed.

The vacuum beyond one loop. The statements about the vacuum use the one-loop potential outside its range of reliability and draw only the conclusion stated in Sec. XII.

Cosmology. The forecasts of Sec. XVI H are for mock data without gravitational lensing; no measured spectrum of the microwave background has been fitted. All bounds that reach back to nucleosynthesis or to recombination assume a constant rate; a rate that was large early and is small today is neither excluded nor described. Only linear perturbations were computed. The consequences of the epoch of the resonance for the formation of halos have not been simulated.

The Higgs hypothesis. That a condensate forms in the vacuum, that the rotation group of the extra dimensions is the electroweak gauge group, the scale and the mass of the scalar particle, and the masses of ordinary fermions: none of these is established. The exponent $n_{\rm EW}$ has not been computed for the model itself; the statement that it is of order one or larger rests on the scale-invariant case and on the general fact that the scale of the condensate vanishes at the critical coupling.

TABLE XXXI. Results obtained in two or more independent ways, with the largest relative deviation between them.

result	obtained by	deviation
only four of the nine invariants of the self-interaction are independent	exact integer arithmetic on the antisymmetrized quartic form (program); Grassmann-valued fields (independent script)	9.4×10^{-16}
the Hartree-Fock couplings	exact traces (program); Wick's theorem with density matrices summed from numerical eigenvectors (independent script)	2.1×10^{-13}
the pair couplings of the 28 pairs at rest	exact diagonalization (program); numerical diagonalization for random couplings (independent script)	7.9×10^{-14}
saturation point of the reference example on flat 7+1 spacetime	a first calculation with a notebook; the program; the independent script	1.7×10^{-10}
states of uniform matter	Kohn-Sham method (gap equation, Gauss-Legendre integrals); CJT effective potential (minimization, Simpson integrals)	1.6×10^{-12}
matter with momentum along the deflating coordinates (7+1)	integral over the ellipsoid of occupied states (program); flat momenta integrated in closed form (independent script)	1.4×10^{-8}
the thresholds at which matter becomes self-bound	the program; a general-purpose minimizer (independent script)	1.0×10^{-9}
matter bound by the Vorton-Scalar (Sec. XII)	the functional of Sec. XIII C without bare mass and contact couplings; the functional of Sec. XII	2.1×10^{-8}
pairing amplitudes of the contact channels	closed forms; direct evaluation with the 256 matrices (program); projectors from numerical eigenvectors (independent script)	1.8×10^{-15}
the exchange of the radial mode as a contact coupling of finite range	the kernel of the exchange; the kernel of the scalar contact coupling with the range $1/m_s$	3.3×10^{-16}
gaps of the paired state without contact couplings	the gap equation of Sec. VIII; the gap equation of Sec. IX with vanishing contact couplings	6.6×10^{-16}
gaps with contact couplings	the program; plain iteration on another grid of momenta (independent script)	1.1×10^{-3}
the Einstein-Lovelock tensors of the orders 1, 2 and 3	sums over pair curvatures (program); the Riemann tensor of the metric and the Euler-Lagrange equations of the reduced action (independent script, computer algebra)	exact
the reference evolution with gravity	a first calculation with a notebook; the program (Runge-Kutta, order 4); the independent script (Runge-Kutta, order 8)	4.8×10^{-8}
the deflating spacetime with matter as a source	the program; a general-purpose integrator with an independent quadrature of the matter (independent script)	6.4×10^{-6}
the deflating spacetime as a solution of Einstein-Gauss-Bonnet gravity	the Lanczos tensor from its definition (Sec. II C); the Einstein-Lovelock polynomials of Sec. XV	exact

XIX. CONCLUSIONS

We asked what becomes of a 16-component Dirac field with a mass term and a self-interaction, and of the superfluid of its pairs, when three of the eight coordinates of a spacetime of signature 7+1 deflate exponentially.

The geometry is simple and rigid. The deflating spacetime is homogeneous, it has 14 to 20 symmetries instead of 36, and it is not a solution of Einstein's equations without matter. With three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for one relation between the rate and the two constants of that theory; but the solution is unstable, the matter of the model cannot be the source that Einstein gravity would need, and matter that is present accelerates the deflation.

The Dirac field separates into two parts. Fields that do not depend on the deflating coordinates obey, after a change of normalization, exactly the flat equations. Fields that do depend on them are populated by the deflation with a Fermi-Dirac spectrum at the temperature of the rate divided by 2π, and their quanta then gain energy without bound.

The self-interaction has nine channels and, because the field anticommutes, four independent couplings. On the deflating spacetime they grow with the inverse volume of the deflating coordinates, in step with the coupling of the Vorton-Scalar. The Kohn-Sham method and the effective action of Cornwall, Jackiw and Tomboulis give the same matter, and the deflation carries it through definite events: a gas becomes self-bound, a massless field acquires a mass, and with a bare Dirac mass the phase mode of the Vorton-Scalar becomes heavy. Matter with momentum along the deflating coordinates is compressed, loses the mass of its particles, and is self-bound

only for about one e-folding of the deflating volume.

The superfluid of hypothesis H1 exists on the deflating spacetime at every moment, with the couplings of that moment, and it acquires a history. The masses do not change; the coupling grows with the inverse volume of the deflating coordinates; the gap grows much faster than the coupling, in weak coupling as the exponential of an exponential; two fermions become bound at a scattering resonance; matter loses its fermion mass and becomes self-bound; and the paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength. A scalar contact attraction raises the gap, and a vector repulsion lowers it. The direction of the order parameter is not selected by the deflation.

For hypotheses H2 and H3 the deflation brings no relief. The rotations that gave the Higgs hypothesis its one structural support on the flat 7+1 spacetime are no longer symmetries of the spacetime. If the superfluid were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would then limit the rate of deflation to about a millionth of the Hubble rate. If gravity is the Einstein theory in eight dimensions, lunar laser ranging limits it to three thousandths of the Hubble rate, and the Hubble constant inferred from the microwave background moves away from the local measurement, in the simplest estimate and in fits of all parameters to mock data alike. In either case nothing of the history just described can have happened within the age of the universe at a constant rate.

The perturbations of ordinary space show most plainly what the hypothesis of deflation leaves open. Whatever keeps the rates constant must carry a tension, and it must take part in the perturbations: with a Newton's constant that depends on the time, the Einstein equations of ordinary space cannot all hold with matter alone. The spectra of the microwave background at all but the largest angular scales do not depend on how the agent does so, and mock data of the quality of the Planck satellite would bound the rate to four per cent of the Hubble rate. The growth of structure does depend on it, at the first order in the rate.

These results are quantitative, and they come with the programs that produced them. They mark what a successful theory of this kind would have to supply: a source or a theory of gravity that sustains the deflation stably and that says how the source responds to matter and to perturbations, a rate that was large early and is small today, and a reason why the constants of ordinary matter do not follow the volume of the deflating coordinates.

DATA AVAILABILITY

The source code of all programs, the data files from which every figure and every computed number of this paper is generated, the instructions for compiling and running the programs (Appendix D), and the records of the provenance of each result are openly available [4].

Appendix A: Symbols

Tables XXXII to XXXV list every symbol of this paper with its meaning and the place of its definition. No symbol has two meanings. Where one letter occurs with different indices or in different type, the tables say how the uses are told apart; a machine check of the registry from which these tables are taken verifies that no symbol and no name in a data file is defined twice (Appendix D).

Appendix B: The Dirac matrices

Table XXXVI prints the eight Dirac matrices of the orthonormal frame and three matrices built from them. All are real, with entries 0, +1 and −1, and each Dirac matrix is a signed permutation matrix. The matrix γ^4 of the time is symmetric and squares to +1; the seven matrices of the space-like coordinates are antisymmetric and square to −1 [Eq. (15)]. Their product $\omega_{1671} = \gamma^0\gamma^1\cdots\gamma^7$ is real and squares to −1; the chirality matrix is $\chi = i\,\omega_{1671}$, and the chiral projectors $P^{L,R} = \frac{1}{2}(1 \mp \chi)$ are complex. The matrix $C^- = \gamma^4\omega_{1671}$ turns each Dirac matrix into minus its transpose, $C^-\gamma^A(C^-)^{-1} = -(\gamma^A)^T$, and enters the pair operators $\psi^T C^-\gamma^A\psi$; the matrix $C^+ = \gamma^4$ turns each into plus its transpose. Two kinds of matrices follow by one multiplication: $\beta = \gamma^4$, which defines the adjoint $\bar\psi = \psi^\dagger\beta$, and the real symmetric matrices $\alpha^A = \gamma^4\gamma^A$ $(A \neq 4)$ of the Hamiltonian. The program verifies every property stated here in exact integer arithmetic (Appendix D).

Appendix C: Curvature of the deflating metric

This appendix derives the Einstein tensor and the Lanczos tensor of Sec. II in a form that can be checked by hand, and gives the coordinates and the additional symmetries of the case of equal rates.

1. Curvatures on pairs of directions

In the orthonormal frame the Riemann tensor (7) of the deflating metric is diagonal on pairs of directions: for each pair of frame directions $A \neq B$ it has one independent component, $K_{AB} = K_{BA} = R^{AB}{}_{AB}$ (no summation), the sectional curvature of the plane that the two directions span. With the Riemann tensor (7) and the flat metric (2) the sectional curvatures are, on the 4+4 and on the 7+1 spacetime alike,

$$K_{4t} = \mathrm{rate}_t^2, \qquad K_{ts} = \mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{C1}$$

and $K_{AB} = 0$ for every pair that contains one of the flat directions $0, 1, 2, 3$. For a Riemann tensor of this kind the mixed components of the Einstein tensor and of the

TABLE XXXII. The symbols of this paper: geometry and the Dirac field.

symbol	meaning	defined in
$x0, \ldots, x7$	the eight coordinates: $x0$ hidden space-like coordinate; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ the deflating coordinates, space-like	Sec. I
xt; xj; a	a deflating coordinate, $t \in \{5, 6, 7\}$; a flat coordinate other than the time, $j \in \{0, 1, 2, 3\}$; index of the four extra axes, $a \in \{0, 5, 6, 7\}$	Sec. I
g55, g66, g77; rate_t	the three rates of deflation (upright, without subscript; not components of the metric); the rate of xt	Eq. (1)
const5, const6, const7; const_t	the three positive constants of the metric: the scale factors at $x4 = 0$	Eq. (1)
gsum, gsq, gprod	the sum of the rates, the sum of their squares, the sum of their products in pairs	Sec. II
b_t	scale factor of the deflating coordinate xt	Eq. (3)
$\eta^{(7+1)}$; $g^{(7+1)}_{\mu\nu}$, $g^{\mu\nu}_{(7+1)}$; $e_A{}^\mu$	flat metric, metric of the deflating spacetime and its inverse, in the sign convention of the hypothesis (space-like components positive, time-like components negative; the superscript marks the spacetime, so that a component of the metric cannot be taken for a rate or a coupling); orthonormal frame	Eqs. (2), (4)
V_d	proper volume of a unit coordinate cell of the deflating coordinates	Eq. (5)
$\Gamma^\rho{}_{\mu\nu}$; $R^\rho{}_{\sigma\mu\nu}$, $R^\mu{}_\nu$, R	Christoffel symbols; Riemann tensor [in Eq. (7) its components in the orthonormal frame], Ricci tensor, curvature scalar	Sec. II B
$\mathcal{G}^\mu{}_\nu$; $\mathcal{H}^\mu{}_\nu$; Λ_8, α_{GB}	Einstein tensor; Lanczos tensor of the Gauss-Bonnet term; cosmological constant and coupling of Einstein-Gauss-Bonnet gravity	Sec. II C
$\mathcal{X}_{\text{scale}}$, $\mathcal{X}_{(s)}$	the scaling Killing vector; the three special Killing vectors of the case of equal rates	Eqs. (12), (C8)
K_{AB}; $\bar{x}4$, $\bar{x}t$	sectional curvature of the plane of the frame directions A and B (two subscripts; K_{4i}, K_{ij} in Sec. XV); coordinates adapted to the case of equal rates	Appendix C
P_μ; ς	conserved momenta of a classical particle; affine parameter	Sec. II E
γ^A, γ^μ; Ω_μ, $\omega_{\mu AB}$	Dirac matrices of the frame (printed in Appendix B) and in coordinate components; spin connection and its coefficients	Sec. III
χ, $P^{L,R}$; C^-	chirality matrix and the two chiral projectors; the matrix that turns each Dirac matrix into minus its transpose	Secs. VI, VIII
ψ_{1671d}; $\hat{\psi}$	the 16-component field on the deflating spacetime; the weighted field $\sqrt{V_d}\,\psi$	Eqs. (23), (25)
h, $\mathsf{h}_\perp$; β, α^A, α^d; $\bar{\psi}$	Hamiltonian of a mode, and its part without the scale factors; γ^4, $\gamma^4\gamma^A$, and the matrix α along the wave vector in the deflating coordinates; the adjoint $\psi^\dagger\beta$	Eqs. (27), (44)
p_j, k_t; $E_\perp$; k_d; u_d	wavenumbers along the flat and the deflating coordinates; $(m^2 + \sum_j p_j^2)^{1/2}$; physical wavenumber along the deflating coordinates; k_d/rate_t	Eqs. (28), (40)
w; Z_+, Z_-	one of the eight constant spinors from which the exact solutions are built; two cylinder functions whose orders differ by one	Eq. (41)
n_{cr}; T_{defl}	number of quanta that the deflation creates in a mode, per internal state; temperature of their spectrum	Eqs. (42), (43)

Lanczos tensor are diagonal, and

$$\mathcal{G}^A{}_A = -\sum_{\{B,C\}} K_{BC}, \tag{C2}$$

$$\mathcal{H}^A{}_A = -4\sum_{\{B,C\},\{B',C'\}} K_{BC}\,K_{B'C'}, \tag{C3}$$

without summation over A. The first sum runs over the pairs of directions that do not contain the direction A. The second runs over the unordered couples of two such pairs that have no direction in common; a set of four directions contributes its three divisions into two pairs. As a test, for a space of constant curvature in eight dimensions, where all K_{BC} equal one number, the first sum has 21 terms and the second 3×35, so that the two formulas give -21 times that number and -420 times its square, the known values $-\frac{1}{2}\cdot 7\cdot 6$ and $-\frac{1}{2}\cdot 7\cdot 6\cdot 5\cdot 4$.

The Einstein tensor. With Eq. (C1), Eq. (C2) gives

$$\begin{aligned}&\mathcal{G}^4{}_4 = -\text{gprod}, \qquad \mathcal{G}^j{}_j = -(\text{gsq} + \text{gprod}),\\ &\mathcal{G}^5{}_5 = -(\text{g66}^2 + \text{g77}^2 + \text{g66}\,\text{g77}),\end{aligned} \tag{C4}$$

and the corresponding expressions for $\mathcal{G}^6{}_6$ and $\mathcal{G}^7{}_7$. Because $\text{gsum}^2 = \text{gsq} + 2\,\text{gprod}$, these are Eq. (8).

The Lanczos tensor. A couple of pairs contributes to Eq. (C3) only if neither pair contains a flat direction, that is, only if the four directions are $4, 5, 6, 7$. For a flat direction $A = j$ these are allowed, and their three divisions into two pairs give

$$K_{45}K_{67} + K_{46}K_{57} + K_{47}K_{56} = \text{g55}\,\text{g66}\,\text{g77}\,\text{gsum}, \tag{C5}$$

which is Eq. (10). For $A = 4, 5, 6, 7$ the direction A is itself among the four, and these components vanish. The

TABLE XXXIII. The symbols of this paper, continued: mass term and self-interaction, the Kohn-Sham and CJT calculations, and gravity.

symbol	meaning	defined in
$\mathcal{L}_{1671d}$; $\mathcal{L}_{\rm kin}$, $\mathcal{L}_{\rm mass}$, $\mathcal{L}_{\rm self}$	Lagrangian of the fermion field, and its three parts: kinetic term, mass term, self-interaction	Eq. (19)
g_r; $g_0, \ldots, g_8$	the nine couplings of the self-interaction, one for each rank (one digit as subscript: not a rate g55, not a component $g^{(7+1)}_{\mu\nu}$ of the metric, not the coupling $g_{\rm VS}$)	Eq. (22)
K, J; r, r', l	strictly increasing lists of indices out of $0, \ldots, 7$; their ranks (numbers of indices), and a summation index	Eq. (21)
Θ^K, Θ_K; ζ_r; sgn_K	the 256 matrices of the self-interaction, $\Theta^K = \zeta_r \gamma^{A_1} \cdots \gamma^{A_r}$, and their inverses $\Theta_K = \mathrm{sgn}_K\, \Theta^K$; the phase 1 or i; the sign ± 1	Eq. (21)
I_r	the nine quartic invariants	Eq. (22)
F, $\mathsf{F}_{rr'}$	the Fierz matrix (sans-serif; the amplitudes F_S, F_P, $F_0, \ldots$ are in italic type)	Eq. (31)
$g_0^{\rm eff}$, $g_1^{\rm eff}$, $g_7^{\rm eff}$, $g_8^{\rm eff}$	the four independent couplings: scalar, vector, axial vector, pseudoscalar	Eq. (35)
n; n, n_S	density matrix of matter; its number density $\langle\psi^\dagger\psi\rangle$ and scalar density $\langle\bar\psi\psi\rangle$	Sec. IV C
$G_S^{\rm HF}$, $G_V^{\rm HF}$	the values that the Hartree-Fock approximation gives to the mean-field couplings	Eq. (37)
$G_{\rm pair}$	contact coupling of a pair of two particles at rest	Eq. (38)
G_S, G_V	the scalar and the vector coupling of the mean-field description of uniform matter (attraction for $G_S < 0$, repulsion for $G_V > 0$)	Sec. VII A
$\mathsf{h}_{\rm KS}$; V_0; $E^*(k)$	Hamiltonian matrix of a particle in matter; the vector mean field $G_V\, n$; the energy $(k^2 + m^{*2})^{1/2}$	Eq. (54)
D; V_D, A_D; ν; $\langle f\rangle$	number of space dimensions in which the particles have momentum; volume and area of the unit ball in D dimensions; number of internal states of a particle, 8; average over the occupied states	Eq. (55)
ε; p_j, $p_{\rm int}$; μ	energy density of matter; its pressure along the coordinate xj and the interaction part of the pressure; chemical potential	Eqs. (56), (57)
$b_t^{\rm ref}$; $n_{\rm ref}$	the scale factor b_t and the density of matter at the reference time $x4_{\rm ref}$	Sec. VII C
$\Gamma[S]$, Γ_2; S, S_0	the CJT effective action and its two-particle-irreducible part; full and free propagator of the fermions	Eq. (77)
$\Omega_{\rm CJT}$	the CJT effective potential of uniform matter with the self-interaction	Eq. (78)
Λ; G_c	cutoff of the momenta of the filled states of negative energy (without index or bar; Λ_8 and $\bar\Lambda$ are cosmological terms); critical coupling of the vacuum	Sec. XIII B
$m_{\rm vac}$, $m_{\rm bare}$; $\langle\bar\psi\psi\rangle_{\rm vac}$	with the filled states of negative energy: mass of a fermion in vacuum, mass parameter of the Lagrangian, scalar density of the vacuum	Eq. (81)
x; f_D	expectation value of the Vorton-Scalar in units of its vacuum value, $\phi/v_{\rm VS}$; fraction m_D/m of the fermion mass that is a bare Dirac mass	Eq. (83)
$\vartheta_{\rm VS}$; m_ϑ	phase of the Vorton-Scalar; mass of its phase mode	Eq. (85)
F_0, F_1, F_7, F_8; Φ, $\Phi(\vec p)$, X	pairing amplitudes of the four contact channels; a state of a pair of two particles (an antisymmetric 16×16 matrix; without suffix, whereas the Vorton-Scalar is $\Phi_{\rm VS}$), the state of the pair along the hidden axis, and its matrix $X = C^- \gamma^0$	Secs. IV D, IX
$V_{\rm ct}$, $V_{\rm VS}$; $m_{\rm ct}$; q	pairing potentials of the contact interaction and of the exchange of the Vorton-Scalar; range parameter of the contact couplings; momentum transfer	Eqs. (67), (68)
a, b; H_a, H_b; H_i	the two scale factors of Sec. XV (sans-serif), for $x0, \ldots, x3$ and for $x5, x6, x7$; their rates; the rate of the direction i	Eq. (90)
$E^{(k)\mu}{}_\nu$; α_1, α_2, α_3; $\bar\Lambda$; κ_8	Lovelock tensor of the order k (the order in parentheses; E_F, E_k, $E_\perp$, E^* are energies); couplings of the Lovelock terms (a digit as subscript; α^j are matrices); cosmological term; gravitational coupling in eight dimensions	Eq. (93)
$T^\mu{}_\nu$; loading; τ_d^0	energy-momentum tensor of matter; $\kappa_8\, \varepsilon(0)/(3\,\mathrm{g55}^2)$; the clock of the undisturbed deflation, $3\,\mathrm{g55}\, x4$	Sec. XV

Lovelock tensor of the third order is built in the same way from sets of six directions, and every such set contains a flat direction: it vanishes.

The field equations. In Eq. (9) the components along $x4$ and along the deflating coordinates read $\mathcal{G}^4{}_4 + \Lambda_8 = 0$ and $\mathcal{G}^t{}_t + \Lambda_8 = 0$. Since $\mathcal{G}^5{}_5 - \mathcal{G}^6{}_6 = (\mathrm{g55} - \mathrm{g66})\,\mathrm{gsum}$, they can hold together only if the three rates are equal, and then $\Lambda_8 = 3\,\mathrm{g55}^2$. The components along the flat coordinates read $-6\,\mathrm{g55}^2 + \Lambda_8 - 12\,\alpha_{\rm GB}\,\mathrm{g55}^4 = 0$, which gives $\alpha_{\rm GB} = -1/(4\,\mathrm{g55}^2)$: Eq. (11). The program verifies Eqs. (C2) and (C3) for the deflating metric in exact arithmetic, and an independent program obtains the

TABLE XXXIV. The symbols of this paper, continued: the model and the superfluid.

symbol	meaning	defined in
$\mathcal{L}^{\rm VS}_{1671d}$; $\Phi_{\rm VS}$, $\hat{\Phi}_{\rm VS}$	Lagrangian of the model; the Vorton-Scalar and its weighted form $\sqrt{V_d}\,\Phi_{\rm VS}$	Eq. (44)
$\lambda_{\rm VS}$, $v_{\rm VS}$, $y_{\rm VS}$	self-coupling of the Vorton-Scalar, the constant that fixes its vacuum value, its Yukawa coupling	Sec. VI
$U(1)_F$, $U(1)_{\rm VS}$; ϵ	the symmetries of fermion number and of chirality; parameter of a symmetry transformation	Sec. VI
m, m_D, m_Y; m_s; $g_{\rm VS}$, $\alpha_{\rm VS}$	mass of a fermion in vacuum, $m = m_D + m_Y$: the bare Dirac mass of the Lagrangian plus the part $m_Y = y_{\rm VS} v_{\rm VS}/\sqrt{2}$ from the Vorton-Scalar; mass of the radial mode; coupling $g_{\rm VS} = m_Y/v_{\rm VS}$ of both modes to the fermions, and $g^2_{\rm VS}/4\pi$	Eq. (45)
$D_{\rm VS}$, C; D_c	the two numbers that decide the state of matter; critical value of $D_{\rm VS}$ for a bound pair	Eq. (49)
τ_d; $\tau_{\rm pair}$	the deflation clock; the clock measured from the binding of the pair	Eqs. (48), (71)
$\varphi_{\rm bos}$, $m_{\rm bos}$, ϖ_b	a mode of the Vorton-Scalar, its mass, the frequency of its weighted form	Sec. VI D
F_S, F_P; $W(k,k')$	exchange amplitudes of the radial and of the phase mode; kernel of the gap equation (always written with its two arguments)	Sec. VIII
λ, λ_S, λ_P; Δ_F; T_c	pairing strength, and its parts from the radial and from the phase mode; gap at the Fermi momentum; critical temperature	Sec. VIII
k_F, E_F, v_F, $N(0)$	Fermi momentum, Fermi kinetic energy, Fermi velocity, density of states at the Fermi surface per internal state	Secs. VIII, XII
$a_{\rm sc}$, E_b	scattering length of two fermions; binding energy of the pair	Sec. X
Ξ_d; $\Delta_{\rm eq}$; $E_{\rm sh}$; $n_{\rm ex}$	the parameter that decides whether the paired state follows the deflation; equilibrium gap; half-width of the shell of states; density of broken pairs	Sec. XI
ξ_k, E_k, κ, n_k	single-particle energy, quasi-particle energy, pair correlation and occupation of a pair of states	Eq. (74)
$V[\Delta]$; ϕ; m^*; $T_{1/2}$	effective potential of the paired state; expectation value of the Vorton-Scalar, $\langle\Phi_{\rm VS}\rangle = \phi/\sqrt{2}$; mass of a fermion in matter; temperature at which the fermions have lost half of their mass	Secs. XII, VII
Δ^a, Δ_t; M; $d_\pm$	order parameter (frame components), and its coordinate components along the deflating coordinates; gap matrix; the two fourfold gaps of a state	Sec. XIV
$SU(2)_A$, $SU(2)_B$, $U(1)_E$	the factors of the internal symmetry of the fields that do not depend on the extra coordinates	Sec. XIV C
P, P_c; w_z; θ_{12}	generator of $U(1)_E$ and the product of the five matrices of the sector; spinor of a zero mode; angle around a vortex line	Sec. XIV D

same Lanczos tensor from its standard definition by computer algebra, for arbitrary rates and constants.

2. Equal rates: a space of constant curvature

Let g55 = g66 = g77, and introduce in place of $x4,\dots,x7$ the coordinates

$$\bar{x}4 = \frac{e^{\mathrm{g55}\,x4}}{\mathrm{g55}}, \qquad \bar{x}t = \mathrm{const}_t\, xt \quad (t = 5,6,7). \tag{C6}$$

The part of the line element that contains $x4,\dots,x7$ becomes

$$-dx4^2 + e^{-2\,\mathrm{g55}\,x4}\sum_t d\bar{x}t^2 = \frac{-d\bar{x}4^2 + \sum_t d\bar{x}t^2}{(\mathrm{g55}\,\bar{x}4)^2}. \tag{C7}$$

This is the line element of de Sitter space of the radius 1/g55, in the half of its flat slicing in which space contracts. De Sitter space is familiar as the geometry of inflation, where the same line element is used with the opposite direction of time. The line element (C7) is a multiple of a flat one. The special conformal transformations of that flat metric along the three axes $\bar{x}5, \bar{x}6, \bar{x}7$ leave the factor $1/\bar{x}4^2$ unchanged and are therefore symmetries. Their generators are, for $s = 5, 6, 7$,

$$\begin{aligned}\mathcal{X}_{(s)} = 2\,\bar{x}s\left[\bar{x}4\,\frac{\partial}{\partial\bar{x}4} + \sum_t \bar{x}t\,\frac{\partial}{\partial\bar{x}t}\right] \\ - \Big(\sum_t \bar{x}t^2 - \bar{x}4^2\Big)\frac{\partial}{\partial\bar{x}s},\end{aligned} \tag{C8}$$

with $\bar{x}4\,\partial/\partial\bar{x}4 = \mathrm{g55}^{-1}\partial_4$. Together with the 14 Killing vectors of the general case and the three rotations of $x5, x6, x7$ they are the 20 Killing vectors of the case of equal rates.

Appendix D: Programs, checks and provenance

This appendix says how every number, table and figure of this paper was computed, how the computation is verified, and how a reader repeats it.

TABLE XXXV. The symbols of this paper, continued: the cosmological part and the Higgs hypothesis.

symbol	meaning	defined in
$a_{\rm sp}$, H, H_0; G_N, G_8	scale factor of ordinary space, Hubble rate, Hubble constant; Newton's constant and the constant of gravitation in eight dimensions	Sec. XVI
ρ, ρ_m; ω_b, ω_c	energy density in ordinary space, and that of matter; the densities of baryons and of cold dark matter of the standard model of cosmology	Sec. XVI
θ_*, r_s; t_0; z_*, $Y_{\rm He}$	angle under which the sound horizon at last scattering is seen, and that sound horizon; age of the universe; redshift of last scattering, mass fraction of helium	Sec. XVI
δ_m, f_g, σ_8	density contrast of matter, its growth rate, amplitude of clustering	Sec. XVI
$p_{\rm agent}$	pressure, in ordinary space, of the agent: of whatever keeps the three rates of deflation constant	Eq. (101)
$\tau_{\rm cf}$; $H_{\rm cf}$, H_e	conformal time of ordinary space (a prime in Sec. XVI E is the derivative with respect to it); conformal Hubble rate, and $H_{\rm cf} - a_{\rm sp}{\rm gsum}/2$	Sec. XVI E
k (Sec. XVI); ℓ, D_ℓ^{TT}	comoving wavenumber of a perturbation of ordinary space; multipole, and the temperature spectrum of the microwave background	Secs. XVI E, XVI G
$h_{\rm sy}$, $\eta_{\rm sy}$; $\alpha_{\rm sy}$	the two functions of the perturbed metric of ordinary space in the synchronous gauge; $(h'_{\rm sy} + 6\eta'_{\rm sy})/2k^2$	Sec. XVI E
$\mathcal{D}$, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{A}$	$8\pi G_N a_{\rm sp}^2$ times the perturbations of the energy density, of the divergence of the momentum density, of the pressure, and the anisotropic stress, of matter and radiation together	Eq. (102)
$\mathcal{E}$, $\mathcal{E}_P$, $\mathcal{E}_m$, $\mathcal{E}_r$; $\mathcal{B}$, $\mathcal{Z}$	$8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe, and the parts of the first from matter and from radiation; two combinations that vanish without deflation	Eq. (107)
$\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$	the gaps: what the momentum constraint, the trace equation and the equation of the shear lack with matter and radiation alone	Eq. (104)
$\Delta\chi^2$; N, z_t; c_{s0}	difference of χ^2 from the best fit of the standard model to the mock data; density of extra radiation in units of that of one neutrino species, and the transition redshift of a stepped fluid; sound speed today of a condensate of bound pairs	Secs. XVI H, XVI I
σ_v; $m_{\rm deg}$, $m_{B,\rm max}$, m_B; $\kappa_{\rm SI}$	velocity dispersion of the dark matter of a halo; the largest masses at which fermions are degenerate and bosons of the mass m_B condensed there; the bound on the cross-section per mass of dark matter	Sec. XVI I
$v_{\rm EW}$, $n_{\rm EW}$; α_c, ϱ_c	electroweak scale and the exponent with which it would follow the coupling under H2; critical coupling of the vacuum condensate, and an abbreviation	Sec. XVII
m_p/m_e; $G_{\rm Fermi}$; $\alpha_{\rm em}$	ratio of the proton mass to the electron mass; Fermi constant; fine-structure constant	Sec. XVI
H1, H2, H3; A1, A2, A3; HV	the three hypotheses about the superfluid; the three assumptions of the cosmological part (A3, on the perturbations, with its closures "Momentum", "Pressure" and "Shear"); the assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	Secs. I, XVI, XVI E

1. The program and how to run it

All physics of this paper is computed by one program, `deflation_cjt_ks`, written in the Rust language (33 source files, 15007 lines with their comments, 68 unit tests). It uses no library other than a second Rust crate of the same repository, which supplies the Dirac matrices, exact integer arithmetic, quadrature, the gap equation and the Einstein-Boltzmann program for static extra coordinates. It is compiled, tested and run from the folder `Revision_Deflation_CJT_and_Kohn-Sham` of the repository [4] with a Rust toolchain alone:

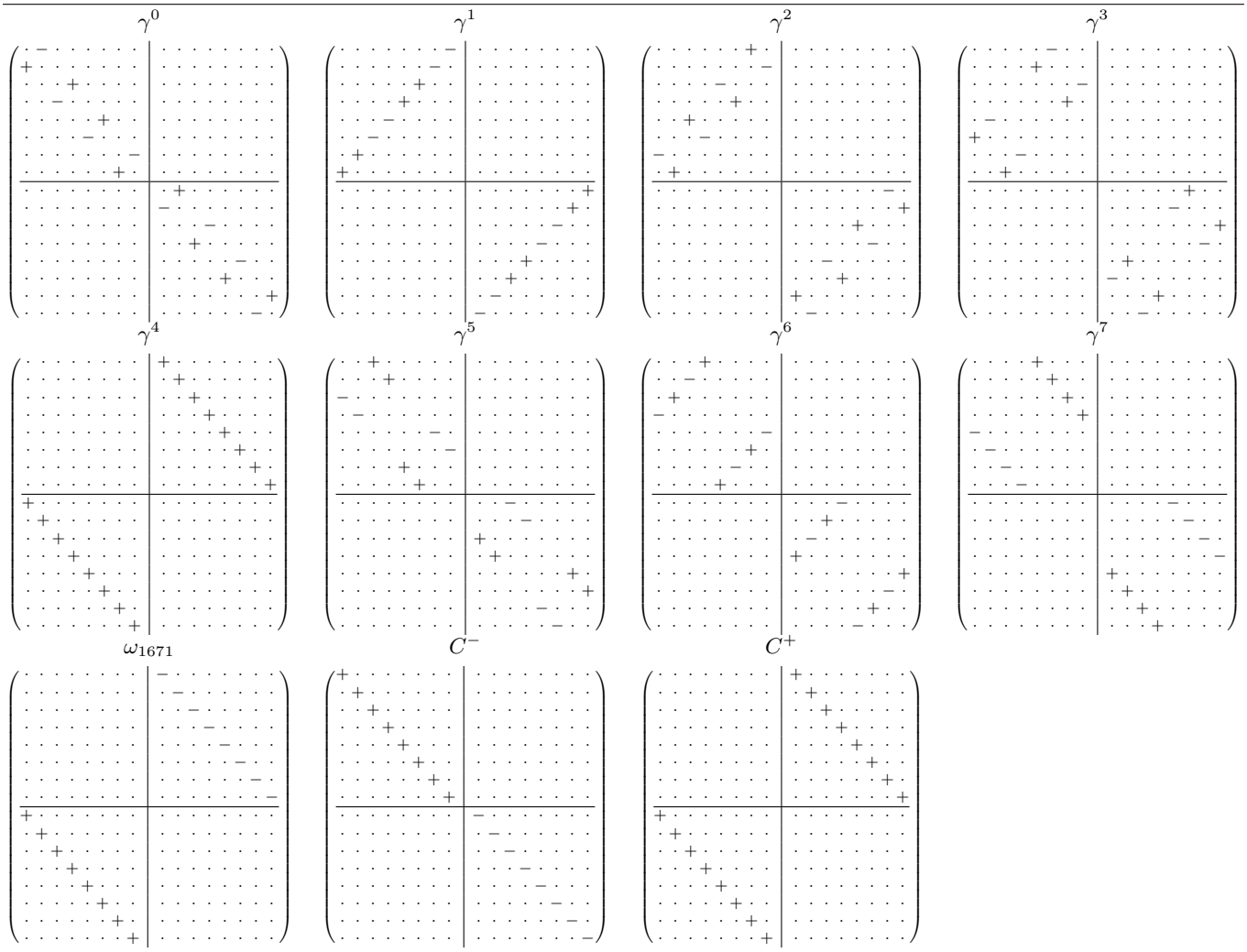
```
M=code/deflation-cjt-ks-rs/Cargo.toml
cargo test  --release --manifest-path $M
cargo build --release --manifest-path $M
B=code/deflation-cjt-ks-rs/target/release
$B/deflation_cjt_ks help
$B/deflation_cjt_ks kohnsham
```

The first and the fourth line only abbreviate two paths. The second line runs the unit tests, the third builds the program, the fifth lists its commands, and the sixth executes one of them. Each command writes its results as tables of numbers into the folder `data`, each table with a description of its content and of its columns at its head, together with a list of *checks*: statements that the command verified, each with its accuracy. A check marked "exact" is an identity verified in exact rational arithmetic; a check marked "numerical" states its largest deviation. Table XXXVII lists the commands. Then

```
python3 code/scripts/make_figures.py
python3 code/scripts/write_doc_tables.py
python3 code/scripts/build_manuscripts.py
```

draw the figures from the data files, write every table and every computed number of the text from the data files, and assemble and typeset this paper. No computed number in a table of this paper was typed by hand.

TABLE XXXVI. The real matrices of the 7+1 spacetime (+ stands for +1, − for −1, a dot for 0; the lines separate the upper and the lower eight components). First and second row: the eight Dirac matrices γ^A of the orthonormal frame. Third row: their product ω_{1671}, and the matrices C^- and C^+.



2. Independent verification

The central results are verified a second time by two scripts in the Python language that share no code with the program and use other methods: computer algebra with symbolic rates and constants instead of rational arithmetic; an implementation of the Grassmann algebra for the invariants of the self-interaction; general-purpose integrators, minimizers and quadratures; the Riemann tensor computed from the metric, and the Lovelock tensors derived as Euler-Lagrange equations of the reduced action, for the equations of gravity; and Peebles' three-level atom [55] for recombination. They are run by

```
S=code/scripts
python3 $S/verify_deflating_independent.py
python3 $S/verify_cjt_ks_independent.py
python3 $S/mutation_test_independent.py
```

The last line tests the two scripts themselves: it introduces deliberate errors into copies of them, one at a time (a wrong sign, a wrong factor, a wrong exponent), and records whether the checks notice. All 36 errors were detected: 20 in the script for the self-interaction, the Kohn-Sham and CJT calculations, pairing and gravity, 16 in the cosmological part of the other script.

3. Where each table and each figure comes from

Tables XXXVIII to XL give, for every table and every figure of this paper that contains computed numbers, the command that computed them, the source files of the program in which the calculation is written, and the data files from which the table or figure was made. The numbers quoted in the running text are taken from the same data files or from the statements of the checks, which the commands write to the files `data/*_checks.txt`; a script (`code/scripts/check_numbers.py`) verifies that every number of the text occurs there.

TABLE XXXVII. The commands of the program: what each computes, and the number of its checks that pass. The checks of both deflating spacetimes are counted, since the program treats the two together. The last four rows are the verifications by other programs.

command	content	checks passed
`selfcoupling`	the nine channels of the self-interaction in exact arithmetic: Fierz matrix, reduction to four couplings, Hartree-Fock couplings, pair couplings, symmetries	34 of 34
`kohnsham`	uniform matter by the Kohn-Sham method and by the CJT effective potential; the filled states of negative energy; bare mass, self-interaction and Vorton-Scalar together	13 of 13
`pairing`	the gap of the pair along the hidden axis with contact couplings and exchange of the Vorton-Scalar	4 of 4
`gravity`	the Einstein-Lovelock equations with matter as a source; stability of the deflation	5 of 5
`matrices`	the Dirac matrices of the frame	106 of 106
`matrices`	geometry, Killing vectors, Einstein and Lanczos tensors, geodesics; matrices in coordinate components, spin connection, Dirac operators	47 of 47
`modes`	solutions of the Dirac equation mode by mode: creation (7+1) and extinction (4+4)	7 of 7
`symmetry`	surviving rotations and boosts, internal symmetry of the sector, zero modes on a vortex line, exponents of the couplings	19 of 19
`channels`	exchange amplitudes in the pair channels; modes of the Vorton-Scalar	7 of 7
`history`	gap, bound pair, matter, vacuum and temperature along the clock; order of events	8 of 8
`quench`	dynamics of the paired state under a growing coupling	5 of 5
`paired`	the paired state and self-bound matter from the effective action	7 of 7
`cosmology`	inferred Hubble constant, bounds on the rate, the exponent n_{EW}	6 of 6
`matter`	dark-matter estimates, the resonance, growth of structure, drifts, temperature of the created quanta	6 of 6
`background`	recombination, sound horizons and helium on the deflating background	2 of 2
`spectrum`	spectra of the microwave background at a fixed angle; the two closures; the gaps and their identities; convergence; growth of structure	8 of 8
`forecast`	mock data against the rate of deflation: Fisher matrices and fits	5 of 5
`families`	the seven families of dark radiation on the deflating background, and together with the rate	4 of 4
`superfluid`	superfluid dark matter: suppression of small-scale structure	2 of 2
independent script 1	geometry, completeness of the Killing vectors, Lanczos tensor, spin connection and Dirac operators by computer algebra; exact solutions; vortex zero modes; Hubble constant; recombination; perturbation equations, identities, closures and growth	43 of 43
independent script 2	self-interaction on Grassmann-valued fields; Kohn-Sham and CJT calculations; pairing amplitudes and gap equation; Lovelock tensors from the metric and from the reduced action; the evolutions with gravity	37 of 37
deliberate errors	errors introduced one at a time into the two independent scripts, and detected by them	36 of 36
registry of symbols	no symbol and no name in a data file has two meanings	7 of 7

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TABLE XXXVIII. Provenance of the tables and figures of this paper, first part. Source files are in `code/deflation-cjt-ks-rs/src`, data files in `data`.

item	command	source files	data files
Fig. 1	`symmetry`	`symmetry_defl.rs`, `cmd_symmetry.rs`	`isometries.csv`
Fig. 2	`matrices`	`metric.rs`, `geometry.rs`, `curved.rs`, `main.rs`	`geodesic_1644d.csv`, `geodesic_1671d.csv`
Table I	`selfcoupling`	`selfint.rs`, `cmd_selfcoupling.rs`	`selfcoupling_fierz.csv`
Fig. 3	`channels`, `modes`	`scalar.rs`, `cmd_channels.rs`, `modes.rs`, `main.rs`	`boson_creation.csv`, `creation_anisotropic.csv`, `creation_history.csv`, `creation_single.csv`
Table II	`symmetry`	`symmetry_defl.rs`, `cmd_symmetry.rs`	`coupling_exponents.csv`
Fig. 4	`history`	`history.rs`, `main.rs`	`events.csv`
Fig. 5	`channels`	`scalar.rs`, `cmd_channels.rs`	`scalar_zero_modes.csv`, `static_source.csv`
Table III	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_flat_reference.csv`
Fig. 6	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_flat_eos.csv`
Fig. 7	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_deflating_71.csv`
Table IV	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_reduced_thresholds.csv`
Fig. 8	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_reduced_saturation.csv`
Table V	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_reduced_gas.csv`
Fig. 9	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_reduced_fixed_density.csv`
Fig. 10	`channels`	`scalar.rs`, `cmd_channels.rs`	`pair_channel_kernel.csv`
Table VI	`history`	`history.rs`, `main.rs`	`gap_history.csv`
Fig. 11	`history`	`history.rs`, `main.rs`	`gap_history.csv`
Table VII	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Fig. 12	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`

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TABLE XXXIX. Provenance of the tables and figures of this paper, second part.

item	command	source files	data files
Table VIII	`history`	`history.rs`, `main.rs`	`dimer_history.csv`
Fig. 13	`history`, `matter`	`history.rs`, `main.rs`, `cosmo_defl.rs`, `cmd_matter.rs`	`dimer_history.csv`, `dm_resonance.csv`
Table IX	`history`	`history.rs`, `main.rs`	`events.csv`
Table X	`quench`	`quench.rs`, `main.rs`	`quench_summary.csv`
Fig. 14	`quench`	`quench.rs`, `main.rs`	`quench_history.csv`, `quench_summary.csv`
Table XI	`paired`	`cmd_paired.rs`	`cjt_history.csv`
Fig. 15	`paired`	`cmd_paired.rs`	`cjt_gl_history.csv`, `cjt_history.csv`
Table XII	`paired`	`cmd_paired.rs`	`matter_saturation.csv`
Fig. 16	`history`, `paired`	`history.rs`, `main.rs`, `cmd_paired.rs`	`matter_saturation.csv`, `medium_history.csv`, `thermal_history.csv`, `vacuum_history.csv`
Table XIII	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`cjt_vs_ks.csv`, `kohnsham_checks.txt`
Table XIV	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`cjt_sea_vacuum.csv`
Fig. 17	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`cjt_sea_saturation.csv`, `cjt_sea_vacuum.csv`
Table XV	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`kohnsham_checks.txt`
Table XVI	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`unified_thresholds.csv`
Fig. 18	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`unified_phase_mode.csv`, `unified_thresholds.csv`
Table XVII	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`unified_phase_mode.csv`
Table XVIII	`paired`	`cmd_paired.rs`	`order_parameter_states.csv`
Table XIX	`gravity`	`gravity.rs`, `cmd_gravity.rs`	`gravity_reference.csv`
Fig. 19	`gravity`	`gravity.rs`, `cmd_gravity.rs`	`gravity_reference_path.csv`

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TABLE XL. Provenance of the tables and figures of this paper, third part.

item	command	source files	data files
Table XX	`gravity`	`gravity.rs`, `cmd_gravity.rs`	`gravity_deflating_summary.csv`
Fig. 20	`gravity`	`gravity.rs`, `cmd_gravity.rs`	`gravity_deflating.csv`, `gravity_vacuum_perturbation.csv`
Table XXI	`gravity`	`gravity.rs`, `cmd_gravity.rs`	`gravity_vacuum_perturbation.csv`
Table XXII	`cosmology`	`cosmo_defl.rs`, `main.rs`	`hubble_deflation.csv`
Fig. 21	`cosmology`, `matter`	`cosmo_defl.rs`, `main.rs`, `cmd_matter.rs`	`growth_deflation.csv`, `hubble_deflation.csv`, `hubble_mass_loss.csv`
Table XXIII	`cosmology`	`cosmo_defl.rs`, `main.rs`	`deflation_bounds.csv`
Fig. 22	`cosmology`, `forecast`	`cosmo_defl.rs`, `main.rs`, `eb_fit.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`deflation_bounds.csv`, `deflation_fisher.csv`
Table XXIV	`spectrum`	`spectra_defl.rs`, `eb_boltzmann.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`cmb_deflation_summary.csv`
Fig. 23	`spectrum`	`spectra_defl.rs`, `eb_boltzmann.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`cmb_deflation_cl.csv`, `cmb_deflation_summary.csv`
Fig. 24	`matter`, `spectrum`	`cosmo_defl.rs`, `cmd_matter.rs`, `spectra_defl.rs`, `eb_boltzmann.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`cmb_deflation_cl.csv`, `cmb_deflation_closure.csv`, `cmb_deflation_gaps.csv`, `cmb_deflation_growth.csv`, `cmb_deflation_summary.csv`, `growth_deflation.csv`
Table XXV	`forecast`	`eb_fit.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`deflation_fisher.csv`
Fig. 25	`cosmology`, `forecast`	`cosmo_defl.rs`, `main.rs`, `eb_fit.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`deflation_fisher.csv`, `deflation_profile.csv`, `deflation_residuals.csv`, `hubble_deflation.csv`
Table XXVI	`families`	`eb_fit.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`dr_deflation_joint.csv`
Fig. 26	`families`	`eb_fit.rs`, `eb_spectra.rs`, `cmd_spectra.rs`	`dr_deflation_joint.csv`, `dr_deflation_profiles.csv`
Table XXVII	`matter`	`cosmo_defl.rs`, `cmd_matter.rs`	`dm_halo_estimates.csv`
Fig. 27	`superfluid`	`eb_boltzmann.rs`, `cmd_spectra.rs`	`sfdm_deflation_summary.csv`, `sfdm_deflation_transfer.csv`
Table XXIX	`cosmology`	`cosmo_defl.rs`, `main.rs`	`higgs_scale_exponent.csv`

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